

A negative solution to the finite lattice representation problem

OpenAI

September 24, 2026

Abstract

We give a negative solution to the finite lattice representation problem. We prove that there is a finite nonempty lattice that is not the full congruence lattice of any finite nonempty algebra of any finite signature.

Contents

1	Introduction	2
1.1	History and the subgroup-interval approach	3
1.2	The proof in two stages	4
1.3	Structure and scope of the proof	5
2	Fences, socles, and coordinate transport	6
2.1	The normal-subgroup test	6
2.2	Subdirect powers and unchanged simple types	8
2.3	Products and one-coordinate intervals	9
2.4	The active chief factor and its core	11
3	Tested chains and the classification reduction	13
3.1	Literature inputs and maximal-step factor actions	14
3.2	A uniform lemma for natural giant actions	16
3.3	The core reduction	20
3.4	Why bounded rank cannot support a long strict chain	21
3.5	The configuration passed to the growth argument	24
4	Fields, indices, and growth along shortcuts	25
4.1	Numerical structure of classical maximal steps	26
4.2	Estimates along a two-edge shortcut	32
4.3	Finite colors for field degrees	34
4.4	Multiplicative inner quotients	35
4.5	A small projective group and a parabolic	36
4.6	Eliminating every crossing path	38
4.7	A common field and the exclusion of radicals	40

5	Components, forms, and rational maps along a path	42
5.1	Extracting actual finite components	43
5.2	Standard finite covers at every field size	43
5.3	Quadratic forms in characteristic two	44
5.4	The retained edge maps	47
5.5	Composing a chosen path	52
6	A uniform bound on natural-module composition length	52
6.1	Semisimplicity and the nonsingular-line exception	53
6.2	Multiplicities and packets	55
6.3	Field shifts and the size of a packet	57
7	Irreducible restrictions and the chain bound	60
7.1	The two rank bounds	60
7.2	From bounded length to natural irreducibility	63
7.3	Restricted natural factors and shortcut tails	64
7.4	The final contradiction	65
8	A finite lattice carrying all the tests	66
8.1	Two original components and their requested tests	67
8.2	The lattice properties required in the reduction	67
8.3	Private insertions and their persistence properties	68
8.4	The shape of one chain test	69
8.5	Combining the forward and reverse tests	71
8.6	Assembly and proof of the lattice properties	73
9	From subgroup intervals to extensions	74
9.1	Minimality in both orientations	75
9.2	The extension dictionary	76
9.3	The detector and the common image	77
10	Boolean gluing of extensions	78
10.1	A criterion for extending over generated domains	79
10.2	An atom whose core is killed	80
10.3	Compatible joins and the contradiction	84

1 Introduction

The congruence lattice of an algebra records all of its quotients at once. The representation theorem of Grätzer and Schmidt says that every algebraic lattice occurs in this way [19, Theorem I, p. 34]. The finite version asks for a finite representing algebra whenever the given lattice is finite. The difference is substantial: a finite congruence lattice may be represented by an infinite algebra without providing any finite representation.

Here a finite algebra is a finite nonempty set equipped with finitely many total operations, each of finite arity. The signature is allowed to depend on the lattice. Nullary operations and the empty signature are allowed. We write $\text{Con}(A)$ for the *entire* lattice of congruence relations on A , ordered by inclusion. Allowing infinitely many finitary operations would give no additional finite representations. Indeed, a finite carrier has only finitely many equivalence relations. For each

relation that is not a congruence, retain one operation witnessing its failure of compatibility. The resulting finite signature has exactly the same congruence lattice.

Theorem 1.1. *There exists a finite nonempty lattice L such that*

$$L \not\cong \text{Con}(A)$$

for every finite nonempty algebra A of every finite signature.

Thus the finite lattice representation problem has a negative answer. The theorem concerns unrestricted finite universal algebras; it is not a restriction to a variety, a bound on the carrier size, or an obstruction to a particular representation method.

Pálffy and Pudlák proved that the following two universal assertions are equivalent [29, Theorem 2]: every finite lattice is the congruence lattice of a finite algebra, and every finite lattice is an interval in the subgroup lattice of a finite group. We establish the failure of the second assertion.

Theorem 1.2. *There exists a finite lattice that is not isomorphic to $[D, G]$ for any subgroup D of any finite group G .*

Here $[D, G] = \{X : D \leq X \leq G\}$ carries subgroup inclusion. The equivalence is global: Theorem 1.2 implies Theorem 1.1, without asserting that a prescribed individual lattice has the two representation properties simultaneously. The one-element lattice causes no exception; it is represented by a one-element algebra and by a trivial subgroup interval.

The constructed subgroup-interval obstruction is self-dual. It also cannot be the entire lattice of intermediate fields of a finite separable extension E/F . Indeed, a finite normal closure N/F gives an order anti-isomorphism with

$$[\text{Gal}(N/E), \text{Gal}(N/F)],$$

by the Galois correspondence; self-duality would turn such a field representation into a forbidden subgroup-interval representation. This is the group-field correspondence discussed in [28, Section 3].

1.1 History and the subgroup-interval approach

The finite problem has several distinct formulations: existence of a representation, finding a representing algebra, and deciding from a finite order table whether a representation exists. McNulty's account separates these questions and places them in their universal-algebraic setting [27, pp. 29–30]. We address the existence question by ruling out a finite subgroup interval. The global transfer just stated is essential: the decorated lattice constructed below is a subgroup-interval obstruction, and we do not identify it pointwise with the lattice asserted by Theorem 1.1.

Two positive representation results clarify the finite requirement. Pudlák and Tůma proved that every finite lattice embeds, preserving meets and joins, into the partition lattice of a finite set [30]. Such an embedding need not realize the given lattice as the entire congruence lattice of operations on that set. On the group side, Repritskiĭ and Tůma represented every finite lattice as an interval in the subgroup lattice of a countable locally finite group [31]. Local finiteness still permits the ambient group to be infinite.

The subgroup-interval approach exposes normal subgroups and primitive sections to lattice constraints. Baddeley and Lucchini developed the reduction of height-two interval questions to finite simple-group problems [3]; Baddeley extended the class of lattice configurations under consideration [4]. Further interval reductions were developed by Börner and Aschbacher [8, 2]. These results have different hypotheses; Pálffy's comparison explains their relationship and the role

of almost simple groups and twisted wreath products [28, Section 4]. In particular, the invariant-subdirect and homomorphism-extension viewpoint in [28] is the starting structure for our final reduction. Aschbacher’s minimum-order and signalizer-lattice viewpoint is a related predecessor of the complementary-socle and extension-kernel arguments.

Lattice shapes can enforce group-theoretic conditions; DeMeo develops this general perspective and the combination of such requirements [16]. Here the particular shapes must do two jobs simultaneously. They force uniform restrictions on every possible simple-group-labelled chain, and they retain the Boolean joins on which the final extensions must agree. Uniformity is the obstacle: every lattice in the construction is finite, but a hypothetical representing group has no prescribed bound on its order, Lie rank, or field size.

1.2 The proof in two stages

Our construction has two stages. The first is a uniform rigidity result for finite configurations of subgroups. We attach a nonabelian simple type T_X to selected vertices X . If D is the common bottom, the unique minimal normal subgroup of $X/\text{core}_X(D)$ is a direct power of T_X ; here $\text{core}_X(D)$ is the largest normal subgroup of X contained in D . Special intervals, called fences, test normal subgroups. Equal labels give subdirect powers, in which selected simple coordinates are linked diagonally. At the comparisons used below, different labels give coordinate products that can be projected to subgroup intervals of almost simple groups. Long chains are supplied with two-edge shortcuts and further interval tests that make these projections available. The crucial conclusion is an *absolute* upper bound on the length of such a chain when its endpoint labels differ. The bound must be independent of the finite group in which the configuration is represented.

The classification of finite simple groups enters this stage through maximal-subgroup structure and representation theory. A subgroup maximal in a maximal subgroup of an almost simple group has at most five nonabelian chief factors [10, Proposition 8.1]. We combine that bound with the fact that coordinate stabilizers in the common bottom already induce the full local permutation actions on the simple factors carrying the lower labels. An elementary bounded-depth action argument then controls the union of nonabelian simple types in the relevant projected coordinate stabilizers, rather than just their number at one vertex. Decreasing bottom cores account for the nonabelian layers lost between socles. These are the uniformity statements needed before applying finite Ramsey theory.

The remaining classical-group comparisons are controlled by field parameters, dimension growth, and irreducible restrictions. The structural starting points are Aschbacher’s maximal-subgroup classification and its classical-group formulations [1, 22], together with Larsen–Pink’s fixed-type envelope theory [24]. Landazuri–Seitz degree bounds separate cross-characteristic growth [23, 20]; Steinberg’s representation theory and Lübeck’s restricted small-degree classification handle the defining characteristic [34, 35, 26]. The paper derives the required bounds on whole chains from these inputs; none of these results alone bounds the lengths of the tested chains. For defining-characteristic representations, the relevant action on absolute constituent types is the difference between a source field shift and an ambient semilinear shift. Their synchronization bounds composition length independently of the field degree. After another Ramsey argument, natural modules at later endpoints restrict irreducibly to earlier ones. The irreducible-triple input comes from Seitz’s classification and the corrections of Cavallin and Testerman, in the formulation of Burness and Testerman [33, 12, 11]. We use it twice: first with a bound on the composition length of the ambient natural module on the smallest subgroup (Lemma 7.1), and then with an intervening simple subgroup of strictly larger rank whose natural module is irreducible on that smallest subgroup (Lemma 7.2). These hypotheses control the twisted-diagonal spin families. The resulting rank

bounds are incompatible with a sufficiently long chain and its two-edge shortcuts.

In the second stage, we build one finite lattice from a Boolean lattice of rank four and a small additional configuration, called the detector. The chain tests are inserted in both required orders while preserving the original Boolean meets and joins. Their length is chosen from the absolute bound, before any representing group is considered.

Suppose the resulting lattice is represented by $[D, G]$, with $|G|$ minimal among representations in either orientation. The global fence gives a socle $S \cong T^I$ and $G = DS$; minimality gives $D \cap S = 1$. Equality of the forward labels forces every proper nonzero original Boolean vertex Y to meet S in a subdirect subgroup, meaning one which projects onto every simple coordinate. Fixing a coordinate gives its stabilizer $A \leq D$ and an action

$$\alpha : A \longrightarrow \text{Aut}(T).$$

A D -invariant subdirect subgroup corresponds to a subgroup U with $A \leq U \leq D$ and a homomorphism $\beta : U \rightarrow \text{Aut}(T)$ extending α . This correspondence reverses inclusion. It realizes the reversed upper filter above Y as the actual group interval $[A, U_Y]$.

The reverse chain tests make the socles of $U_Y / \text{core}_{U_Y}(A)$ powers of one simple group R , which need not equal T . The detector ensures $\alpha(A) \geq \text{Inn}(T)$. Read the original Boolean copy in reverse order. Proper Boolean unions give extensions on generated domains that agree on their overlaps. For a full union, the corresponding forward vertices meet at D , so their intersections with S have trivial intersection. A common extension would produce a nontrivial subdirect subgroup in this intersection and is therefore forbidden.

The final argument forces that forbidden extension. If the core of A in an atom domain had trivial image, compatible lifts of socle factors on the three rank-two faces through that atom would already produce a common extension. Hence every atom core has image containing $\text{Inn}(T)$. Intersections give the same property for larger proper faces. A least normal subgroup of A with this inner image is then normal in all the relevant domains; conjugation on its quotient isomorphic to T supplies their common extension. This is the contradiction.

1.3 Structure and scope of the proof

Section 2 develops the lattice tests and factor transport. The central group-theoretic result is Theorem 3.2, stated in Section 3 and proved in Sections 3–7. These sections control, successively, factor actions and simple types, field and dimension growth, algebraic lifts along paths, natural-module composition length, and irreducible restrictions. Section 8 constructs a finite lattice realizing all the tests. Sections 9 and 10 reduce a hypothetical representation to an extension configuration and rule it out by Boolean gluing, proving Theorem 1.2.

The finite lattice is specified by finitely many private-branch insertions with a sufficiently large finite chain parameter. An explicit numerical value of that parameter is unnecessary for the existence conclusion. Its existence follows from bounds that are uniform over all possible finite-group representations.

All groups are finite unless called algebraic groups. Algebraic groups occur only in the representation-theoretic part of the proof and are taken over algebraically closed fields of positive characteristic, with central covers and the stated special isogenies when needed. We use the classification of finite simple groups and identify each additional classification or representation input at its point of application. The new uniformity, lifting, lattice bookkeeping and gluing deductions are proved explicitly.

2 Fences, socles, and coordinate transport

All groups in this section are finite. A vertex of a subgroup interval is written as a capital letter and identified with the corresponding subgroup. The common bottom is denoted by D . We first record the normal-subgroup test that will give the vertices their simple-group labels. We then explain the two different kinds of factor transport: diagonal merging when the simple type is unchanged, and coordinate products when it changes.

2.1 The normal-subgroup test

Definition 2.1. A bounded lattice interval $[a, b]$ is *fenced* if it has an interior element and, for each $a < c < b$, there are $a < u < v < b$ such that

$$c \wedge u = c \wedge v = a, \quad c \vee u = c \vee v = b.$$

Thus the two comparable elements u, v are both complements of c in the interval. The requirement of an interior element excludes vacuous fences on covering intervals. Length always counts edges.

Lemma 2.2 (Normal-subgroup test). *Suppose that $[D, X]$ is fenced. If $R \trianglelefteq X$, then either $R \leq D$ or $DR = X$. In particular, $[D, X]$ is not modular. The same assertion applies to a normal subgroup of X/F whenever $F \trianglelefteq X$ and $F \leq D$, with bottom D/F .*

Proof. For $D \leq U \leq V \leq X$, the normality of R and the subgroup modular identity give

$$(DR \vee U) \wedge V = (RU) \cap V = U(R \cap V) = U \vee (DR \wedge V). \quad (2.1)$$

If $D < DR < X$, apply the fence to DR and its comparable complements $U < V$. Equation (2.1) would give $V = U$. Thus DR is an endpoint. In a modular lattice the same identity holds with any element in place of DR , so the existence of a fence also excludes modularity. Finally, taking inverse images identifies $[D/F, X/F]$ with $[D, X]$. $\square$

We will repeatedly use the following explicit form of the subgroup correspondence. If $R \trianglelefteq X$ and $X = DR$, then

$$[D, X] \longrightarrow \{E \leq R : D \cap R \leq E, \text{ } E \text{ is normalized by } D\}, \quad U \longmapsto U \cap R \quad (2.2)$$

is an order isomorphism, with inverse $E \mapsto DE$. Indeed $U = D(U \cap R)$ for $D \leq U \leq DR$, and $(DE) \cap R = E$ when $D \cap R \leq E$. This correspondence includes every overgroup of D .

Lemma 2.3 (Labels). *Suppose that $[D, X]$ is fenced, and put*

$$F_X = \text{core}_X(D) = \bigcap_{g \in X} D^g.$$

Then X/F_X has a unique minimal normal subgroup

$$S_X = N_X/F_X \cong T_X^{m_X}, \quad (2.3)$$

where T_X is nonabelian simple and N_X is its full inverse image in X . Moreover,

$$X = DN_X.$$

Every nontrivial normal subgroup of X/F_X contains S_X , and

$$C_{X/F_X}(S_X) = 1.$$

In particular, X/F_X has no nontrivial soluble normal subgroup. If I_X is the set of simple direct factors of S_X , then D and X induce the same transitive permutation group on I_X .

Proof. Factor out F_X , so that the action of X on the cosets of D is faithful. Lemma 2.2 says that every nontrivial normal subgroup supplements D , or equivalently is transitive in this action. Let M be a minimal normal subgroup. If M were abelian, (2.2) would identify $[D, X]$ with an interval of D -invariant subgroups of an abelian group. The latter lattice is modular: intersection is its meet and subgroup product is its join. This contradicts Lemma 2.2.

If M' were another minimal normal subgroup, then $[M, M'] \leq M \cap M' = 1$. Every D -invariant subgroup of M would therefore be normalized by $DM' = X$, and so would be either 1 or M . The correspondence (2.2) would leave at most two elements in $[D, X]$, again a contradiction.

A finite minimal normal subgroup is characteristically simple, hence a direct product of isomorphic simple groups. Since M is nonabelian, these simple groups are nonabelian. Conjugation by X is transitive on the direct factors: otherwise the product of one orbit of factors would be a proper nontrivial normal subgroup of X contained in M . This proves (2.3). Every nontrivial normal subgroup contains a minimal normal subgroup and therefore contains M . Its centralizer is normal and intersects M in $Z(M) = 1$, so that centralizer is trivial. The assertion about soluble normal subgroups follows as well.

Finally $X = DM$. Inner automorphisms of M fix each of its simple direct factors. Thus D and X have the same image on their factor set. Restoring F_X proves the statements as written. $\square$

We call an X satisfying Lemma 2.3 a *labelled vertex*, and call the isomorphism type of T_X its *simple type*. The preceding argument is also the strongly nonmodular normal-subgroup argument in [28, Lemma 3.6].

Corollary 2.4 (Bottlenecks). *Let $D < B \leq Z$, suppose that $[D, B]$ is fenced, and suppose that B is comparable with every member of $[D, Z]$. If $R \trianglelefteq Z$ and $R \not\leq D$, then*

$$B = D(R \cap B).$$

Consequently a soluble normal subgroup of Z is contained in D . If $R, Q \trianglelefteq Z$ and

$$[R, Q] \leq F \trianglelefteq Z, \quad F \leq D,$$

then at least one of R, Q is contained in D . These conclusions also hold with $B = Z$.

Proof. The subgroup $R \cap B$ is normal in B . Suppose first that $R \cap B \leq D$. If $B \leq DR$, then

$$B = B \cap DR = D(B \cap R) = D,$$

a contradiction. Comparability therefore gives $DR \leq B$, whence $R = R \cap B \leq D$. Its contrapositive, followed by Lemma 2.2 in B , gives the asserted supplementation. The image of $R \cap B$ modulo F_B then contains the nonabelian socle S_B , by Lemma 2.3. This is impossible if R is soluble. For the final assertion, $F \leq F_B$ because F is normal in B and contained in D . If both R and Q escaped D , their intersections with B would both map over S_B , while their images would commute. This would make S_B abelian. $\square$

2.2 Subdirect powers and unchanged simple types

For a direct product $T^I = \prod_{i \in I} T_i$, with each $T_i \cong T$, a subgroup is *subdirect* if all coordinate projections are surjective. A *full diagonal strip* on a nonempty subset $J \subseteq I$ is a subgroup obtained by identifying all the T_j , $j \in J$, by isomorphisms, and putting identity elements outside J . We recall the standard diagonal-strip description [28, Lemma 2.1], including its behavior under inclusion.

Lemma 2.5 (Subdirect powers). *Let T be nonabelian simple. Every subdirect subgroup $E \leq T^I$ is a direct product of full diagonal strips on the parts of a partition of I . Both the partition and the coordinate identifications within each part are determined by E . An inclusion of subdirect subgroups corresponds to a refinement of these parts, with the existing identifications restricted to the smaller parts.*

Proof. Induct on $|I|$, separating the last coordinate. The projection to the other coordinates is, by induction, a product T^r of diagonal strips. We have a subdirect subgroup of $T^r \times T$. The kernels of its two projections are normal in the respective factors. The kernel on the last T is either T , in which case this is the full product, or 1, in which case the subgroup identifies a quotient of T^r with T . Normal subgroups of T^r are products of its simple direct factors, so that quotient is one of the factors. The last coordinate is therefore joined to exactly one existing strip.

Two coordinates belong to the same strip precisely when the projection of E to their product is the graph of an isomorphism rather than the full product. This characterizes the partition and its links, and also proves the assertion about inclusion. $\square$

Proposition 2.6 (Equal-type transport). *Let $D < X < Y$ be labelled vertices. Then $F_Y \leq F_X$, and T_X is isomorphic to a section of T_Y . Put*

$$E_X = (X \cap N_Y)/F_Y \leq S_Y.$$

The following conditions are equivalent:

- (i) $T_X \cong T_Y$;
- (ii) E_X is subdirect in S_Y .

When they hold, the quotient map induces an isomorphism

$$E_X \xrightarrow{\sim} S_X, \quad N_X = F_X(X \cap N_Y), \quad F_X \cap N_Y = F_Y. \quad (2.4)$$

Moreover $m_X < m_Y$, and the strip supports give a canonical X -equivariant surjection

$$\sigma_{YX} : I_Y \longrightarrow I_X. \quad (2.5)$$

For three comparable labelled vertices of the same simple type these maps compose: $\sigma_{ZX} = \sigma_{YX}\sigma_{ZY}$.

Proof. The normal subgroup F_Y of X is contained in D , so $F_Y \leq F_X$. Since $Y = DN_Y$ and $D \leq X$,

$$X = D(X \cap N_Y).$$

The image of E_X in X/F_X is a nontrivial normal subgroup and therefore contains S_X . Every simple composition factor of a subgroup of a direct product is a section of one of the factors: projection to one coordinate gives an exact sequence whose kernel lies in the product of the remaining coordinates, and induction applies. Thus T_X is a section of T_Y .

If $T_X \cong T_Y$, some coordinate projection of E_X is all of T_Y ; otherwise none of those smaller groups could have a composition factor of order $|T_Y|$. The transitive action of D on I_Y makes all the projections surjective. Conversely, suppose E_X is subdirect. Lemma 2.5 expresses it as a product of diagonal copies of T_Y . The action of D on those copies is transitive. Since $X/F_Y = (D/F_Y)E_X$, the subgroup E_X is minimal normal in X/F_Y . Its intersection with F_X/F_Y is either trivial or all of E_X . The latter would imply $X = D$, which is excluded. Its image is therefore the unique minimal normal subgroup S_X , proving the equivalence and (2.4).

There are m_X strips on m_Y coordinates. If $m_X = m_Y$, every strip is a singleton, so $E_X = S_Y$ and $X = Y$. Thus $m_X < m_Y$. The quotient identification with S_X assigns to every strip its actual simple factor in I_X and defines (2.5). The subgroup E_X is normal in X/F_Y , and its quotient identification with S_X commutes with conjugation by X ; hence this support map is X -equivariant.

For composition, factor out F_Z and let $E_Y = Y \cap N_Z$. We have $N_Y = F_Y E_Y$ and $F_Y \cap E_Y = 1$. Since $F_Y \leq X$,

$$X \cap N_Y = F_Y(X \cap E_Y) = F_Y(X \cap N_Z).$$

Thus the subdirect subgroup used for $X < Y$ is the image of the one used for $X < Z$ under the already specified identification $E_Y \cong S_Y$. Refinement of its strip partition is exactly composition of the support maps. $\square$

In particular, a change of simple type is strict in order: $T_X \not\cong T_Y$ implies $|T_X| < |T_Y|$. A section of a finite group having its full order must be the group itself.

2.3 Products and one-coordinate intervals

Fix a labelled upper vertex Y . Work modulo F_Y when discussing its socle $S_Y = T^{I_Y}$. Conjugation by D is transitive on I_Y . Choose a reference coordinate and transport identifications of the other coordinates by elements of D . Notation such as P^{I_Y} always means the product of these transported copies. The subgroup in the reference coordinate must be invariant under the corresponding factor stabilizer; this is precisely the condition that makes the transported product invariant under the whole group in question.

Call $D \leq Z \leq Y$ *saturated relative to Y* if $(Z \cap N_Y)/F_Y$ is the product of its coordinate projections. The proof below uses the product–subdirect intersection arguments of [28, Lemmas 2.12 and 2.13] to obtain the required saturation.

Proposition 2.7 (Strict products and saturation). *Let $D \leq X < Y$, where Y is labelled. Suppose that the coordinate projections of*

$$B = (X \cap N_Y)/F_Y$$

are proper and nontrivial. If $[X, Y]$ is fenced, or if X is a coatom of $[D, Y]$, then X is saturated relative to Y .

Once $B = P^{I_Y}$ is such a product, no proper subdirect subgroup of S_Y contains B . Every coatom of $[X, Y]$, and every intersection of these coatoms, is saturated relative to Y . These assertions do not require X to be labelled. For labelled $X < Y$ of different simple types the projection hypothesis holds.

Proof. Factor out F_Y , and let C be the product of the coordinate projections of B . It is X -invariant. The correspondence (2.2), with bottom X and normal subgroup S_Y , identifies $[X, Y]$ with the X -invariant subgroups of S_Y containing B .

Suppose $B < C < S_Y$. If $U < V$ are comparable complements of XC , put $E_U = U \cap S_Y$ and $E_V = V \cap S_Y$. Their projections already contain the projections of C . Hence projecting $\langle E_U, C \rangle = S_Y$ shows that E_U is subdirect; the same holds for E_V .

Every link within a strip of E_U carries the projection of B in one coordinate onto its projection in the other, because $B \leq E_U$. Consequently $E_U \cap C$ is a product of independent diagonal copies of those nontrivial projections, one for each strip of E_U . The proper inclusion $E_U < E_V$ splits at least one strip, by Lemma 2.5. Intersecting with C then permits an additional independent nonidentity coordinate value, and gives $E_U \cap C < E_V \cap C$. This contradicts the required equality of both intersections with B . A fence therefore forces $B = C$. If X is a coatom, $B < C < S_Y$ directly contradicts maximality.

A proper subdirect subgroup has a link between two distinct coordinates. The full product P^{I_Y} contains an element which is identity in the first of these coordinates and nonidentity in the second, contradicting that link. Now let Z be a coatom above X . If its projections were full, it would be such a proper subdirect subgroup. Otherwise its product closure is proper, and maximality forces it to equal that closure. Intersections of products are products.

Finally, for different labelled simple types, the projections are nontrivial because their image contains S_X , and transitivity makes all of them nontrivial. If one were full, all would be full, contradicting Proposition 2.6. $\square$

Proposition 2.8 (One-coordinate projection). *Let $D \leq X < Y$, with Y labelled, and suppose that*

$$(X \cap N_Y)/F_Y = P_X^{I_Y}, \quad 1 < P_X < T_Y.$$

Suppose also that X is an intersection of coatoms of $[X, Y]$; a coatom itself is allowed. Fix $i \in I_Y$ and let

$$\rho : Y_i \longrightarrow \text{Aut}(T_Y)$$

be the action of its factor stabilizer on that coordinate. Identify T_Y with its inner automorphism group, and put $H_ = \rho(X_i)$. Then*

$$H_* \cap T_Y = P_X. \tag{2.6}$$

The saturated overgroups of X form a sublattice of $[X, Y]$, and

$$Z \mapsto K_Z = \rho(Z_i) = H_* P_Z, \quad (Z \cap N_Y)/F_Y = P_Z^{I_Y}, \tag{2.7}$$

is an isomorphism from that sublattice onto the entire ordinary subgroup interval $[H_, H_* T_Y]$. Original covering relations between saturated vertices remain covering relations after projection. In particular, a strict change of simple type at a cover yields a maximal subgroup of an almost simple group, supplementing its simple socle.*

Proof. For a coatom M above X , Proposition 2.7 writes its intersection with S_Y as $P_M^{I_Y}$, with $1 < P_M < T_Y$. Let $J = H_* \cap T_Y$. It normalizes P_M , so

$$P_M J \leq N_{T_Y}(P_M) < T_Y.$$

The last inequality follows from simplicity and $1 < P_M < T_Y$. Both P_M and J are H_* -invariant, and their product therefore transports to a saturated overgroup of M which is still proper in Y . Maximality gives $J \leq P_M$. Intersecting over the coatom representation of X gives $J \leq P_X$; the reverse inclusion is immediate. This proves (2.6).

Products above $P_X^{I_Y}$ correspond precisely to H_* -invariant subgroups $P_Z \leq T_Y$ containing P_X . By (2.6), these in turn correspond precisely to the subgroups K_Z of $H_* T_Y$ containing H_* : the two maps are $P_Z \mapsto H_* P_Z$ and $K_Z \mapsto K_Z \cap T_Y$. Since $Z = X P_Z^{I_Y}$ and the product fixes the coordinate set, $Z_i = X_i P_Z^{I_Y}$, proving (2.7). Intersections and generated subgroups of coordinate products are again coordinate products, proving the sublattice assertion. Any intermediate projected subgroup would lift to an original intermediate saturated subgroup, which proves the assertion about covers. At a strict cover Proposition 2.7 supplies the product hypothesis, and the bottom is itself a coatom. The ambient group $H_* T_Y$ is a subgroup of $\text{Aut}(T_Y)$ containing T_Y , hence is almost simple. $\square$

2.4 The active chief factor and its core

We will need the action on a specified nonabelian chief factor in a projected group. It is not necessary to assert that every projected coset action is quasiprimitive. The following proposition keeps the original cores and the projected cores distinct.

Proposition 2.9 (Active-factor transport). *Let $D < Z \leq Y$ be labelled vertices, and suppose that Z is saturated relative to Y :*

$$(Z \cap N_Y)/F_Y = P_Z^{I_Y}.$$

Let K_Z be the automorphism group induced by a factor stabilizer in Z on a reference coordinate of S_Y . There are K_Z -normal subgroups $Q_Z \leq R_Z \leq P_Z$ such that

$$(F_Z \cap N_Y)/F_Y = Q_Z^{I_Y}, \quad (2.8)$$

$$(N_Z \cap N_Y)/F_Y = R_Z^{I_Y}, \quad (2.9)$$

$$R_Z/Q_Z \cong T_Z^{a_Z}, \quad m_Z = a_Z m_Y. \quad (2.10)$$

Here a_Z is a positive integer. The section R_Z/Q_Z is a chief factor for the action of K_Z on P_Z : there is no K_Z -normal subgroup strictly between Q_Z and R_Z . Its factors give a Z -equivariant surjection

$$\tau_{ZY} : I_Z \longrightarrow I_Y \quad (2.11)$$

whose fibers have size a_Z . The stabilizers in D and Z of a coordinate in I_Y induce the same transitive action on its fiber.

In addition, suppose $X < Z$ and the hypotheses of Proposition 2.8 hold for $X < Y$. Set

$$H_* = \rho(X_i), \quad K_Z = H_* P_Z, \quad C_Z = \text{core}_{K_Z}(H_*).$$

Then the subgroups in (2.8) satisfy

$$Q_Z = C_Z \cap P_Z, \quad K_Z = H_* R_Z. \quad (2.12)$$

Thus $R_Z C_Z / C_Z \cong R_Z / Q_Z$ is a transitive minimal normal subgroup in the coset action of K_Z / C_Z on H_ / C_Z . The groups H_* and K_Z induce the same action on its simple factors. No assertion that this is the entire projected socle is required.*

If $a_Z = 1$, the quotient P_Z / R_Z is soluble.

Proof. Factor out F_Y throughout the proof. Put

$$E = Z \cap N_Y = P_Z^{I_Y}, \quad F = F_Z, \quad W = F \cap E.$$

Let Q_i be the projection of W to the i th coordinate, and let $\hat{Q} = \prod_i Q_i$. These projections are transported into one another by D , and each is invariant under its coordinate stabilizer. Thus $\hat{Q}$ is normal in Z .

We claim that $W = \hat{Q}$. For $q, q' \in Q_i$, choose $w \in W$ whose i th coordinate is q and let $e_i(q') \in E$ be supported only on coordinate i . Since $W \trianglelefteq E$,

$$[w, e_i(q')] = e_i([q, q']) \in W.$$

It follows that $[\hat{Q}, \hat{Q}] \leq W$. The image of $\hat{Q}$ in Z/F is therefore the abelian normal subgroup $\hat{Q}/W$. Lemma 2.3 forces that image to be trivial. Hence $W = \hat{Q}$, proving (2.8). This argument also covers

$Z = X$ at a covering comparison; it does not use a projected core with bottom equal to the whole projected group.

The normal subgroup EF/F of Z/F is nontrivial, since $Z = DE$ and $Z > D$. It contains S_Z . Therefore, if $L = N_Z \cap E$, then

$$N_Z = FL, \quad L/W \cong S_Z.$$

In particular L/W is a perfect normal subgroup of $E/W = \prod_i (P_i/Q_i)$.

Every perfect normal subgroup V of a direct product $\prod_i B_i$ is the product of its intersections with the factors. To see this, for $v, w \in V$ the element $[v, e_i(w_i)]$ lies in $V \cap B_i$ and is the i th coordinate of $[v, w]$. Hence $V' \leq \prod_i (V \cap B_i) \leq V$, and $V = V'$ gives equality. Apply this observation to L/W . It splits as a product of its coordinate intersections, each a direct product of copies of T_Z . Taking full inverse images gives $L = R_Z^{I_Y}$ and (2.9)–(2.10). Conjugation by Z permutes these coordinate intersections and commutes with $L/W \cong S_Z$, so the resulting support map is Z -equivariant. Transitivity of D on I_Y makes the multiplicity a_Z the same in every coordinate.

Any K_Z -normal subgroup strictly between Q_Z and R_Z would transport to a Z -normal subgroup strictly between W and L . Its image would be a proper nontrivial normal subgroup of Z/F inside S_Z . This proves the chief-factor assertion and also the transitivity of K_Z on the a_Z simple factors of R_Z/Q_Z .

We have $Z = DN_Z = DL$. Since $L \leq N_Y$ fixes I_Y , for a chosen coordinate i this gives

$$Z_i = D_i L, \quad K_Z = \rho(D_i) R_Z.$$

The group R_Z acts by inner automorphisms on R_Z/Q_Z and fixes its simple factors. Thus $\rho(D_i)$ and K_Z have the same image on that factor set. The kernel of ρ centralizes the ambient coordinate and so acts trivially on this section. This identifies the action with the original fiber action and proves (2.11) and its assertions about stabilizers.

For the additional projected-core assertion, let $C_Z = \text{core}_{K_Z}(H_*)$. The subgroup Q_Z is normal in K_Z and lies in P_X , because $F_Z \leq D \leq X$. By (2.6) it lies in H_* , and therefore $Q_Z \leq C_Z \cap P_Z$. Conversely, transport $C_Z \cap P_Z$ over all coordinates. Its product is normal in Z and is contained in $P_X^{I_Y} \leq X < Z$. Lemma 2.2 for the original interval $[D, Z]$ forces this product into D , hence into F_Z . Equation (2.8) now gives the reverse inclusion.

Since $\rho(D_i) \leq H_* \leq K_Z$ and $K_Z = \rho(D_i) R_Z$, we obtain $K_Z = H_* R_Z$. Also $R_Z \cap C_Z = Q_Z$. The resulting image of R_Z modulo C_Z is minimal normal by the chief-factor assertion and is transitive by supplementation. Its factor action is the same for H_* and K_Z , because it is already induced by $\rho(D_i)$.

Finally suppose $a_Z = 1$. The support map (2.11) is then a bijection. The subgroup E fixes I_Y and consequently fixes I_Z . Conjugation embeds EF/F faithfully in $\text{Aut}(S_Z)$, since $C_{Z/F}(S_Z) = 1$ by Lemma 2.3. Its image therefore lies in $\text{Aut}(T_Z)^{I_Y}$ and contains the inner subgroup S_Z . The quotient $E/L \cong (P_Z/R_Z)^{I_Y}$ embeds in $\text{Out}(T_Z)^{I_Y}$. Schreier's Theorem, a consequence of the classification of finite simple groups, says that $\text{Out}(T_Z)$ is soluble; see also the explicit use in [28, Section 1]. Hence P_Z/R_Z is soluble. $\square$

Remark 2.10. The projected bottom in Proposition 2.8 is the image of X_i , not necessarily the image of D_i alone: in general $H_* = \rho(D_i) P_X$. The smaller image already induces the full active factor actions, so the actual projected bottom does also. Likewise, the identity $Q_Z = C_Z \cap P_Z$ in Proposition 2.9 is asserted only for $X < Z$. The product-core and chief-factor statements preceding it apply also to the lower endpoint of a covering comparison. These distinctions allow us to use maximal-subgroup information without assuming more quasiprimitivity in the projected diagram than has been proved.

3 Tested chains and the classification reduction

All constants called absolute in this section are independent of the finite group, the lattice, the Lie rank, the characteristic, the field degree, and the length of the chain. A bound depending on a fixed rank bound is stated as such. We use the labels and factor transport of Lemma 2.3 and Proposition 2.9.

For $k \geq 2$, write M_k for the height-two lattice with k atoms; thus M_{16} has eighteen elements.

Definition 3.1 (A tested chain). Let $D < a < b$ belong to a finite group interval, and let

$$a = x_0 < x_1 < \cdots < x_N = b.$$

A tested chain consists of these vertices and vertices v_{ij} for $0 \leq i < j \leq N$, with the following properties.

- (T1) Every x_i and v_{ij} is labelled: its interval above D is fenced. Moreover, $x_i \prec v_{ij} \prec x_j$, where both relations are covers in the entire interval.
- (T2) Every $[x_i, x_j]$, $i < j$, is fenced.
- (T3) Every x_i with $i < N$, and every v_{ij} with $j < N$, is an intersection of coatoms of $[a, b]$. The v_{iN} are coatoms by T1.
- (T4) Put

$$\mathcal{U} = \{x_j : 1 \leq j \leq N\} \cup \{v_{ij} : 1 \leq i < j \leq N\}.$$

For each $u \in \mathcal{U}$, the interval $[a, u]$ is fenced, a is an intersection of its coatoms, and there is $a < c_u < u$ such that the *entire* interval $[c_u, u]$ is M_{16} .

- (T5) For $i < j$ and $x_i \leq z \leq x_j$, either $[D, z]$ is fenced, or there is a labelled z' with $x_i \leq z' < z$ which is comparable with every vertex of $[D, z]$.

No additional requirement is imposed on $[a, v_{0j}]$. In particular, T4 never subdivides the cover $a \prec v_{0j}$.

Condition T5 ensures the following consequences at every overgroup between chain vertices: every soluble normal subgroup is contained in D , and, of two normal subgroups whose commutator lies in a normal subgroup contained in D , at least one is contained in D . At a fenced vertex these follow from Lemma 2.3; at a labelled bottleneck they follow from Corollary 2.4. The same consequences hold modulo a higher labelled core, which is normal in each intermediate vertex and contained in D . No full supplementation dichotomy at an unlabelled bottleneck is asserted. The construction in Section 8 will verify exactly Definition 3.1. Figure 1 isolates the shortcut comparison; the remaining tests make this comparison effective uniformly over finite groups.

Theorem 3.2 (Chain rigidity). *There is an absolute integer B such that every tested chain of length $N \geq B$ satisfies $T_a \cong T_b$.*

The proof proceeds by contradiction through arbitrarily long tested chains whose endpoint types differ. This section gives the classification and uniformity reductions. Proposition 4.16 removes the geometric and cross-characteristic alternatives; the proof of Theorem 3.2 is completed in Section 7. Passing to a subchain always retains the originally supplied shortcuts and tests. It does not assert that the subchain, with a new initial endpoint, is a new tested chain.

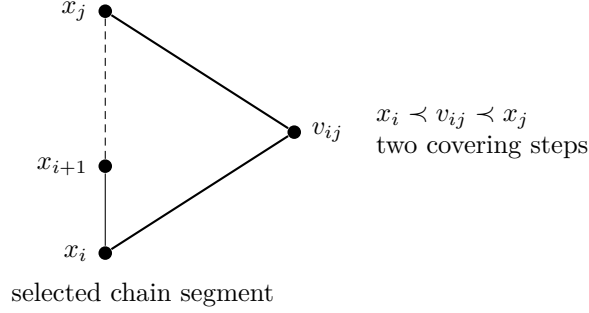


Figure 1: One shortcut in a tested chain. The two thick edges are covers in the entire subgroup interval; the selected chain on the left can have arbitrarily many intermediate vertices and is not required to be saturated. Fences, coatom witnesses, and the M_{16} tests are omitted. The proof compares the restrictions imposed by every short route with those imposed by traversing a long chain.

3.1 Literature inputs and maximal-step factor actions

Proposition 3.3 (Maximal-subgroup input). *The following results are used from the classification and representation theory of finite simple groups.*

- (L1) *A maximal subgroup of an almost simple classical group, not containing its socle, is a stabilizer in one of Aschbacher’s eight geometric collections, or has an almost simple irreducible component. The geometric collections are subspace or flag stabilizers, direct-sum decompositions, extension-field structures, tensor products, subfield structures, symplectic-type groups, tensor-induced decompositions, and forms. Graph automorphisms may interchange the relevant subspaces or flag members; triality and exceptional graph–field automorphisms occur only in bounded rank. We use [1] and the almost simple formulation and descriptions in [22, Table 1.2.A and Theorem 1.2.1, pp. 3–4], with the geometric descriptions in Chapters 3–5; the connected tensor structures and their finite fixed-point versions are also given in [25, Theorem 2 and Section 4(5)].*
- (L2) *The maximal subgroups of alternating and symmetric groups have the intransitive, imprimitive, affine, diagonal, product-action, or almost simple alternatives of the maximal-subgroup form of the O’Nan–Scott Theorem. The relevant formulation and all the alternatives used here are recorded in the proof of [10, Proposition 3.1].*
- (L3) *For a fixed algebraic type, sufficiently general finite subgroups lie between the derived group and the inner-diagonal group of a finite fixed-point group. The complementary alternative is containment in a proper algebraic subgroup of bounded complexity. The bounded-envelope version permits passage to the reductive simple quotients of that subgroup; see [24, Theorems 0.5 and 12.3].*
- (L4) *A subgroup maximal in a maximal subgroup of an almost simple group has at most five nonabelian chief factors [10, Proposition 8.1]. The number refers to chief factors, not to their simple direct factors.*
- (L5) *If $[H, A]$ is a height-two interval and A is alternating or symmetric of degree at least five, it has at most eleven interior vertices [5, Theorem D]. Here the entire interval is required to*

have height two: H is maximal in every maximal overgroup. This is the hypothesis imposed whenever the result is used.

- (L6) *Cross-characteristic irreducible projective degrees satisfy the Landazuri–Seitz lower bounds [23]. In particular, outside bounded rank, a source of natural dimension n and field parameter q requires degree at least q^{cn} for an absolute $c > 0$; the applicable classical bounds are recorded in [20, Tables 3–6]. Above an absolute classical rank, every perfect central cover of a finite simple classical group is a central quotient of its usual simply connected fixed-point group, uniformly over all finite fields, including the standard graph twists [36, Theorem 1.1 and p. 528]. The central-cover deduction and preservation of representations are proved in Lemma 5.3. After passing to that cover, defining-characteristic irreducibles have the Steinberg highest-weight and tensor-product parametrization [34, Theorems 1.1 and 1.3], including the twisted fixed-point groups [35, Theorem 43].*

The clauses of Proposition 3.3 are literature inputs. We first control the actions on simple factors and the resulting chains of projected cores. Numerical field and dimension estimates will enter in Section 4, after the reduction obtained here. No classification of arbitrary subgroup chains is assumed.

For a transitive action of a group K on a factor set, with point stabilizer K_ω , a tower $K_\omega = K_r \leq \cdots \leq K_0 = K$ describes successive invariant block systems. We call the induced coset actions of K_{j-1} on K_j/K_j its *action levels*. The action kernels remain in these stabilizer groups.

Proposition 3.4 (Factor actions at a maximal step). *Consider a strict labelled cover and its one-coordinate maximal subgroup M in an almost simple group, as supplied by Proposition 2.8. Mark its active nonabelian chief factor. There are absolute positive integers b_0, r_0 with the following property.*

- (C1) *The action on the simple factors of the marked chief factor is obtained through at most r_0 transitive action levels. Each level either has degree at most b_0 , or is the natural action of A_d or S_d .*

Proof. We spell out the factor actions. A subspace or paired-flag stabilizer has at most a bounded number of Levi components. A tensor product of two spaces has two component groups. Extension-field, subfield, form, and almost simple cases have one, except for the bounded low-dimensional coincidences. A direct-sum decomposition or a tensor-induced decomposition has the full symmetric action on its blocks, with a possible alternating restriction from the determinant or spinor condition. A block contributes at most a bounded number of simple factors; for instance an orthogonal plus-type four-space contributes two. Taking a point inside a block and then the block itself accounts for this extra bounded level. In the alternating case, the intransitive and affine alternatives have bounded component actions, while the imprimitive, diagonal, and product-action alternatives have this same full block action. Exceptional types have bounded rank and hence bounded numbers of components. Central, determinant, and form quotients do not split a simple component into more factors. This proves (C1); the explicit $\mathcal{C}_2$ and $\mathcal{C}_7$ component quotients can also be read in [10, proofs of Lemmas 5.4 and 5.9]. $\square$

Several structural consequences of the maximal-subgroup descriptions will be needed when we reduce to classical groups of large rank. For a classical target and a classical active source type of sufficiently large rank, a cover with more than one active simple factor belongs to the geometric block or tensor cases. A soluble classical component has absolutely bounded natural dimension: a linear group in growing dimension contains a growing special linear simple section, and the corresponding

assertion for a form group follows from its natural hyperbolic subspaces. For a fixed nonabelian simple type, the possible natural classical realizations have boundedly many dimensions. Outside the bounded list of low-order isomorphisms, its field, family, and dimension are determined by its type; dual, form, and special-isogeny conventions add only boundedly many choices. Finally, a geometric stabilizer has boundedly many component slots apart from repeated blocks. Thus fixing the inactive simple labels, repetition degrees, and bounded structural choices leaves boundedly many possible omitted dimensions. These facts use the full geometric stabilizers in Proposition 3.3(L1), not a bound on the orders of the components. Their numerical field and order data will be established in Section 4.

3.2 A uniform lemma for natural giant actions

The shortcuts will put the relevant factor stabilizers inside one fixed projected bottom group. We need to control their composition types and indices simultaneously. The next two lemmas do this for bounded stabilizer towers, including the overgroups that arise when equal-type strips are merged.

For a finite group H , write $\text{cf}_{\text{na}}(H)$ for its set of nonabelian simple composition-factor types. Only these types will be bounded; the set of cyclic-prime factors need not be.

Lemma 3.5 (Bounded routes). *Fix positive integers r, b . Suppose*

$$A = H_r \leq H_{r-1} \leq \cdots \leq H_0 = H$$

and every induced action on H_{j-1}/H_j is either of degree at most b or is a natural alternating or symmetric action. There are constants h, c, k , depending only on r, b , such that:

- (i) *the interval $[A, H]$ has height at most h ;*
- (ii) *the set $\{[H : U] : A \leq U \leq H\}$ has cardinality at most k ;*
- (iii) *for every such U , its nonabelian composition types belong to*

$$\text{cf}_{\text{na}}(H) \cup \{A_{d-j} : A_d \in \text{cf}_{\text{na}}(H), 0 \leq j \leq c\} \cup \mathcal{E}_{r,b},$$

where $\mathcal{E}_{r,b}$ is a fixed finite set of types.

The conclusions also hold for stabilizers of all invariant quotient blocks of the action on H/A .

Proof. We induct on r . The assertion for one step follows from primitivity of a natural giant, or from the bounded index in the other case. For the induction put $N = \text{core}_H(H_1)$. For an arbitrary $U \in [A, H]$, the group $A(U \cap N)$ lies in $[A, H_1]$, and its intersection with N is $U \cap N$. The induction hypothesis therefore controls the intersection chain and its types. A strict increase in U must increase $U \cap N$ or UN/N , since equality of both gives equality of orders. Furthermore

$$[H : U] = [H/N : UN/N] [N : U \cap N],$$

and the possible second factors are a fixed multiple of the possible indices $[H_1 : A(U \cap N)]$. Thus it remains to control the images.

If H/N has bounded degree, its order is at most $b!$ and there is nothing more to prove. Otherwise it is A_d or S_d , and H_1/N is its natural point stabilizer. Project the remaining route into H_1/N . A natural-giant step projects either to the same action or to a trivial action, because it is primitive; a bounded-degree step retains bounded degree. Starting with the natural A_{d-1} in H_1/N , retain an alternating group normal in the current projected group and supported on all but at most r

points. A natural-giant quotient either kills this alternating group, which is then retained in the next subgroup, or maps it onto the natural alternating socle of its image. In the latter case equality of orders gives the same degree, and its point stabilizer removes one supported point. A step of bounded degree retains the alternating group once its order exceeds $b!$. After a step, take the union of all conjugates of the retained support in the new subgroup. The overlapping-support argument in the next paragraph supplies a normal alternating group on that union, without further loss of points. Apart from a possible sign on the large support, the residual quotient acts on the omitted points and has order at most $2r!$. For $d > 2r + b + 10$, each step therefore loses at most one supported point, and the terminal image contains the alternating group supported on at least $d - r$ points. Here the identification of the next point stabilizer uses $\text{Aut}(A_e) = S_e$ for $e \geq 7$; the degree-six exception and all smaller first degrees are absorbed in the fixed finite universe. Thus we may take $c = r$ in the large-degree argument.

An overgroup of an alternating group supported on $d - c$ points has bounded height and boundedly many possible indices. To see this directly, assume $d > 2c + 4$. The supports of any two conjugates overlap in at least three points, so their alternating groups generate the alternating group on the union of their supports. The union of all conjugate supports is an orbit of the overgroup; its conjugate alternating groups generate the alternating group on that orbit. The orbit omits at most c points. Modulo the alternating group on that orbit, the overgroup has order at most $2c!$. Its large simple type is therefore A_{d-j} with $0 \leq j \leq c$, and its possible indices belong to the finite list of falling factorials $(d)_j$ divided or multiplied by integers bounded in terms of c . The excluded small d add only bounded groups. Combining this with the intersection induction proves all three conclusions. Invariant quotient blocks correspond exactly to the overgroups U already considered. $\square$

Lemma 3.6 (A common index palette). *Fix positive integers R, b, K . Consider any collection of the routes in Lemma 3.5, each of length at most R and bounded degree at most b , such that all their natural giant degrees belong to one set $\mathcal{D}$ with $|\mathcal{D}| \leq K$. The union, over all groups and routes in the collection, of all indices $[H : U]$ with $A \leq U \leq H$ has cardinality bounded by a function of R, b, K . The same holds through a bounded number of operations, each consisting of a permitted route followed by an arbitrary invariant quotient-block choice, with a fresh route inside the chosen block stabilizer permitted at the next operation.*

Proof. Put $d_0 = 2R + b + 10$ and $M = d_0!$, and define the positive rational atoms

$$\Gamma(\mathcal{D}) = \{u/v : 1 \leq u, v \leq M\} \cup \{(d)_j : d \in \mathcal{D}, d > d_0, 0 \leq j \leq R\}.$$

We claim that every index on a route of length r is a product of at most $L_r = r(r+3)/2$ atoms or their inverses. Induct on r , putting $N = \text{core}_H(H_1)$ and $B = A(U \cap N)$. Since $U \cap N \trianglelefteq U$, we have $A \leq B \leq H_1$ and $B \cap N = U \cap N$. Consequently

$$[H : U] = [H/N : UN/N] \frac{[H_1 : B]}{[H_1 : AN]}. \quad (3.1)$$

The image factor is bounded by M when the first action is small. Otherwise the supported alternating-group calculation in Lemma 3.5 gives

$$[H/N : UN/N] = \frac{\varepsilon(d)_j}{t}, \quad 0 \leq j \leq r, \quad \varepsilon \in \{1, 2\}, \quad 1 \leq t \leq 2j!.$$

It therefore uses at most two atoms. The numerator $[H_1 : B]$ uses at most L_{r-1} atoms by induction.

The denominator is controlled across the whole collection, not just held fixed within one route. Project the suffix route into H_1/N . Its degree is exactly

$$[H_1 : AN] = \prod_{j=2}^r [H_{j-1}N/N : H_jN/N]. \quad (3.2)$$

Under any homomorphism with kernel C , a natural step $L_0 > L_1$ projects to the same full natural action or to a trivial action: maximality gives $L_1C = L_1$ or L_0 . A bounded step keeps bounded degree. Thus every factor on the right of (3.2) is an allowed natural degree, a bounded integer, or one, and is a single atom. In particular no arbitrary scalar-kernel divisor enters the denominator. Equation (3.1) gives $L_r \leq L_{r-1} + 2 + (r - 1)$, proving the claim.

There are at most $M^2 + K(R + 1)$ atoms, so the union of all index palettes has cardinality at most

$$(1 + 2(M^2 + K(R + 1)))^{L_R}.$$

Only integral values of these rational recipes occur as actual indices. Invariant quotient blocks correspond to the overgroups U already allowed. If H' is any such block stabilizer, apply the argument anew to the next full route inside H' ; it is not necessary that H' itself admit a natural-step route from H . Indices multiply through a bounded number of these operations, giving the final assertion. $\square$

Lemma 3.7 (Uniform active actions). *Suppose a tested chain has $T_a \not\cong T_b$, and project at one coordinate of the socle of b . Let H_* be the image of the factor stabilizer in a . Then:*

- (i) H_* has at most five nonabelian chief factors;
- (ii) there is one set of nonabelian types, of absolutely bounded cardinality, which contains the types of every whole projected stabilizer group in H_* of an active factor, of an intermediate fiber, or of an invariant quotient block used along a two-edge shortcut, and the types of all their induced images;
- (iii) equal-type stretches of chain vertices have absolutely bounded length;
- (iv) the set of ratios m_{x_i}/m_{x_j} , $i < j$, has absolutely bounded cardinality after equal-type repetitions are removed.

Proof. By Propositions 2.7 and 2.8, the starting group is a nontrivial proper coordinate product, and all designated intersections of coatoms are saturated. The shortcut $a \prec v_{0N} \prec b$ projects to a two-step maximal chain in an almost simple group. Thus Proposition 3.3 gives the five-chief-factor bound for the actual projected bottom H_* . The image of the original D -stabilizer need not equal H_* as a group: in general H_* is that image multiplied by the starting coordinate product. What is required is equality of the induced active permutation actions, as follows.

For every label z , the equality $z = DN_z$ implies that D and z have the same image on I_z , because N_z fixes its individual simple factors. If $I_z \rightarrow I_w$ is a product-step factor map and $g \in z$ fixes $i \in I_w$, write $g = dn$ with $d \in D$ and $n \in N_z$. Then d fixes i and has the same action as g on the fiber over i . Thus the bottom stabilizer induces the *full* local factor action at each level, not an arbitrary subgroup of the natural giant. This property survives the initial projection by Proposition 2.9.

We now account for all the shortcuts, including middles below a proper endpoint. An endpoint x_j is reached from b by the original shortcut $b > v_{jN} > x_j$; take an empty route for $j = N$. The first downward step is strict, because a proper subdirect group cannot contain the nontrivial full

coordinate product at the bottom. Two strict covers give a concatenated route of length at most $2r_0$. A strict cover followed by an equal-type merge instead makes the endpoint factor stabilizer an overgroup of the first route's terminal stabilizer.

Starting at x_j , the middle v_{ij} and the lower endpoint x_i are reached by at most two further original covers. An equal-type operation takes an invariant quotient block. A strict operation supplies a fresh full local route inside the actual current projected factor stabilizer $U \leq H_*$. Indeed the image of the original D -stabilizer lies in U and already has the full local permutation image, whereas U acts through the labelled-node stabilizer because a lies below that node. The kernel of the original-coordinate projection centralizes that coordinate and acts trivially on its sections. Thus these are equal local permutation images; no equality of the stabilizer groups themselves is needed.

The factor maps and equal-type strips respect the original b coordinates. Here is the intersection identity behind this assertion. For labels $x \leq z$ used in a route, work modulo F_b and write

$$x \cap N_b = P_x^{I_b}, \quad N_z \cap N_b = R_z^{I_b}, \quad F_z \cap N_b = Q_z^{I_b}.$$

These product descriptions follow from saturation and Proposition 2.9. Since $F_z \leq D \leq x$ and $N_z = F_z(N_z \cap N_b)$, intersection and the quotient map give

$$\frac{x \cap N_z}{F_z} \cong \frac{x \cap N_z \cap N_b}{F_z \cap N_b} = \prod_{i \in I_b} \frac{P_x \cap R_z}{Q_z}.$$

In an equal-type comparison this is the subdirect subgroup of S_z whose strips describe the merge. Because it is a full product over I_b , no strip can link factors in different original coordinates: such a link would prohibit independent elements in the two factors. For a strict split followed by a merge inside $[x_i, x_j]$, repeat this identity with x_j as upper vertex. The fenced endpoint comparison makes x_i saturated in those coordinates, and the strict shortcut cover makes its middle saturated there. Thus the merge cannot cross those local starting fibers either. Altogether, every required factor, fiber, and quotient-block stabilizer uses at most four original covers, interpreted as full local routes or quotient operations. Four successive operations, each with route length at most $2r_0$, give a conservative common bound, including all intermediate block stabilizers.

We establish the type pool before the degree and index pools. For fixed route parameters let

$$\Phi(\mathcal{P}) = \mathcal{P} \cup \{A_{d-j} : A_d \in \mathcal{P}, 0 \leq j \leq c, d-j \geq 5\} \cup \mathcal{E},$$

where c and the finite exceptional set $\mathcal{E}$ are supplied by Lemma 3.5. If the whole current projected group has nonabelian composition types in $\mathcal{P}$, that lemma puts the types of every route prefix and every quotient-block overgroup in $\Phi(\mathcal{P})$. This includes any kernel of the corresponding permutation action. Retain the original types at every iteration. Starting with $\text{cf}_{\text{na}}(H_*)$, of cardinality at most five, a bounded number of iterations gives one pool of absolute bounded cardinality for all these projected groups. Their induced images have no additional composition types. This proves (ii); it makes no claim about entire unprojected stabilizers in the original group.

For an equal-type stretch, fix its highest vertex. Equal-type transport identifies all earlier vertices with strip subgroups of that socle, whose support partitions strictly change along the stretch. Equal support partitions and subgroup inclusion would force equality of orders and hence equality of subgroups, including the diagonal twists. The preceding coordinate compatibility restricts these to invariant partitions of a single original b -coordinate fiber. Transitivity ensures that a strict change appears on that chosen fiber. All their block stabilizers are overgroups of a single route terminal supplied by the shortcut from b to the highest vertex. The height assertion of Lemma 3.5 proves (iii).

Each full giant of large degree d contributes the composition type A_d to its actual starting projected group. Every such group is a route prefix already covered by the preceding pool. Hence the set $\mathcal{D}$ of giant degree values occurring anywhere in these bounded operations has absolute bounded cardinality. The values may be large. Apply Lemma 3.6 with this entire set $\mathcal{D}$, and multiply indices through the bounded number of operations. This yields a common index palette across all the actions, including arbitrary quotient-block choices followed by fresh full routes. In particular it is a union bound, not merely a separate cardinality bound for each action.

More explicitly, put $q_i = m_{x_i}/m_b$. The original shortcut from b to x_i realizes q_i as the degree on its active-factor fiber: a strict/strict path gives a route degree, and a strict/merge path gives a quotient-block degree. The common recipe palette bounds the cardinality of $\{q_i\}$. Since $m_{x_i}/m_{x_j} = q_i/q_j$, its quotient palette also has bounded cardinality. This proves (iv). $\square$

3.3 The core reduction

Lemma 3.8 (Killed layers and core gaps). *In the initial projection, consider chain vertices strictly above the original starting vertex a , and write*

$$K_i = H_* P_i, \quad C_i = \text{core}_{K_i}(H_*), \quad Q_i = C_i \cap P_i,$$

and let R_i/Q_i be its active nonabelian chief factor. For $i < j$, the inner comparison and its killed kernel are

$$J_{ij} = (P_i \cap R_j)/Q_j, \quad B_{ij} = (Q_i \cap R_j)/Q_j.$$

There is an embedding $B_{ij} \hookrightarrow C_i/C_j$. After deleting at most five nonsoluble gaps, an arbitrarily long subchain remains on which all the B_{ij} , including the analogous shortcut kernels, are soluble. If the chain multiplicities are equal, every endpoint inner comparison, projected to one factor of its upper active socle, has just its active simple factor as a nonabelian composition factor.

Proof. The exact product/core identity in Proposition 2.9 gives, before projection,

$$(F_i \cap N_b)/F_b = Q_i^{m_b}, \quad (N_i \cap N_b)/F_b = R_i^{m_b}, \quad R_i/Q_i \cong T_i^{m_i/m_b}.$$

Since $F_j \leq D \leq x_i$, we have $Q_j \leq P_i$. Intersecting with the lower group and then with its core gives the displayed formulas for J_{ij} and B_{ij} . Also $C_j \leq C_i$ and $R_j \leq P_j$, whence

$$C_j \cap R_j = Q_j.$$

Thus $uQ_j \mapsto uC_j$ embeds B_{ij} in C_i/C_j .

The C_i form a decreasing chain of normal subgroups of the one group H_* . Each nonsoluble successive quotient consumes at least one nonabelian chief factor of H_* . There are at most five, by Lemma 3.7. Cutting at these transitions leaves at most six stretches; one is arbitrarily long. The core of a shortcut middle lies between the cores of its two endpoints, so its killed layers are controlled by the same soluble quotient. No separate bound on a collection of unrelated shortcut cores is being assumed.

For the final assertion work with the original labelled vertices $x_i < x_j$ and put

$$E = x_i \cap N_j, \quad W = F_i \cap N_j, \quad L = N_i \cap N_j.$$

These subgroups are normal in x_i , with $F_j \leq W \leq L$. The kernel of the conjugation action of E on $S_i = N_i/F_i$ is exactly W , because $C_{x_i/F_i}(S_i) = 1$. When $m_i = m_j$, the product support map

$I_i \rightarrow I_j$ is a bijection. Since $E \leq N_j$ fixes I_j , it fixes every factor of S_i . Its socle preimage is L , and therefore

$$E/L \hookrightarrow \text{Out}(T_i)^{m_j},$$

which is soluble by Schreier's Theorem. In the x_j coordinates write

$$E/F_j = P_{ij}^{m_j}, \quad W/F_j = Q_{ij}^{m_j}, \quad L/F_j = R_{ij}^{m_j}.$$

Proposition 2.9 gives $R_{ij}/Q_{ij} \cong T_i$, and the preceding embedding makes P_{ij}/R_{ij} soluble.

The killed kernel is exactly the one controlled in the initial b projection. Indeed $N_j = F_j(N_j \cap N_b)$ and $F_j \leq F_i$ give

$$F_i \cap N_j = F_j(F_i \cap N_j \cap N_b), \quad (3.3)$$

$$W/F_j \cong ((Q_i \cap R_j)/Q_j)^{m_b}. \quad (3.4)$$

It is soluble on the retained stretch, so Q_{ij} is soluble as well. Thus P_{ij} has exactly its one active simple factor as a nonabelian composition factor. The same intersection calculation identifies the killed kernels for either half of a shortcut. This argument takes place in the original labelled x_i ; it uses neither projected quasiprimitivity nor a supplementation dichotomy at an unlabelled bottleneck. A shortcut middle which still has several active factors can retain a nonabelian factor-permutation quotient; this is not asserted to be soluble. Its degrees are controlled by the common type pool, and it occurs only in the split-merge case considered separately below. $\square$

Lemma 3.9 (Exclusion of alternating upper types). *For $u \in \mathcal{U}$ with $T_a \not\cong T_u$, the type T_u is not alternating of degree at least seven.*

Proof. Project the strict comparison from a to u using T4. The bottom is a full product and an intersection of coatoms. The coatoms and their intersection c_u are saturated, so Proposition 2.8 carries the entire interval $[c_u, u]$ to an M_{16} interval ending at an almost simple group with socle T_u . For an alternating socle of degree at least seven that group is alternating or symmetric, contradicting Proposition 3.3. The exceptional automorphisms in degree six concern only bounded-order types. $\square$

3.4 Why bounded rank cannot support a long strict chain

The bounded-rank argument needs a normalized subfield group, rather than just a numerical resemblance between two field sizes. We give the details, including the automorphisms of the almost simple ambient group.

Lemma 3.10 (Bounded envelopes). *Fix a connected adjoint simple algebraic type Φ . There is a constant c_Φ such that a finite subgroup P of an algebraic group X of type Φ has one of the following descriptions.*

- (i) *For a Frobenius map F_0 ,*

$$E = (X^{F_0})' \leq P \leq X^{F_0},$$

where E is simple.

- (ii) *Every nonabelian composition factor of P of order greater than c_Φ is of Lie type in the characteristic of X , with algebraic dimension strictly less than $\dim X$. The total number of nonabelian composition factors, counted with multiplicity, is at most c_Φ .*

In particular, the ranks of nonabelian simple sections of finite groups of Lie type of rank at most R are bounded by a function of R ; simple sections of bounded order are included in this bound.

Proof. Apply [24, Theorem 0.5]. Outside (i), P lies in a proper algebraic subgroup $B < X$ from a fixed constructible family. Apply [24, Theorem 12.3] to that family. It supplies an envelope $H \leq B$, with $P \leq H$, whose connected reductive quotient has simple adjoint factors $X_1, \dots, X_t$ such that the image L_i of $P \cap H^\circ$ satisfies

$$E_i = (X_i^{F_i})' \leq L_i \leq X_i^{F_i},$$

with E_i simple. Each $\dim X_i \leq \dim H^\circ < \dim X$. The kernel of the map from H° to the product of its simple adjoint quotients is soluble: it is an extension of the unipotent radical by the full center of the connected reductive quotient. That center can be disconnected, but is abelian. If L is the image of $P \cap H^\circ$ in $\prod_i L_i$, then L' is a subdirect subgroup of $\prod_i E_i$, because each L_i/E_i is abelian. The subdirect product Lemma makes L' a product of full diagonals. Its simple factors are therefore among the E_i , counted at most once per position. There are at most $\dim X/3$ positions.

The component bound in [24, Proposition 1.4] bounds $|H/H^\circ|$ throughout this constructible family. Its contribution has bounded order and bounded composition length. This proves (ii). Applying the same argument to every subgroup of a finite group of rank at most R proves the last assertion: the ambient algebraic dimensions range over a finite set, and the finitely many small orders and characteristics in the exceptional alternative are absorbed in a function of R . $\square$

Lemma 3.11 (Normalized subfield steps). *Fix an adjoint simple type Φ . Let $T = (X^F)'$ be simple, $T \leq A \leq \text{Aut}(T)$, and let $M < A$ be maximal with $T \not\leq M$. Suppose that $P = M \cap T$ is in alternative (i) of Lemma 3.10, and that the parameter of F_0 is large enough in terms of Φ . Then $F = F_0^r$ for a prime r . Every element of M normalizes the derived fixed-point groups for all intermediate powers of F_0 .*

Proof. First, the pointwise centralizer of $E = (X^{F_0})'$ in $\text{Aut}_{\text{alg}}(X)$ is trivial when the field parameter is sufficiently large. Indeed the fixed groups X^a , for nonidentity algebraic automorphisms a , form a constructible family of proper closed subgroups: take the equalizer of the action and projection maps over the finite-type algebraic automorphism group, remove the identity section, and stratify. Now use [24, Proposition 3.5].

Here is the isogeny calculation needed to include field and exceptional graph operations. Let $\mathcal{I}(X)$ be the group of bijections of $X(\overline{\mathbb{F}}_p)$ generated by self-isogenies and their inverses. The isogeny Theorem [15, Theorem 6.1.16(1) and Proposition 6.1.13(2)] gives a homomorphism

$$\nu : \mathcal{I}(X) \longrightarrow \mathbb{Z}, \quad \ker \nu = \text{Aut}_{\text{alg}}(X).$$

For completeness, after fixing a Borel pair, a self-isogeny permutes the simple roots by π and raises their parameters to powers p^{e_i} . Use the row-coroot convention $C_{ij} = \langle \alpha_j, \alpha_i^\vee \rangle$ for the Cartan matrix. The compatibility condition is

$$p^{e_j} C_{ij} = p^{e_i} C_{\pi(i)\pi(j)}.$$

The labeled undirected Coxeter diagram is preserved. Unless a multiple edge is reversed, connectedness forces all e_i equal. An edge reversal is possible only for B_2, F_4 in characteristic two, or G_2 in characteristic three. The exponents on the two root lengths then differ by one. Thus every self-isogeny is $a\theta^m$, where a is algebraic, $m \geq 0$, and $\theta = F_p$ except in those three cases, when $\theta^2 = F_p$; see also [15, Remark 6.2.10]. Root-group constants are absorbed by the algebraic automorphism, using the pinned uniqueness in the isogeny Theorem. Conjugation by θ preserves algebraic automorphisms, so the same normal form holds with $m \in \mathbb{Z}$ in $\mathcal{I}(X)$. Inseparability degree proves uniqueness of m , and $\nu(a\theta^m) = m$ is the asserted homomorphism. Positive ν gives a Frobenius map: Lang–Steinberg removes its inner factor, leaving a positive diagram–field map [25, Proposition 2.7(i), p. 435]. Its field parameter is p^ν , or $p^{\nu/2}$ in the three exceptional cases.

Let $C_E \leq \mathcal{I}(X)$ fix E pointwise. The preceding centralizer observation makes $\nu|_{C_E}$ injective. Its image is a nonzero subgroup of $\mathbb{Z}$, since it contains $\nu(F_0)$. Thus $C_E = \langle F_1 \rangle$ for a Frobenius map F_1 , and $F_0 = F_1^b$ for some $b \geq 1$. Put $d = \dim X$ and let q_0 be the parameter of F_0 . The fixed-point order bounds and bounded diagonal index in [24, Theorem 3.4(b),(d)] give

$$\frac{(q_0 - 1)^d}{z_\Phi} \leq |E| \leq |X^{F_1}| \leq c_\Phi q_0^{d/b}.$$

Enlarging c_Φ includes the finitely many possibilities for the parameter of F_1 below four. For sufficiently large q_0 , this forces $b = 1$. Consequently $C_E = \langle F_0 \rangle$. Since F fixes E , it follows that $F = F_0^r$ with $r > 0$.

The quotient P/E is abelian and E is perfect, so $P' = E$. Every $m \in M$ therefore normalizes E . The finite automorphism Theorem [35, Theorems 30 and 36] lifts m to an element $\tilde{m} \in \mathcal{I}(X)$. The commutator $\tilde{m}F_0\tilde{m}^{-1}F_0^{-1}$ has $\nu = 0$ and fixes E pointwise. It is therefore the identity. Thus m normalizes $E_e = (X^{F_0^e})'$ for every divisor e of r .

We have $r > 1$, since $r = 1$ would give $T \leq M$. If r were composite, choose $1 < e < r$ dividing r . Fixed-point order estimates, after increasing the original field threshold, give $E_e \not\leq P$ and $1 < E_e < T$. Hence

$$M < ME_e \leq N_A(E_e) < A.$$

The last inequality follows because T is simple and cannot normalize a nontrivial proper subgroup of itself. This contradicts maximality and proves that r is prime. $\square$

Lemma 3.12 (Bounded-rank chains). *For every R there is $b(R)$ such that at most $b(R)$ distinct Lie types of rank at most R occur in a tested chain after the alternating exclusion. In an arbitrarily long family of such chains with distinct types, one may therefore require the ranks of all retained endpoints and all their shortcut middles to exceed any prescribed constant.*

Proof. Consider a subsequence of distinct endpoint types of ranks at most R . A lower simple type is a section of each later one by Proposition 2.6; in particular their orders increase strictly. Remove the boundedly many types below the order thresholds of Lemma 3.10, for the finitely many relevant algebraic types. Every two-edge shortcut has intermediate types of rank bounded in terms of R , by the last assertion of that lemma. Increase the threshold accordingly. On each strict cover read downwards, the characteristic is then unchanged and the algebraic dimension does not increase. The only case of equal algebraic dimension is the sufficiently general alternative. An equal-type cover changes neither characteristic nor dimension.

There are finitely many possible dimensions and algebraic families. If the subsequence were arbitrarily long, it would have an arbitrarily long further subsequence with one dimension and family. On every prescribed two-edge shortcut between its endpoints the dimension cannot drop. Each strict cover is therefore a normalized prime subfield step by Lemma 3.11; a possible equal-type cover contributes no field change. With one fixed family at the endpoints, the conversion between its usual exponent f and the Frobenius parameter exponent cancels. Thus, for every $i < j$,

$$f_j/f_i \in \mathbb{Z}_{>1}, \quad \Omega(f_j/f_i) \leq 2,$$

where Ω counts prime factors with multiplicity. Four successive endpoints would give

$$\Omega(f_4/f_1) = \sum_{i=1}^3 \Omega(f_{i+1}/f_i) \geq 3,$$

a contradiction. All discarded sets and pigeonhole choices have bounds depending only on R , proving the first assertion.

For any desired rank threshold, remove the boundedly many endpoints below a sufficiently larger threshold. If a remaining shortcut middle had bounded rank, its lower endpoint, being a simple section of it, would have rank bounded by Lemma 3.10. The larger threshold was chosen to exclude this. Arbitrarily long chains remain, proving the second assertion. $\square$

3.5 The configuration passed to the growth argument

Proposition 3.13 (Reduction to classical one-factor comparisons). *If Theorem 3.2 is false, then for every s and R there is a retained chain of at least s vertices, with all its original two-edge shortcuts and normal-group tests, such that:*

- (i) *all endpoint types are distinct classical types; their ranks, and the ranks of all shortcut middle types, exceed R ;*
- (ii) *all endpoint multiplicities are equal. Thus each comparison, projected to one upper active coordinate, has one active simple factor;*
- (iii) *the killed kernels of all endpoint and shortcut comparisons are soluble. Every endpoint inner comparison has exactly its one active simple factor as a nonabelian composition factor;*
- (iv) *there is a pool of absolutely bounded cardinality containing all nonabelian simple types in every projected active-factor and fiber stabilizer in the fixed H_* , and in their induced images, used in these comparisons. No assertion concerns the entire unprojected original stabilizers. The splitting multiplicities and invariant-block indices have boundedly many possible values;*
- (v) *each shortcut is either two strict steps, both with one active simple factor and all other inner composition factors soluble, or a strict split followed downwards by an equal-type merge. The latter is geometric. In geometric steps the inactive dimensions have boundedly many possible values once their simple labels and bounded structural choices are fixed; when all inactive components are soluble these dimensions are absolutely bounded.*

All bounds described as absolute are uniform over the original groups. This proposition introduces no new lattice tests at the retained bottom.

Proof. Take counterexamples of lengths tending to infinity, retaining the original endpoints initially. A lower simple type is a section of an upper one, so equality of the endpoint types of a segment forces equality throughout that segment. Lemma 3.7 bounds the lengths of these equal-type stretches. Choose one vertex from each stretch and discard the original initial stretch. There are arbitrarily many remaining vertices, and none has the type of the original a . Their shortcut middles are above a retained lower vertex, so also cannot have type T_a . Lemma 3.9 now excludes all large alternating types at both endpoints and middles. Sporadic groups and all other bounded-order coincidences form a finite list. Bounded-rank Lie types can be removed using Lemma 3.12. This gives (i), with arbitrarily large minimum rank.

All later coordinate projections still use the original tests. For a retained upper endpoint or middle u , condition T4 applies to $a < u$, with $T_a \not\cong T_u$. Proposition 2.8, with original lower vertex a , gives $H_a \cap T_u = P_a$. A retained vertex z saturated in these coordinates then has

$$K_z = H_a P_z, \quad K_z \cap T_u = P_z,$$

because $P_a \leq P_z$. Saturation for a distinct-type endpoint comparison comes from its fence, and saturation at a strict shortcut cover comes from its coatom status. Thus the required inner-intersection identity never relies on inheriting a local coatom-meet representation from the global one below b , or on a new test at a retained bottom.

Use the common cardinality bound on ratios in Lemma 3.7 to color pairs by m_i/m_j . Finite Ramsey gives an arbitrarily long homogeneous subchain. On a homogeneous triple the common ratio t satisfies $t^2 = t$; since $t > 0$, it is one. This proves (ii). Next apply Lemma 3.8 in the original fixed projection: deleting the at most five nonsoluble transitions leaves an arbitrarily long stretch. All shortcut cores between its endpoints lie in the corresponding soluble gap. Passing to any later active coordinate takes subgroups and images of these soluble killed kernels, and hence preserves solubility. Lemma 3.8 and Proposition 2.9 give (iii).

The common type pool and index bounds are precisely Lemmas 3.7 and 3.6; none of these bounds counts abelian factors. They remain valid under all the preceding thinning, giving (iv).

Read a shortcut downwards from its upper endpoint. Its first cover cannot be a proper subdirect merge, because the lower endpoint is a nontrivial full product. If both covers are strict, their positive integral splitting multiplicities multiply to the endpoint ratio one. Both are therefore one. Their killed kernels are soluble by (iii), and their tails are soluble by Proposition 2.9. If the second cover is an equal-type merge, the first must split; the only unbounded-rank maximal-subgroup cases with such an active split are the geometric block and tensor cases described after Proposition 3.4. The merge stays inside each original upper coordinate. Its possible indices are already covered by (iv).

The inactive nonabelian labels in a shortcut belong to the pool in (iv). To see this, write its local inner group as $P \trianglelefteq K$, with soluble killed kernel Q and active chief factor $R/Q \cong T^a$. The faithful conjugation argument in Proposition 2.9 gives $P/Q \hookrightarrow \text{Aut}(T^a)$. The kernel of the factor-permutation map on P/R lies in $\text{Out}(T)^a$ and is soluble. Since P is normal in K , its permutation image is normal in the full local permutation image. Lemma 3.7 identifies that full image with an image of one of the projected stabilizers counted in (iv). Its normal subgroup therefore has its nonabelian composition types in the same pool. Thus all nonabelian composition types outside the active chief factor R/Q lie in this pool. The active type T can still vary along the chain.

Apply now the structural observations following Proposition 3.4. There are boundedly many choices of inactive simple labels by (iv), and the repetition indices also have boundedly many values. The resulting omitted dimensions therefore have boundedly many possible values. When all inactive components are soluble, their individual natural dimensions are absolutely bounded. This proves (v) and all the assertions passed to the next section. $\square$

4 Fields, indices, and growth along shortcuts

We continue the reduction of Theorem 3.2, starting with the configurations supplied by Proposition 3.13. Their distinct classical endpoint types have arbitrarily large rank, their multiplicities agree, and their killed kernels are soluble. We now compare the numerical restrictions imposed by a two-edge shortcut with the growth obtained by traversing several successive comparisons. The three quantities are field exponents, natural dimensions, and inner order quotients. We first establish the maximal-step estimates needed for these comparisons. They will exclude wholly geometric paths and characteristic crossings, leaving common-field paths with quadratic dimension growth and controlled inner order.

Write $z_0 < \dots < z_\ell$ for the retained original vertices, in their original order, and abbreviate $F_i = F_{z_i}$, $N_i = N_{z_i}$, $T_i = T_{z_i}$, $m_i = m_{z_i}$, and $I_i = I_{z_i}$. The common bottom D and lower endpoint a remain those of the original tested chain; every shortcut and normal-group test is inherited from it.

Write

$$T_i = T_i(q_i), \quad q_i = p_i^{f_i}, \quad n_i = \dim V_i, \quad a_i = |T_i|.$$

Here V_i is the natural simple module. In characteristic two the type B_r parameter is $n_i = 2r$, using its simple symplectic natural quotient. For a unitary group the parameter is $q_i = p_i^{f_i}$, although its natural matrix field is $\mathbb{F}_{q_i^2}$. Put $\epsilon_X = 2$ for a unitary type and $\epsilon_X = 1$ otherwise, so the natural matrix field has degree $\epsilon_X f_X$ over its prime field. We retain this conversion for restriction of scalars; twisted-tensor steps will have the exact usual-field relation $f_s = b f_t$.

For each $i < j$ choose the prescribed two-edge shortcut. We write its ascending steps as B_{ij}, A_{ij} and its intermediate simple label as Y_{ij} ; a prime denotes an intermediate parameter. The letters A_{ij} and B_{ij} denote steps, not subgroups. On an individual ascending step, subscripts s, t denote source and target parameters.

All uses of “arbitrarily long” below have the same quantifiers: for every integer s there are such configurations with at least s chain vertices, and their minimum rank can be required to exceed any prescribed constant. A finite coloring followed by finite Ramsey preserves this property. So does retaining every d th vertex for a fixed d . Constants in this section are independent of the configuration and of its length. An explicitly mentioned pool of parameter values may depend on the configuration, but its cardinality has a uniform bound.

4.1 Numerical structure of classical maximal steps

The factor actions needed for the whole-chain reduction were proved in Proposition 3.4. We now derive the field, dimension, and order estimates from the maximal-subgroup descriptions in Proposition 3.3. The proofs retain the actual finite normalizers, since their field actions will enter both the growth argument and the later constituent calculation.

Proposition 4.1 (The maximal-step catalogue). *Consider a strict labelled cover and its one-coordinate maximal subgroup M in an almost simple group with socle T , as supplied by Proposition 2.8. Mark its active nonabelian chief factor. The non-geometric irreducible category below means that its simple source acts absolutely irreducibly on the ambient natural module and that the maximal subgroup is outside all geometric collections, in particular the subfield collection. All growing-rank assertions refer to growing source rank. There are absolute constants $C, c > 0$ with the following properties.*

- (C2) *For unbounded-rank classical active types, a geometric defining-characteristic step changes natural dimension by taking a subspace or quotient, a repeated natural module, a tensor product, or restriction of scalars. Subfield steps leave natural dimension unchanged and multiply the field exponent by a prime. An extension-field step, read upwards from the active factor, has*

$$n_t = b n_s, \quad \epsilon_s f_s = b \epsilon_t f_t, \quad f_t = \frac{\epsilon_s}{b \epsilon_t} f_s, \quad (4.1)$$

where b is the prime degree between natural matrix fields. A twisted-tensor step has

$$n_t = n_s^b, \quad f_s = b f_t, \quad (4.2)$$

exactly; here b is prime, after bounded source-rank cases are removed. All other changes of field exponent belong to a fixed finite set of positive rational factors.

- (C3) *If there is one active simple component and the other geometric components are soluble, their natural dimensions are absolutely bounded. The active dimension in a subspace case is*

comparable with the ambient dimension; a totally singular summand and its dual may represent the same active component twice. A nontrivial parabolic radical in these cases has order between q^{cn_t} and $q^{Cn_t^2}$. For a fixed ambient group, once the active slot, other component dimensions, and the bounded flag/form choices are fixed, there are boundedly many possible orders of the full inner stabilizer divided by the order of its active simple factor, with multiplicity included.

- (C4) A symplectic-type step of unbounded active rank is cross-characteristic. Its source is symplectic or orthogonal over a prime field of order r , of natural dimension $2d + O(1)$, and its target natural dimension is $n_t = r^d$. If $P = M \cap T$, then

$$cn_t^2 \leq |P|/|T_s| \leq Cn_t^2.$$

Its target field exponent is at most $|T_s|^C$.

- (C5) After the twisted-tensor case in (C2) is separated off, a defining-characteristic almost simple irreducible step has a single nonzero restricted, nonnatural Frobenius digit, up to a common Frobenius twist. This restricted module may have further tensor factors in a very special characteristic. Its dimension is at least cn_s^2 . Its source and target field exponents differ by a rational factor from a fixed finite set. Natural, dual-natural, and natural-size special-isogeny representations are included among the geometric cases instead.

- (C6) For a cross-characteristic almost simple step,

$$q_s^{cn_s} \leq n_t \leq |T_s|^C, \quad f_t \leq |T_s|^C.$$

The upper degree bound also holds in defining characteristic. For this and the preceding irreducible case, the homomorphism from the ambient automorphism normalizer of the source simple group to its automorphism group has kernel of absolute bounded order. In particular, a subgroup of that normalizer has order at most C times the order of its induced automorphism group of the source.

Proof. For the dimension and field statements, take the full stabilizers of the indicated structures. On a nondegenerate direct summand the active group has its natural module; on a totally singular summand the dual summand has the contragredient module. A tensor stabilizer acts as the tensor product of the component natural modules. Restriction of scalars multiplies dimension by its degree b and gives $\epsilon_s f_s = b\epsilon_t f_t$ for the natural matrix fields. For example, the mixed $\mathrm{GU}_m(q)$ row in $\mathrm{Sp}_{2m}(q)$ has $b = 2$ and equal usual exponents, since $\epsilon_s = 2$ and $\epsilon_t = 1$. It is a restriction of scalars, with a direct sum after scalar extension, and is not a twisted-tensor row. A subfield inclusion changes only the field. If an ordinary field degree were composite, its proper intermediate field structure would be preserved by the normalizer, contradicting maximality. The classical stabilizer descriptions with prime degree are given explicitly in [10, proofs of Lemmas 5.5 and 5.7]. Unitary conjugation and the diagram maps have bounded order, which accounts for the permitted bounded ratios in the remaining field operations. The twisted-tensor assertion is proved separately in Lemma 4.3 below.

The only soluble classical component groups have bounded natural dimension: a linear group in growing dimension contains a growing special linear simple section, and the corresponding assertions for the form groups follow by their natural hyperbolic subspaces. Thus the omitted dimensions in a one-active-component subspace or tensor stabilizer are bounded. The explicit block description of a parabolic gives a radical of order q^e , with $e \leq Cn_t^2$ and $e \geq cn_t$ whenever that radical is nontrivial in the retained large-rank cases. For example, the linear radical attached to a k -space has exponent $k(n_t - k)$; the symplectic and orthogonal exponents count the off-diagonal blocks subject to one

symmetric or alternating relation. The smallest nonzero retained exponent is linear in n_t . The other factors are the displayed classical groups and scalar groups. Consequently their orders are determined by the dimensions, field, form signs, and the bounded choices of interchanged flags. Intersecting the full stabilizer with the ambient socle adds only its prescribed determinant or spinor condition; it does not introduce a freely chosen subgroup of a scalar group. This proves (C3).

For a symplectic-type normalizer put $N = n_t = r^d$, and let W be the projective image of its normal r -group, of order r^{2d} . The full inner-normalizer rows, including their minimal-field conditions, are given in [9, Section 2.2.6, pp. 70–71, Table 2.9]. Their small exceptions disappear on retaining $d \geq 4$. Ordinary linear conjugation fixes scalars pointwise, so it preserves the nondegenerate scalar commutator pairing $W \times W \rightarrow \mu_r$. Moreover the projective linear centralizer of W is W : a centralizing operator twists the representing matrices by a character of W , that character is implemented by an element of W through the pairing, and the remaining centralizer is scalar by irreducibility.

In the linear and unitary rows, the normal group is an exponent- r extraspecial group for odd r , and a central product $C_4 \circ 2^{1+2d}$ for $r = 2$. In both cases the ordinary projective normalizer has quotient $\mathrm{Sp}_{2d}(r)$ on W . Indeed every symplectic change of generators lifts to a center-fixing automorphism and is implemented by the irreducible representation with its specified central character. For $r = 2$, multiplying a generator by the central element of order four restores its required square; thus a square form attached to a pure extraspecial subgroup is not an invariant of this scalar-extended normalizer. In the unitary case an implementing operator preserves the Hermitian form up to a multiplier, which is removed by a scalar using the surjective finite-field norm. The standard tensor generators of W have determinant one: for odd r this holds for the cyclic shift and diagonal root-of-unity generators, and for $r = 2$ their determinants are $(-1)^{2^{d-1}} = 1$. The map from this normalizer to the abelian quotient of the projective linear or unitary group by its simple socle therefore factors through $\mathrm{Sp}_{2d}(r)$. That group is perfect in the retained range, so the map is trivial. Hence the full ordinary projective normalizer is already the inner intersection P . Quotienting by the entire ambient scalar center leaves kernel exactly W , and

$$P/W \cong \mathrm{Sp}_{2d}(r), \quad |P|/|T_s| = (2, r-1)N^2.$$

This uses the quotient, without requiring the extension to split.

The pure symplectic and orthogonal form rows have $r = 2$. Here the extraspecial square form is invariant. The cited inner rows give $P/W = O_{2d}^\varepsilon(2)$ or $\Omega_{2d}^\varepsilon(2)$, with active simple group $T_s = \Omega_{2d}^\varepsilon(2)$ and index at most two. These are the inner rows after the determinant and spinor conditions; the scalar kernel of the natural cover is removed in W . Consequently $N^2 \leq |P|/|T_s| \leq 2N^2$ in every retained row. There is no extra factor from $r-1$ or the ambient field exponent.

The target-field bound uses the minimal-field condition of the maximal row. For odd r , the degree $e = \mathrm{ord}_r(p)$ divides $r-1$; the usual target exponent is e in the linear row and $e/2$ in the unitary row. The binary and pure form rows have usual exponent one. Thus $f_t \leq \max(1, r-1) \leq |T_s|$. The representation characteristic is different from r , and the active source is over the prime field of order r . This proves (C4).

For an absolutely irreducible representation of a central cover $\tilde{T}_s$, the dimension is at most $|\tilde{T}_s|^{1/2}$: the square of each simple-module dimension is bounded by the dimension of the group algebra. The field-of-definition bound can be obtained directly, including in modular characteristic. For a finite group K in characteristic p , there are at most $|K|$ absolute simple module types, by the semisimple quotient of its group algebra. Frobenius permutes these types, so an absolute simple representation has a Frobenius orbit of length $a \leq |K|$. An isomorphism with its p^a -Frobenius twist descends the representation to $\mathbb{F}_{p^a}$: choose a finite field containing its matrices and an intertwiner,

scale the intertwiner to have norm one, and apply matrix Hilbert's Theorem 90. The scalar norm adjustment is possible because finite-field norms are surjective. Apply this with $K = \tilde{T}_s$. The Schur multipliers of the large classical groups have order at most a polynomial in $|T_s|$; the finitely many exceptional multipliers do not change the bound. Forms cost an extension of degree at most two. Maximality outside the subfield collection makes these bounds applicable to the target field itself. For the cross-characteristic lower bound, the classical tables in [20, Section 3, Tables 3–6] give degree at least $q_s^r/3$ once the source rank $r \geq 6$, including the smaller Weil degrees. Their excluded families have bounded rank. As $n_s \leq 2r + 1$, this is at least $q_s^{n_s/4}$ after increasing the rank threshold. This proves the numerical assertions of (C6).

For the defining-characteristic quadratic bound, use [26, Theorem 5.1 and Table 2] in rank $r > 11$. Modules outside that table have dimension greater than $r^3/8$ in type A , and greater than r^3 in types B, C, D . Every listed nontrivial, nonnatural, nondual degree is at least $r^2/2$. Consequently the retained restricted module has dimension at least $r^2/2 \geq n_s^2/10$. Type B in characteristic two is read through its type- C natural simple quotient, as in our parameter convention; the natural-size special-isogeny cases are already assigned to geometry.

A single occupied ordinary Frobenius digit has full source period: any shift strictly between zero and f_s moves it to a different position in the unique finite restricted label. Returning through f_s applies at most the bounded diagram operation. An internal tensor decomposition of that digit changes neither its occupied position nor this uniqueness argument. Thus the minimal scalar field degree is f_s times one of the bounded diagram periods. The matrix degree is separately $\epsilon_t f_t$. The form-compatible minimal-field argument in Lemma 4.2 bounds the ratio of this matrix degree to the minimal degree outside the geometric collections. This gives the fixed finite rational set of usual exponent ratios and proves (C5). Lemma 4.2 below supplies the final normalizer assertion. $\square$

Lemma 4.2 (Irreducible normalizers). *In a non-geometric absolutely irreducible step of Proposition 4.1, let Y denote the simple source inside the ambient projective classical group T . In particular, the maximal subgroup is not in a subfield collection. Then*

$$|\ker(N_{\text{Aut}(T)}(Y) \longrightarrow \text{Aut}(Y))| \leq C$$

for an absolute C . Moreover $|\text{Out}(Y)| \leq C n_Y f_Y$ for a large-rank classical source with parameters (n_Y, p^{f_Y}) .

Proof. Lift the irreducible projective representation to the perfect central cover of Y . A projective linear element centralizing its image commutes with that representation up to a linear character of the perfect cover, hence commutes exactly. Schur's Lemma makes it scalar, so it disappears projectively.

If a semilinear element in this kernel has scalar-field action σ , the representations ρ^σ and ρ are isomorphic. If σ has order a on the scalar field E , its intertwiner A satisfies

$$A\sigma(A) \cdots \sigma^{a-1}(A) = cI, \quad c \in E^{\sigma,*}.$$

Surjectivity of the scalar norm adjusts A so that the product is one; matrix Hilbert's Theorem 90 then gives a model over E^σ . An invariant bilinear form is a line defined by linear equations over the minimal field, and invariant quadratic-form coefficients satisfy linear equations there as well. In characteristic two, two invariant quadratic forms with the same polarization differ by the square of an invariant linear form. Absolute irreducibility makes that difference zero, so the prescribed quadratic form itself descends up to scalar. Matching finite-field form classes costs at most a quadratic extension. For a unitary group the involution on the minimal field is either nontrivial,

giving Hermitian descent and surjective norms, or trivial, giving a preserved bilinear structure and hence a geometric form case. A larger field period would therefore provide a proper subfield structure normalized by the whole irreducible normalizer, contrary to this step being outside the subfield collection.

To justify the assertion about the whole normalizer, put an absolutely irreducible model over its minimal field E_0 . A normalizing matrix intertwines this model with the model obtained by a source automorphism and a field automorphism; both have entries in E_0 . The intertwining equations therefore have coefficients in E_0 . Their solution space over the algebraic closure is one-dimensional by absolute irreducibility. It consequently has a nonzero E_0 -rational solution, necessarily invertible. Up to a scalar, the normalizer thus preserves the projective E_0 -structure. The same argument applies after dualizing in a graph component, and the uniqueness of the invariant form gives the already allowed quadratic extension. A graph operation contributes at most another factor of two in unbounded rank. This bounds the kernel independently of the ambient field degree.

Finally the diagonal, field, and graph description of automorphisms gives respectively factors at most n_Y , f_Y , and an absolute constant in the outer group. $\square$

Twisted tensor subgroups and their normalizers were studied in detail by Schaffer [32]; the algebraic and finite tensor structures used here are those of [25, Section 4(5)]. The next lemma proves the precise maximality consequence and field relation needed for these tested chains.

Lemma 4.3 (Twisted tensor maximality). *An almost simple irreducible maximal subgroup with a nontrivial Steinberg tensor decomposition is, in growing source rank, a geometric twisted-tensor step with the exact dimension and field relations in (C2). Its tensor-orbit degree is prime.*

Proof. Let S be the simple source, M the almost simple maximal subgroup, and T the ambient simple classical group. Work over an algebraic closure on the cover supplied by Lemma 5.3, and write the irreducible module as

$$V = \bigotimes_{i \in I} V_i, \quad \mathcal{A}_i = \text{End}(V_i) \otimes 1 \leq \text{End}(V),$$

where V_i are the nonzero ordinary restricted Frobenius digits, with their indicated twists. The dimensions of these nontrivial digits tend to infinity with the source rank.

We first verify preservation of the actual tensor structure. A source field automorphism translates ordinary digit positions, applying the diagram operation when a position passes the source period; inner, diagonal, and diagram automorphisms act on the restricted modules at those positions. Ambient semilinear operations give the same shifts, and a graph operation dualizes the modules. Uniqueness of the finite restricted highest-weight parametrization matches the resulting whole digit modules. Tensor their individual intertwiners, with the matching permutation, to obtain an intertwiner of V . The discrepancy between a projective and linear intertwining identity is a linear character of the perfect cover and is trivial. Absolute irreducibility makes the space of whole-module intertwiners one-dimensional. Every actual normalizing operator therefore differs from this tensor intertwiner by a scalar and permutes the embedded algebras $\mathcal{A}_i$. For a graph operation the corresponding map on endomorphisms is an anti-isomorphism, which has the same conclusion. This argument preserves a whole ordinary digit even if that restricted module has an internal tensor decomposition in a very special characteristic.

Replace each digit by its full connected special linear group, or by its full connected bilinear-form group when required. Bilinear self-duality makes each ordinary digit self-dual separately. Unitary descent acts on the special linear factors, permitting duals. In characteristic two every nontrivial irreducible self-dual digit has an alternating form: a nonalternating invariant form would give an

invariant linear functional by taking square roots of its diagonal. The tensor product of at least two nondegenerate alternating spaces has a quadratic form with the product polarization and value zero on pure tensors [25, Proposition 2.2(iv)]. It agrees with the prescribed finite quadratic form, after a scalar normalization of the polarization: their difference is the square of an invariant linear form, which vanishes by nontrivial absolute irreducibility. These constructions give a connected semisimple tensor subgroup K in the ambient form group.

The groups are intrinsic to the algebras $\mathcal{A}_i$ and their restricted adjoint involutions. Hence the actual ambient Frobenius F and the full finite normalizer preserve K . In growing classical rank, finite ambient automorphisms have lifts commuting with F ; diagonal lifts satisfy this projectively because $F(x)x^{-1}$ is central. Graph and field lifts also commute, with only the bounded-rank triality and exceptional graph cases removed. This is the setup of [25, Theorem 2 and Section 4]. Henceforth view K , and later K' , in the ambient adjoint group. Thus M normalizes the terminal derived subgroup $H = (K^F)^{(\infty)}$. Since H is perfect it lies in T , and $S \leq H$ because S is a perfect subgroup of K^F .

For each orbit of F on the factors, fixed points are obtained by choosing one coordinate fixed by the corresponding power of F ; see [25, Proposition 2.8(ii)]. Consequently H is a central product of classical components, one per orbit. It is proper in T also as a finite group. To check this uniformly, put $q = p^{f_t}$, let the individual digit dimensions be $d_1, \dots, d_k$, put $d_{\min} = \min_i d_i$, and write $n_t = \prod_i d_i$. The earlier bound for nontrivial digits gives $d_{\min} \rightarrow \infty$ with the source rank. Elementary classical order bounds give

$$|H| \leq q^2 \sum_i d_i^2, \quad |T| \geq q^{n_t^2/8}.$$

For $k \geq 2$, $\sum_i d_i^2/n_t^2 \leq k/d_{\min}^{2(k-1)} \leq 2/d_{\min}^2$. Thus $H < T$ for all sufficiently large source ranks. If MH were the entire almost simple ambient group, its normal subgroup $H \leq T$ would equal T . Hence MH is proper, and maximality gives $H \leq M$. But the perfect group H/S is a subgroup of the soluble group M/S , so $H = S$. There is therefore one Frobenius orbit, and its full natural classical component is the source. If its length is b , the digit dimension is n_s and $n_t = n_s^b$ exactly.

The field relation is read from this actual natural-factor cycle. Put $t = f_t$. For a linear target, F^b on one factor is the standard field map with exponent bt , up to an inner conjugacy. For a unitary target, write $F = F_{p^t} \tau$ with τ the inverse-transpose operation. On one factor the return map is $F_{p^{bt}} \tau^b$, again up to an inner conjugacy. Its source is unitary when b is odd and linear when b is even, in both cases with usual exponent bt . An orthogonal graph choice changes the form sign, not this exponent. The characteristic-two quadratic construction likewise changes no field exponent. Thus

$$f_s = b f_t \tag{4.3}$$

exactly, in the convention of this paper. The natural factors have positions $0, t, \dots, (b-1)t$ modulo bt , which are distinct. For the odd unitary convention using positions $2jt$, multiplication by two permutes the same residues modulo b ; the prime two-cycle in a unitary target has a linear source. These are precisely the usual-field alternatives in [25, Section 4(5)].

Finally suppose $b = uv$, with $u, v > 1$. Number the factors by $\mathbb{Z}/b\mathbb{Z}$, and group the cosets of its unique subgroup of order u . The normalizer's permutation normalizes the regular cyclic descent action and preserves this block system. The algebra for a block B is the algebra generated by its $\mathcal{A}_i$, namely $\text{End}(\bigotimes_{i \in B} V_i) \otimes 1$. Replace each block by its full special linear or bilinear-form group. For a characteristic-two quadratic target use the full symplectic group on each grouped alternating space. The resulting quadratic tensor form is unchanged: it has the same polarization and vanishes on every fully pure original tensor, and such tensors span V . Hence this grouped subgroup K' still preserves the prescribed ambient form and is invariant under F and M .

Its terminal derived fixed-point subgroup H' has one classical component on dimension n_s^u over the field with usual exponent vf_t , with the unitary parity or orthogonal sign prescribed by the return map. For $n_s \geq 16$, the elementary bounds $Q^{d^2/8} \leq |\text{Cl}_d(Q)| \leq Q^{2d^2}$ give

$$|S| \leq q^{2uvn_s^2} < q^{vn_s^{2u}/8} \leq |H'|, \quad |H'| \leq q^{2vn_s^{2u}} < q^{n_s^{2uv}/8} \leq |T|.$$

The strict inequalities follow from $n_s^{2u-2} > 16u$ and $n_s^{2u(v-1)} > 16v$. As before M normalizes H' and $S \leq H'$. Its properness in T forces $H' \leq M$, and perfectness then forces $H' = S$, contradicting the first inequality. Therefore b is prime. $\square$

4.2 Estimates along a two-edge shortcut

We record how the catalogue applies to the reduced configurations. In the estimates below, a *nonlinear* cover means a non-geometric defining-characteristic irreducible cover as in (C5); a *crossing* cover means an almost simple cross-characteristic cover or a symplectic-type cover. The field estimates constrain prime degrees, the dimension estimates can be iterated along the chain, and the inner order quotients will multiply through intermediate vertices.

- (G1) For a geometric or defining-characteristic almost simple cover, (C2) and (C5) express the field ratio as a fixed rational factor times the contribution of a subfield, extension-field, or twisted-tensor degree. An exceptional degree is prime or belongs to a fixed bounded set of positive integers. All bounded rational corrections, including two, are fixed before any prime coloring is made. The restriction-of-scalars and twisted-tensor formulas are (4.1) and (4.2), respectively. The mixed unitary restriction of degree two is a bounded geometric case, with $\epsilon_s/\epsilon_t \in \{1/2, 1, 2\}$. A natural space and its dual give a separate dimension operation, not an additional restriction degree. Other form and special-isogeny rows retain their own fixed rational field factors; no telescoping of ϵ factors through those rows is used. Lemma 4.3 supplies distinct tensor positions modulo f_s , including the unitary alternatives. Thus there is no matrix-field factor of two in (4.2), and the dimension formulas for a two-step subfield/extension or subfield/twisted-tensor path are exact.
- (G2) The geometric dimension operations in (C2), together with Proposition 3.13(v), give boundedly many omitted dimensions once the active slot, permutation data, and inactive simple labels are fixed. In paths with no replication and no inactive nonabelian factor, these dimensions are bounded integers; ordinary tensor factors other than the active one then have bounded dimensions. An unbounded replication degree belongs to the bounded-cardinality pool in Proposition 3.13.
- (G3) Clause (C5) gives $n' \geq cn^2$ on a nonlinear cover, with nontrivial Steinberg tensor decompositions already assigned to geometry. Clauses (C4) and (C6) give the common crossing estimate

$$n' \geq \exp(cn \log q), \quad n', f' \leq |T|^C. \quad (4.4)$$

For an almost simple cover the upper degree bound also holds in defining characteristic. Every ascending same-characteristic geometric step decreases natural dimension by at most a fixed factor, by G1–G2. It also decreases $n \log q$ by at most a fixed factor. For restriction of scalars, (4.1) gives

$$n_t \log q_t = \frac{\epsilon_s}{\epsilon_t} n_s \log q_s \geq \frac{1}{2} n_s \log q_s, \quad n_t = bn_s \geq n_s.$$

For a twisted-tensor step, (4.2) gives $n_t \log q_t = (n_s^b/b) \log q_s \geq n_s \log q_s$ when $n_s, b \geq 2$. The other geometric rows have fixed positive lower factors. A same-characteristic nonlinear step has both lower-bound properties as well, by its quadratic degree bound and fixed positive field ratio.

- (G4) In a one-factor almost simple cover the inner order quotient is at most $|\text{Out}(T)|$. Clause (C4) bounds the symplectic-type quotient between cN^2 and CN^2 , where N is the target natural dimension; the target dimension and field exponent are at most $|T|^C$ for source T . For a parabolic or unipotent subspace cover with one active factor, no inactive nonabelian factor, and bounded field ratio, (C3) makes the target dimension n' comparable to the source dimension n and places the inner quotient between $q^{cn'}$ and q^{Cn^2} . For fixed target type and parameters that quotient has boundedly many possibilities. This is a bound on cardinality; the quotient itself is usually very large.

A strict split followed by an equal-type merge is geometric. Every shortcut containing an almost simple irreducible or a symplectic-type cover has one active simple factor at both covers and no other nonabelian inner composition factor, by Proposition 3.13. For those shortcuts, the next lemma transfers the normalizer bound to the actual lower coordinate stabilizer.

Lemma 4.4 (Transport of a coordinate stabilizer). *Suppose that the last cover is an almost simple irreducible cover with source Y , and that all three shortcut multiplicities agree, as in every application below. Denote the lower node's full upper-coordinate stabilizer image by M_{ij} , and let P_B be the inner projection in the first cover. Then*

$$|M_{ij}| \leq C |P_B| n'^C (1 + f')^C. \quad (4.5)$$

Proof. Write the actual shortcut as $U = z_i \prec V \prec W = z_j$, and work modulo F_W . In one W -coordinate the last cover has $Y \trianglelefteq P_A \leq \text{Aut}(Y)$. Its killed kernel from the middle is soluble by Proposition 3.13 and normal in P_A , hence is trivial. The active subgroup in that coordinate is therefore the actual simple group Y . Proposition 2.9 gives a V -equivariant bijection $\beta : I_V \rightarrow I_W$ because the multiplicities agree. For $\alpha \in I_V$ and $t = \beta(\alpha)$, the actual stabilizers in U coincide: $U_\alpha = U_t$. Let ρ_V, ρ_W be their middle and upper coordinate actions, and let $\phi : N_{\text{Aut}(T_j)}(Y) \rightarrow \text{Aut}(Y)$ be conjugation. The quotient-induced actions satisfy

$$\phi \circ \rho_W|_{U_t} = \rho_V|_{U_\alpha}, \quad \phi(M_{ij}) = M_B,$$

where M_B is the actual first-cover coordinate stabilizer image.

The inner-intersection assertions use the original lower endpoint a from Definition 3.1. All retained comparisons start strictly above this a , so none uses an excluded middle v_{0j} . Condition T4 makes a an intersection of coatoms in $[a, V]$ and $[a, W]$ for every retained middle and upper endpoint. Proposition 2.8 in these original- a projections identifies the inner part of a saturated image with its transported inner projection. For $U \prec V$ the coatom case of that Proposition also applies directly. In particular,

$$M_B \cap \text{Inn}(Y) = P_B.$$

There is no assumption that a meet of coatoms in the original global interval is a meet of coatoms in every smaller interval. The bounded kernel from Lemma 4.2 now gives

$$|M_{ij}| \leq C |M_B| \leq C |P_B| |\text{Out}(Y)| \leq C |P_B| n' f',$$

which implies (4.5). This includes diagonal, field, and graph operations. All higher-core losses are already kernels of the displayed actual coordinate actions; no uncontrolled factor from the field of the final target is inserted. $\square$

4.3 Finite colors for field degrees

For a positive rational x , let $v_\ell(x)$ be its valuation at a prime ℓ . If S is a finite set of primes, put

$$\Omega_S^+(x) = \sum_{\ell \notin S} \max\{v_\ell(x), 0\}, \quad \Omega_S^-(x) = \sum_{\ell \notin S} \max\{-v_\ell(x), 0\}.$$

Both sums have finite support.

Lemma 4.5 (Finite prime complexity). *Let $x_1, \dots, x_s$ be positive real numbers whose pair ratios are positive rationals. Suppose that, for every $i < j$,*

$$x_j/x_i = c_{ij} u_{ij}^\varepsilon, \quad c_{ij} \in \mathcal{C}, \quad \varepsilon \in \{1, -1\},$$

where $\mathcal{C}$ is a fixed finite set of positive rationals, u_{ij} is a positive integer with at most h prime factors counted with multiplicity, and ε is the same for all pairs. There is a number of colors depending only on $\mathcal{C}, h$ such that on every homogeneous triangle all three x 's are equal. Consequently arbitrarily long such arrays have arbitrarily long constant subarrays.

Proof. Let S contain the primes occurring in the numerators or denominators of members of $\mathcal{C}$. For $\ell \in S$, the integer $v_\ell(x_j/x_i)$ lies in a fixed bounded interval: the contribution of c_{ij} is bounded, and that of u_{ij} has absolute value at most h . Outside S all valuations have sign ε , and the sum of their absolute values is at most h . Color (i, j) by the valuation vector on S and this last sum. This is a finite coloring with the asserted uniformity.

For $i < j < k$ we have

$$v_\ell(x_k/x_i) = v_\ell(x_j/x_i) + v_\ell(x_k/x_j).$$

On a homogeneous triangle each coordinate of the common valuation vector therefore equals twice itself, so it is zero. Outside S there is no cancellation because all valuations have the same sign. Their total absolute value likewise equals twice itself, hence is zero. Every valuation of every pair ratio is zero, so all ratios equal one. Finite Ramsey gives the last assertion. $\square$

Lemma 4.6 (Opposite field moves). *Arbitrarily long homogeneous shortcut chains cannot have, on every comparison, one proper subfield step and one proper extension step, including the twisted-tensor extension alternative.*

Proof. Color first by the order and kind of the two steps. By G1, their combined dimension formula is either

$$n_j = b_{ij} n_i \quad \text{for every } i < j, \quad \text{or} \quad n_j = n_i^{b_{ij}} \quad \text{for every } i < j,$$

where $b_{ij} > 1$ is prime or belongs to a fixed bounded set of integers. In the first case put $x_i = n_i$, and in the second put $x_i = \log n_i$. In either case $x_j/x_i = b_{ij}$ has uniformly bounded prime complexity. Lemma 4.5, with $\mathcal{C} = \{1\}$, gives a homogeneous triangle with $b_{ij} = 1$, a contradiction. Notice that field ratios alone would not suffice: ratios r_j/r_i of distinct primes are compatible on arbitrarily many vertices. It is the exact dimension formula that excludes this possibility. $\square$

Proposition 4.7 (Purely geometric paths). *An arbitrarily long reduced chain cannot have exclusively geometric two-edge shortcuts. This includes strict split/equal-type merge shortcuts.*

Proof. Color by the two cover categories, the endpoint family, and whether a split and merge occur. There are finitely many colors. The endpoint characteristic is common because every step is in defining characteristic. If both an increasing and a decreasing exceptional field move occur, Lemma 4.6 applies. Otherwise all exceptional primes in a pair ratio have the same sign. The product

of the two bounded rational factors belongs to a fixed finite set, and at most two exceptional degrees occur. A bounded exceptional degree has uniformly bounded prime complexity as well. Apply Lemma 4.5 to $x_i = f_i$. We may therefore assume that all endpoint fields and families agree.

The exceptional degrees at the individual covers are now bounded integers. Indeed, with no opposite field moves their product, or its reciprocal, is a member of the fixed finite set of bounded twisting ratios, because $f_j/f_i = 1$. Each positive degree divides the numerator or denominator of one of these finitely many rationals.

Fix a lower endpoint i . We show that its upper natural dimension has only boundedly many possible values. For block and parabolic shapes, the ambient dimension is the active dimension, or twice that dimension when its dual is present, plus the dimensions of the omitted blocks. The dimensions of nonabelian omitted blocks are determined, with bounded ambiguity, by the bounded-cardinality pool of inactive types. Soluble omitted blocks have bounded dimensions. A tensor shape multiplies the active dimension by a product of the omitted dimensions. A replicated base multiplies it by a replication degree from its bounded-cardinality pool. A form change has the same natural dimension, with the bounded natural-module exceptions included in the family color. Field restriction multiplies dimension by one of the bounded degrees just obtained, and a twisted-tensor step raises it to one of those degrees. Thus one geometric cover gives a bounded list of possible outputs from a fixed input. Two covers give a list whose size is at most the product of the two bounds. For a split followed by a merge the second step has the same simple label, so gives no new natural dimension at all.

All labels now have the same family and field. Their strict growth in order makes their natural dimensions distinct. There cannot be more upper endpoints than the bounded number of possible outputs from the fixed lower endpoint. This contradicts arbitrary length. $\square$

4.4 Multiplicative inner quotients

From now on each comparison contains a nonlinear or crossing step. There is no splitting. The multiplicity is the common integer m from Proposition 3.13, also at the two covers of every chosen shortcut. Let $P_{ij} \leq T_j$ be the saturated inner projection of the lower node at an upper coordinate, and let $M_{ij} \leq \text{Aut}(T_j)$ be its full induced coordinate stabilizer. Applying Proposition 2.8 with the original a and condition T4, as explained in Lemma 4.4, gives $P_{ij} = M_{ij} \cap T_j$.

Lemma 4.8 (Index quotients). *The positive integers*

$$e_{ij} = |P_{ij}|/|T_i|$$

satisfy $e_{ik} = e_{ij}e_{jk}$ for every $i < j < k$. They multiply in the same way through each selected shortcut.

Proof. The active factor T_i is a composition factor of P_{ij} , so its order divides $|P_{ij}|$. Saturation and supplementation give

$$[z_j : z_i] = [T_j^m : P_{ij}^m] = \left(\frac{|T_j|}{|P_{ij}|} \right)^m.$$

Multiplying group indices through z_j and taking the positive m th root gives

$$\frac{|T_k|}{|P_{ik}|} = \frac{|T_k|}{|P_{jk}|} \frac{|T_j|}{|P_{ij}|}.$$

Rearranging and dividing by $|T_i|$ proves the formula. The same calculation applies at a shortcut because its three multiplicities agree. No identification of its intermediate label with a chain label is required. $\square$

We use the soluble-normal and commuting-normal conclusions of Lemma 2.3 and Corollary 2.4. Fix retained labelled endpoints $z_i < z_j$ and put $F_j = \text{core}_{z_j}(D)$. This original higher core is normal in z_j and contained in D , hence normal in every intervening group. Write

$$S_j = N_j/F_j = \prod_{t \in I_j} T_{j,t}, \quad (z_i \cap N_j)/F_j = P_{ij}^{I_j}.$$

For each M_{ij} -invariant subgroup $R \leq T_j$, write $\hat{R} \leq S_j$ for its transported coordinate product. It is normalized by z_i/F_j .

Lemma 4.9 (Coordinate normal tests). *Let*

$$z_i/F_j \leq \bar{U} < z_j/F_j$$

be an actual subgroup, and let U be its full inverse image in z_j . A transported product normal in $\bar{U}$ is called killed if it is contained in D/F_j ; it then lies in $\text{core}_{\bar{U}}(D/F_j)$.

(N1) *If $\hat{R} \trianglelefteq \bar{U}$ is soluble, then*

$$\hat{R} \leq D/F_j, \quad R \leq P_{ij}. \quad (4.6)$$

(N2) *Let $R, Q \leq T_j$ be M_{ij} -invariant, with $\hat{R}, \hat{Q}$ normal in the same $\bar{U}$. Suppose that*

$$[\hat{R}, \hat{Q}] \leq \bar{K} \trianglelefteq \bar{U}, \quad \bar{K} \leq D/F_j. \quad (4.7)$$

Then at least one of $\hat{R}, \hat{Q}$ is killed, and its coordinate subgroup is contained in P_{ij} .

Proof. For $U > z_i$, the original test T5 provides a fenced interval or a labelled bottleneck at U . If $U = z_i$, the endpoint itself is labelled. Since $F_j \trianglelefteq z_j$ and $F_j \leq D$, apply Lemma 2.3 or Corollary 2.4 in the quotient by F_j , with bottom D/F_j . The soluble-normal conclusion gives (4.6). For (N2), the full inverse image of $\bar{K}$ is normal in this same U and contained in D , so the commuting-normal conclusion kills one product. Since $D \leq z_i$, a killed product has its coordinate subgroup in P_{ij} . $\square$

For (N1) we use the enlargement

$$\bar{U} = (z_i/F_j)\hat{R}; \quad (4.8)$$

for (N2) we use $\bar{U} = \langle z_i/F_j, \hat{R}, \hat{Q} \rangle$. Properness and normality in this same enlargement are checked in each application. If one tested product contains a transported active subgroup known to map onto the original lower labelled socle, it is not killed, so the other product is killed. Containment in P_{ij} alone does not imply that a product is killed and cannot replace the normality and bottom-containment hypotheses on $\bar{K}$. Solubility here is a property of the quotient product; its full inverse image need not be soluble. No normality of the lower core F_i in the enlargement, or full supplementation dichotomy at an unlabelled bottleneck, is used.

4.5 A small projective group and a parabolic

Lemma 4.10 (Small coordinate stabilizers). *There is an absolute constant $c_0 > 0$ with the following property. Let T be a classical simple group of sufficiently large natural dimension n , over parameter $q = p^f$, and let $M \leq \text{Aut}(T)$ have order m . If*

$$n > 2(m^2 + m),$$

then M normalizes a proper parabolic subgroup of T whose unipotent radical has order at least $q^{c_0 n}$. Consequently the coordinate normal test excludes $M = M_{ij}$ whenever, in addition, $q_j^{c_0 n_j} > |M_{ij}|$.

Proof. Use the natural-space field $E = \mathbb{F}_{p^e}$; thus $e = f$, or $e = 2f$ in the unitary case. Apart from the bounded-rank graph exceptions already excluded, automorphisms in the form cases are represented by projective semilinear similarities. In the linear case there may also be correlations, which interchange the natural space and its dual. The ordinary orthogonal graph operation is a similarity in this description; characteristic-two type B uses its symplectic natural quotient.

First suppose that M acts on the natural-space side. Choose one semilinear representative s_g for each projective element, with $s_1 = 1$. The products satisfy $s_g s_h = c(g, h) s_{gh}$ for nonzero scalars $c(g, h) \in E$. For an unknown vector $v \in E^n$ impose

$$B(s_g v, s_h v) = 0 \quad (g, h \in M)$$

in a bilinear or Hermitian form case, and impose also $Q(s_g v) = 0$ in a quadratic form case. In a plain linear space no equations are necessary. Field automorphisms and semilinear maps are $\mathbb{F}_p$ -linear. These are therefore at most $e(m^2 + m)$ scalar polynomial equations of degree at most two in en prime-field variables. Their sum of degrees is less than en .

For completeness, the elementary polynomial count giving a nonzero solution is as follows. If $F_1, \dots, F_t$ have total degree sum less than the number N of prime-field variables, the number of their common zeros, reduced modulo p , is

$$\sum_{x \in \mathbb{F}_p^N} \prod_{a=1}^t (1 - F_a(x)^{p-1}).$$

Every expanded monomial has degree less than $N(p-1)$, so some variable has exponent less than $p-1$. The sum over that variable is zero in $\mathbb{F}_p$, including exponent zero. The displayed sum is zero. All our equations vanish at zero, so their common zero set has another element. This is the needed Chevalley–Warning argument [13, 38].

Choose such a nonzero v and put

$$U = \text{span}_E \{s_g v : g \in M\}.$$

Then $1 \leq \dim U \leq m$. Scalar discrepancies between representatives do not change this span, so U is M -stable. The imposed equations make U totally singular in the form cases. In particular, in characteristic two we used the quadratic equations as well as the polar equations. Thus the stabilizer of U is a proper parabolic normalized by M .

Now let T be linear and suppose M contains correlations. Put $H = M \cap \text{PTL}(V)$ and choose $d \in M \setminus H$. On subspaces the correlation has the form

$$d(S) = \text{ann}(F(S))$$

for an invertible semilinear map $F : V \longrightarrow V^*$. Choose representatives s_h , $h \in H$, and impose

$$F(s_h v)(s_k v) = 0 \quad (h, k \in H).$$

Again there are at most $e|H|^2$ quadratic scalar equations. The same count gives $v \neq 0$. Let $U = \text{span}_E \{s_h v : h \in H\}$ and $W = d(U)$. The equations say $U \leq W$. Since H is normal in M , both subspaces are H -stable. Moreover $d(W) = d^2(U) = U$, because $d^2 \in H$. Thus the flag $U < W$ is preserved by all of M , as an unordered pair under correlations. Its dimensions are $r, n-r$, where $1 \leq r \leq |H| < n/2$, so it is a proper flag. Its parabolic stabilizer is normalized by M . No involution assumption on d has been used.

In these constructions the small subspace has dimension $1 \leq r \leq m < n/4$ once n is sufficiently large. The upper triangular block matrices in the radical supply at least $n/4$ independent prime-parameter root coordinates, each of order at least q . For example the linear case has $r(n-r)$ coordinates, and the flag case has at least $r(n-2r)$; in the polar cases the block from the nondegenerate middle space to the singular r -space supplies at least $r(n-2r-2)$ coordinates. These lower bounds are at least $n/4$ after enlarging the fixed dimension threshold. We may take a smaller absolute c_0 to cover all the families. Projectivization does not remove unipotent coordinates: the scalar center has order prime to p .

Let J be this proper parabolic and R its radical. The group R is soluble and normalized by M . In the coordinate application, take the actual enlargement (4.8). Its inner part is

$$\bar{U} \cap S_j = (P_{ij}R)^{I_j}, \quad P_{ij}R \leq N_{T_j}(J) < T_j.$$

The last inequality holds because J is a nontrivial proper subgroup of the simple group T_j . Thus $U < z_j$; properness only in $\text{Aut}(T_j)$ would not suffice. The soluble coordinate normal test gives $\hat{R} \leq D/F_j$, hence $R \leq P_{ij} \leq M_{ij}$. Its order contradicts $q_j^{c_0 n_j} > |M_{ij}|$. $\square$

4.6 Eliminating every crossing path

Put $\mathcal{E}_0(x) = x$ and $\mathcal{E}_{h+1}(x) = \exp(\mathcal{E}_h(x))$.

Lemma 4.11 (Repeated crossing growth). *Suppose every comparison in a reduced homogeneous chain has a crossing. For every fixed h, C and every sufficiently distant pair $i < j$,*

$$n_j > \mathcal{E}_h(a_i^C).$$

The required distance is a constant depending only on h, C and the constants in the catalogue. Also $n_j > a_i^C$ for every fixed C after a fixed separation and a sufficiently large minimum rank.

Proof. Every ascending same-characteristic geometric step preserves both natural dimension and $n \log q$ up to fixed positive lower factors, as established in G3. The same holds for a same-characteristic nonlinear step by its quadratic degree bound and fixed positive field ratio. If the last cover crosses, a preceding same-characteristic cover leaves $n' \log q'$ at least a fixed multiple of $n_i \log q_i$. A preceding crossing has the same consequence from $n' \geq \exp(cn_i \log q_i)$ and $\log q' \geq \log 2$ at sufficiently large rank. The last crossing then gives the endpoint lower bound. If only the first cover crosses, it gives $n' \geq \exp(cn_i \log q_i)$, and the last cover preserves natural dimension up to a fixed factor. Decreasing c and increasing the fixed minimum-rank threshold, we obtain in every case

$$n_{i+1} \geq \exp(cn_i \log q_i).$$

Write $u = n_i \log q_i$. Classical order formulas give

$$\log a_i \leq C_1 n_i^2 \log q_i \leq C_2 u^2.$$

After the first step every additional comparison gives at least the map $x \mapsto F(x) = \exp(c \log(2)x)$. For fixed C_3 ,

$$\frac{F(\exp(cu))}{\exp(C_3 u^2)} \longrightarrow \infty \quad (u \longrightarrow \infty).$$

If $L(u)/R(u) \longrightarrow \infty$ and $R(u) \longrightarrow \infty$, then $F(L(u))/\exp(R(u)) \longrightarrow \infty$ as well. Induction therefore shows that $h+2$ comparisons dominate $\mathcal{E}_h(\exp(C_3 u^2))$, and hence $\mathcal{E}_h(a_i^C)$ when $C_3 \geq CC_2$. This also proves the last assertion. The target prime characteristic need not be bounded in this argument: its only use in the iteration was $\log q \geq \log 2$. $\square$

Proposition 4.12 (No characteristic crossings). *After passage to a reduced homogeneous chain, neither cover of any comparison changes the defining characteristic.*

Proof. Suppose otherwise, and color by the two step categories. Every pair in the resulting arbitrarily long chain has a crossing. There are three possibilities for its last step A .

The last step is geometric in the same characteristic. The first step B crosses. Hence $n', f' \leq a_i^C$ by (4.4). An ordinary geometric last step changes dimension by bounded factors, except for an extension degree. By (4.1) that degree is $b = \epsilon' f' / (\epsilon_j f_j) \leq 2f'$. A last subfield degree can be arbitrarily large but leaves dimension unchanged. A twisted-tensor last step has exponent $b = f' / f_j \leq f'$ by (4.2). Therefore

$$n_j \leq (Cn')^{C(1+f')} + C(1+f')n' + C \leq \exp(a_i^{C'})$$

for a fixed C' . Lemma 4.11 contradicts this bound on a sufficiently distant pair.

The last step is symplectic type. Its source Y is a symplectic or orthogonal group over a prime field, say of order r , and its target dimension is r^d with $2d + O(1) = n'$. First suppose B is in the same characteristic. Then $r = p_i$ and $f' = 1$. In a subfield first step the formula $1 = \rho f_i b$, with ρ in a fixed finite set of positive rationals, bounds both f_i and b ; in the untwisted formula such a proper step is impossible. An ordinary extension degree in B satisfies $b = \epsilon_i f_i \leq 2f_i$, because the target here has $f' = 1$ and $\epsilon' = 1$; a twisted-tensor exponent satisfies $b = f_i$. The ordinary geometric formulas give a fixed-height tower bound on n' in terms of a_i . If B is nonlinear, its irreducible degree is at most a_i^C , giving the same conclusion. Since $p_i \leq a_i$, the formula $n_j = p_i^d$ gives $n_j \leq \mathcal{E}_h(a_i^{C'})$ for fixed h, C' . This contradicts Lemma 4.11.

It remains in this case that B also crosses. Let b_{ij}, s_{ij} be the inner multipliers of B, A . By G4,

$$1 \leq b_{ij} \leq a_i^C, \quad cn_j^2 \leq s_{ij} \leq Cn_j^2, \quad e_{ij} = b_{ij}s_{ij}.$$

The bound for b_{ij} is the outer-order bound when B is almost simple; in the symplectic-type case it follows from $b_{ij} \leq C(n')^2 \leq a_i^{C'}$. For $i < j < k$, Lemma 4.8 now gives

$$cn_j^2 \leq e_{ij} = \frac{b_{ik}s_{ik}}{b_{jk}s_{jk}} \leq \frac{C}{c} a_i^C.$$

The factors involving the common upper dimension n_k have canceled. This contradicts Lemma 4.11 on a sufficiently distant pair $i < j$.

The last step is almost simple irreducible. Let its source be Y . If B crosses, then $n', f' \leq a_i^C$ and $|P_B| \leq a_i^C$: for an almost simple B this uses $|P_B| \leq a_i |\text{Out}(T_i)|$, and for a symplectic-type B it uses the preceding bound on its multiplier. Equation (4.5) gives

$$|M_{ij}| \leq a_i^{C'}. \tag{4.9}$$

It is irrelevant here whether A itself is a crossing.

Suppose next that B is in the same characteristic, so that A crosses. If B is not a subfield step, its geometric dimension formulas, or its irreducible degree bound when nonlinear, bound n' by a fixed-height tower in a_i . Its field satisfies $\log q' \leq C \log q_i$: restriction of scalars contributes the bounded factor ϵ_i / ϵ' , and the other non-subfield rows have their fixed field factors. Thus $|Y|$ is bounded by a fixed-height tower in a_i , and $n_j \leq |Y|^C$ contradicts Lemma 4.11.

Finally suppose B is subfield. Its rank equals the lower rank and its characteristic equals p_i . Its inner projection has order at most a_i^C , but f' can be unbounded. The last crossing gives

$$\log n_j \geq cn' f' \log p_i, \quad f' \leq C \log n_j.$$

Equation (4.5), with n' comparable to n_i , therefore gives

$$|M_{ij}| \leq a_i^{C'} (1 + \log n_j)^{C'}. \quad (4.10)$$

The same upper bound includes (4.9). By Lemma 4.11, after a fixed separation we have $a_i^{C'} \leq n_j^{1/16}$; and, for sufficiently large rank, $(1 + \log n_j)^{C'} \leq n_j^{1/16}$. Hence $|M_{ij}| \leq n_j^{1/8}$. This satisfies the dimension inequality of Lemma 4.10, and $q_j^{c_0 n_j} > n_j^{1/8} \geq |M_{ij}|$. That Lemma gives the final contradiction.

The three possibilities exhaust the last cover in a crossing path. $\square$

4.7 A common field and the exclusion of radicals

Lemma 4.13 (Common defining field). *On the remaining arbitrarily long chains, the endpoints can be taken to have one common family and one common field $q = p^f$. The intermediate field ratios belong to a fixed finite set of positive rationals. In particular, all extension and twisted-tensor degrees in these paths are bounded integers. Every comparison satisfies*

$$n_j \geq c n_i^2. \quad (4.11)$$

Proof. Propositions 4.7 and 4.12 leave a defining-characteristic nonlinear cover on every comparison, with the common characteristic of the endpoints. Color by endpoint family and cover categories. A nonlinear cover has a bounded rational field ratio; only the other cover can carry an exceptional prime degree. Lemma 4.5 applied to f_i therefore gives equal endpoint exponents. The exceptional degree at the remaining cover must cancel a member of a fixed finite set of rational twisting ratios, so it is bounded. The intermediate ratios consequently belong to a fixed finite set.

The nonlinear cover increases natural dimension at least quadratically. All remaining geometric steps increase it by at least a fixed positive factor. Thus its position as first or last cover makes no difference to (4.11), after decreasing c . $\square$

Lemma 4.14 (No parabolic cover). *After a further homogeneous reduction, no selected shortcut contains a parabolic or subspace cover with a large unipotent radical.*

Proof. Use Lemma 4.13, and color by the position of the parabolic cover. The other cover is nonlinear.

If the parabolic cover is first, its active dimension is n_i and its ambient dimension is comparable to n_i . Its inner quotient is at most $q^{C n_i^2}$. The multiplier of the last nonlinear cover is at most $|\text{Out}(Y)|$, which is absorbed by this bound, since n' is comparable to n_i and f'/f is bounded. Hence

$$e_{ij} \leq q^{C_1 n_i^2} \quad \text{for every } i < j. \quad (4.12)$$

For $j < k$ the radical in its first cover also gives $e_{jk} \geq q^{c_1 n_j}$, with the bounded field ratio absorbed in c_1 . By Lemma 4.8,

$$q^{c_1 n_j} \leq e_{jk} \leq e_{ik} \leq q^{C_1 n_i^2}.$$

Iterating (4.11) through a fixed number of intermediate chain vertices makes n_j/n_i^2 tend to infinity as the minimum rank grows. The displayed inequalities are incompatible.

If the parabolic cover is last, write $e_{ij} = b_{ij} s_{ij}$, where $b_{ij} \leq |\text{Out}(T_i)|$ is the nonlinear multiplier and s_{ij} is the parabolic multiplier. For a fixed upper endpoint k , the values s_{ik} have a uniformly bounded number of possibilities by G4. List those values in increasing order and color (i, k) by its

position in that list. This gives a uniform finite coloring even though the lists for different k may be different. On a homogeneous triple, $s_{ik} = s_{jk}$, and therefore

$$e_{ij} = \frac{e_{ik}}{e_{jk}} = \frac{b_{ik}}{b_{jk}} \leq |\text{Out}(T_i)|.$$

But the parabolic radical on the pair (i, j) gives $e_{ij} \geq q^{c_2 n_j}$. The classical outer-order bound is

$$|\text{Out}(T_i)| \leq C n_i f \leq q^{C_2(1+\log n_i)},$$

so even (4.11) contradicts these inequalities at sufficiently large rank. $\square$

Lemma 4.15 (Inner order bound). *For the remaining shortcuts there is an absolute C such that*

$$|P_{ij}| \leq q^{C n_i^2}. \quad (4.13)$$

Proof. There is at least one nonlinear cover. For a remaining geometric cover with active label T and natural dimension n , the other classical factors have bounded natural dimension and are soluble. Their orders, scalar factors, and bounded component-permutation factors are bounded by

$$q^{C_0}(1+n)^{C_0}(1+f)^{C_0}.$$

Field and twisted-tensor degrees are bounded by Lemma 4.13; their semilinear factors satisfy the same bound. These are precisely the block, form, tensor, and field structures left after Lemma 4.14; no unipotent radical or extra nonabelian factor is being omitted from the estimate.

If the geometric cover is first, its intermediate natural dimension is at most $C_1(1+n_i)^{b_1}$ for fixed constants. Its multiplier has the displayed bound in the lower parameters, and the last nonlinear multiplier is at most $|\text{Out}(Y)| \leq C n' f'$. Since $|T_i| \leq q^{C n_i^2}$ and $f \leq q$, their product satisfies (4.13).

If the nonlinear cover is first, its intermediate degree is at most $a_i^{C_2}$ and f'/f is bounded. The geometric multiplier is bounded by $q^{C_3}(1+n')^{C_3}(1+f')^{C_3}$, which is at most $q^{C_4 n_i^2}$. Multiplying by a_i and the first nonlinear outer multiplier proves the same bound. With two nonlinear covers only the two outer multipliers occur, and the latter argument applies without the geometric factor. $\square$

Proposition 4.16 (The remaining paths). *If the reduced chains of Proposition 3.13 have arbitrary length, then they still have arbitrary length subject to all of the following properties.*

- (i) *The endpoints have a common classical family and field $q = p^f$, their natural dimensions tend to infinity, and their common factor multiplicity is m . Every selected shortcut has one active factor at each cover and no other nonabelian factor in its inner projection.*
- (ii) *Every selected two-edge shortcut contains a defining-characteristic nonlinear cover. Its representation is a single Frobenius digit after the twisted-tensor geometric alternative has been separated. The other cover is another such nonlinear cover, or a block, form, tensor, extension-field, subfield, or twisted-tensor cover. No crossing, symplectic-type cover, or parabolic cover with a large unipotent radical remains. The characteristic-two nonsingular-line form inclusion is retained, with its natural simple quotient convention.*
- (iii) *There are fixed constants b, C and a fixed finite set $\mathcal{R} \subset \mathbb{Q}_{>0}$ such that every intermediate exponent satisfies $f'/f \in \mathcal{R}$, every field-extension or twisted-tensor degree is at most b , and every inactive geometric natural dimension is at most b . In particular each remaining geometric cover satisfies $n_{\text{target}} \leq C(1+n_{\text{source}})^b$.*

(iv) *There are fixed $c, C > 0$ such that*

$$n_j \geq cn_i^2, \quad |P_{ij}| \leq q^{Cn_i^2}.$$

For any prescribed exponent d , fixed-spacing thinning gives also $n_j \geq n_i^d$ for every retained pair, after raising the minimum rank.

(v) *The coordinate normal tests of Lemma 4.9 remain available for every tested proper coordinate overgroup. After one more fixed-spacing thinning, M_{ij} normalizes no proper parabolic of T_j .*

Proof. Parts (i)–(iv) collect Propositions 4.7 and 4.12 and Lemmas 4.13, 4.14, and 4.15. For the spacing assertion put $x_i = cn_i$. Then (4.11) says $x_j \geq x_i^2$ on every consecutive comparison. After s comparisons, $n_j \geq c^{-1}(cn_i)^{2^s}$. Choose $2^s > d$ and then increase the minimum rank to obtain $n_j \geq n_i^d$.

The coordinate normal test is inherited from the original tested intervals and is not altered by deleting chain vertices. Every proper parabolic of a large-rank classical group has unipotent radical of order at least $q^{c_3 n_j}$ for an absolute $c_3 > 0$. One can see the uniform bound by choosing a maximal parabolic containing it: the omitted simple root has at least the relative rank many positive-root coordinates outside the Levi subsystem. Their product contributes at least $q^{c_3 n_j}$; unitary root orbits can only increase the corresponding root-group orders. The radical of the smaller parabolic contains that of the maximal one. Fixed-spacing thinning makes $c_3 n_j > Cn_i^2$ uniformly. If M_{ij} normalized such a parabolic J with radical R , use (4.8). Since $P_{ij}R \leq N_{T_j}(J) < T_j$, its full inverse image is proper in z_j . The soluble product $\hat{R} \trianglelefteq \bar{U}$ then satisfies (4.6). In particular $R \leq P_{ij}$, contrary to $|R| \geq q^{c_3 n_j} > q^{Cn_i^2} \geq |P_{ij}|$. This proves (v). $\square$

The Proposition makes no assertion that a unique nonabelian composition factor alone produces a quasisimple subgroup. The actual components and their compatible rational maps will be obtained from the retained geometric structures in the next section. In particular, the characteristic-two form case is still among those structures and requires its own form-preservation argument.

5 Components, forms, and rational maps along a path

We now work in the remaining situation of Proposition 4.16. In particular the characteristic is fixed, the classical ranks tend to infinity, and the geometric steps with a nontrivial unipotent radical have been excluded. All statements about classical groups below are made beyond the finitely many small parameters already discarded in that proposition. We write $P^{(\infty)}$ for the terminal member of the derived series of a finite group P . A group is *quasisimple* if it is perfect and its quotient by its centre is nonabelian simple. Soluble groups can occur beside an active component; their presence must be distinguished from a noncentral soluble extension underneath it.

We need two kinds of structure from the remaining finite comparisons. The normal-subgroup tests will be applied to actual quasisimple components and their centralizers. Later, the irreducible-triple classification will be applied to connected algebraic groups containing the finite images. Proposition 5.8 supplies both the components and rational lifts of their finite inclusions. Lemma 5.9 composes these lifts along one chosen path: irreducibility on its finite image then implies irreducibility on the containing algebraic image. Independently chosen rational paths need not agree, even when their finite inclusions do.

The preparatory lemmas extract components from the retained subgroup structures, choose standard finite covers uniformly over the field, and lift the prescribed quadratic form in characteristic two.

5.1 Extracting actual finite components

Lemma 5.1 (A two-step component lemma). *Suppose $P_0 \leq P_1$, $E_1 \triangleleft P_1$ is quasisimple, and P_1/E_1 is soluble. Let $\pi : P_1 \rightarrow \text{Aut}(E_1)$ be conjugation. Suppose that $\bar{P}_0 = \pi(P_0)$ has a normal quasisimple subgroup $\bar{E}_0$ with soluble quotient. Then*

$$E_0 = P_0^{(\infty)} \leq E_1$$

is quasisimple and P_0/E_0 is soluble. In particular E_0 is normalized by every group normalizing P_0 .

Proof. Set $Q = P_0 \cap E_1$ and $A = \pi(Q)$. The group $\bar{P}_0/A$ is a quotient of P_0/Q , which embeds in P_1/E_1 , so it is soluble. A perfect group has trivial image in a soluble group. Consequently $\bar{E}_0 \leq A$. Let B be its full inverse image in Q . Then $B \triangleleft P_0$ and

$$B/Z_0 \cong \bar{E}_0, \quad Z_0 = B \cap \ker \pi = B \cap Z(E_1) \leq Z(B).$$

Because B/Z_0 is perfect, $B = B'Z_0$, and therefore $B' = [B'Z_0, B'Z_0] = B''$. Thus B' is perfect. The inverse image in B of $Z(\bar{E}_0)$ has central commutators, and its intersection with B' is central in B' : for an element of this inverse image, commutation with B' is a homomorphism from the perfect group B' to the abelian group Z_0 . It is trivial. It follows that B' is a perfect central extension of the nonabelian simple group $\bar{E}_0/Z(\bar{E}_0)$, hence is quasisimple.

The quotients P_0/Q , $Q/B \cong A/\bar{E}_0$, and B/B' are soluble. Hence P_0/B' is soluble. Since B' is perfect, it equals $P_0^{(\infty)}$. Finally $B' \leq Q \leq E_1$, and the terminal derived subgroup is characteristic. $\square$

Remark 5.2. The hypothesis about $\bar{E}_0$ concerns its structure as a normal subgroup. The assertion that $\bar{P}_0$ has just one nonabelian composition factor would not suffice: a semidirect product of a nontrivial module by a simple group can have that property without having a component. The explicit structures in Proposition 4.1 supply the stronger hypothesis used here.

5.2 Standard finite covers at every field size

Lemma 5.3 (Standard finite covers). *There is an absolute rank bound, independent of the finite field, with the following property. Let $\mathbf{G}$ be a simply connected classical algebraic group above this bound, let σ be a standard split or graph-twisted Steinberg map, and put $U = \mathbf{G}^\sigma$. Then U is perfect, $H_2(U, \mathbb{Z}) = 0$, and its standard simple quotient map $u : U \rightarrow S$ is central. For any quasisimple central extension $a : L \rightarrow S$, there is a unique homomorphism $\pi : U \rightarrow L$ satisfying $a\pi = u$. It is surjective and has central kernel. A specified projective representation of L also lifts uniquely to a linear representation of U over an algebraically closed field.*

Proof. Steinberg's multiplier Theorem has only a finite list of exceptions for the split groups, and the accompanying twisted statement has another finite list [36, Theorem 1.1 and the opening paragraph of p. 528]. Their ranks are bounded, independently of the field size. We supply perfectness separately: the root-presentation statement [35, Lemma 32', original pp. 51–53] applies to the universal split groups, and [35, Theorem 34 and its proof, original pp. 187–188] gives the twisted case outside its finite list of exceptions. The root-generated groups used there are the full fixed-point groups for simply connected groups over finite fields: see [35, Corollary 3 to Theorem 7, original p. 65, and the Corollary after Lemma 64, original p. 186]. Increase the discarded rank bound beyond these finite exception lists. Thus $H_1(U, \mathbb{Z}) = H_2(U, \mathbb{Z}) = 0$ and u is the usual central simple quotient. This argument retains every field size, including $q = 2, 3, 4$.

Let $A = \ker a \leq Z(L)$. Pull back a along u to obtain the central extension

$$1 \rightarrow A \rightarrow U \times_S L \rightarrow U \rightarrow 1.$$

The action on A is trivial. The universal coefficient sequence

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}^1(H_1(U, \mathbb{Z}), A) \longrightarrow H^2(U, A) \longrightarrow \text{Hom}(H_2(U, \mathbb{Z}), A) \longrightarrow 0$$

gives $H^2(U, A) = 0$. The extension splits, and projection of a splitting to L supplies π . Two such maps differ by a homomorphism $U \rightarrow A$, which is zero because U is perfect. If $H = \pi(U)$, then $L = HA$ and A is central. Therefore

$$L = [L, L] = [HA, HA] = [H, H] \leq H,$$

so π is onto. Its kernel is contained in $\ker u \leq Z(U)$.

For a projective representation, compose with π and pull back $\text{GL}(V) \rightarrow \text{PGL}(V)$. This is a central extension of U by $k^\times$. The same cohomology argument splits it; perfectness gives uniqueness of the resulting linear lift. $\square$

5.3 Quadratic forms in characteristic two

Here and below a Frobenius twist $M^{[a]}$ means pullback by the p^a -power Frobenius morphism. In particular $[1]$ denotes the first Frobenius twist; we write $[f]$ when $q = p^f$. For a dominant weight λ , the symbols $L(\lambda)$, $\nabla(\lambda)$, and $\Delta(\lambda)$ denote the simple, costandard, and Weyl rational modules of highest weight λ , respectively; we use the highest-weight, simple-head and simple-socle properties in [21, Part II, Chapter 2]. Write X^+ for the dominant weights and w_0 for the longest Weyl-group element, and put $\lambda^* = -w_0\lambda$.

Lemma 5.4 (Restriction of first cohomology). *Let $\mathbf{G}$ be a simply connected simple algebraic group over an algebraically closed field k of characteristic p . Let $\sigma = F_q\gamma$, where $q = p^f \geq 2$ and γ is a standard diagram automorphism commuting with F_q . For every nonzero q -restricted dominant weight ν , restriction is injective:*

$$H^1(\mathbf{G}, L(\nu)) \longrightarrow H^1(\mathbf{G}^\sigma, L(\nu)).$$

Proof. Put $I = \text{ind}_{\mathbf{G}^\sigma}^{\mathbf{G}} k$. The induction filtration of [6, arXiv version, Proposition 3.1.2], obtained from the coordinate-ring filtration and the Lang map, gives I a filtration with sections

$$\nabla(\lambda) \otimes \nabla(\lambda^*)^\sigma, \quad \lambda \in X^+,$$

each once. Write such a section as $\nabla(\lambda) \otimes \nabla(\mu)^{[f]}$, where μ is obtained from λ^* by the diagram automorphism. Choose the regular height

$$h(\xi) = \left\langle \xi, \sum_{\alpha > 0} \alpha^\vee \right\rangle.$$

It is invariant under diagram automorphisms and duality, is positive on every nonzero dominant weight and every positive root, and satisfies $h(\mu) = h(\lambda)$.

We first justify removing the actual constants from this filtration. A nonconstant section cannot contain a trivial submodule. Indeed a map from k into it would transpose to a nonzero map $\Delta(\lambda^*) \rightarrow \nabla(\mu)^{[f]}$. The image contains the simple socle $L(q\mu)$, so its highest weight must occur in the source. This gives $qh(\lambda) \leq h(\lambda)$, impossible when $\lambda \neq 0$. Let I_j be the first filtration stage containing the actual one-dimensional space of constants. That space injects into I_j/I_{j-1} , so this section is the unique section for $\lambda = 0$. The constants split it and $I_j = I_{j-1} \oplus k$. The images of the

earlier stages in I/k , followed by I_i/k for $i \geq j$, give a filtration of I/k ; omitting the repeated stage removes exactly the constant section. Thus its sections are precisely the nonconstant ones.

We prove that a nonzero q -restricted simple $L(\nu)$ cannot map to a nonconstant section. Suppose otherwise. Transposing the second tensor factor gives a nonzero homomorphism

$$L(\nu) \otimes \Delta(\mu^*)^{[f]} \longrightarrow \nabla(\lambda).$$

Its image contains the simple socle $L(\lambda)$. Filter the Weyl module by simple composition factors and apply Steinberg's tensor product Theorem [34, Theorems 5.1 and 6.1]. Since ν is q -restricted, each simple composition factor of the source has highest weight $\nu + q\eta$ for some dominant η . Consequently

$$\lambda = \nu + q\eta, \quad h(\lambda) \geq h(\nu). \quad (5.1)$$

Transposing the first tensor factor instead gives a nonzero homomorphism

$$\Delta(\lambda^*) \otimes L(\nu) \longrightarrow \nabla(\mu)^{[f]}. \quad (5.2)$$

Its image contains $L(q\mu)$, so the weight $q\mu$ occurs in its source. The maximum height of a weight of that source is $h(\lambda^*) + h(\nu) = h(\lambda) + h(\nu)$. Thus

$$qh(\lambda) \leq h(\lambda) + h(\nu). \quad (5.3)$$

Equations (5.1) and (5.3) are inconsistent when $q > 2$.

When $q = 2$, both inequalities are equalities. Hence $\eta = 0$ and $\lambda = \nu$. The only weight of maximum height in the source of (5.2) is $\lambda^* + \nu$, and its weight space is the one-dimensional span of the tensor $a \otimes b$ of highest-weight vectors. If $2\mu \neq \lambda^* + \nu$, the required weight does not occur. Otherwise the homomorphism must be nonzero on $a \otimes b$. Choose a simple root α with $\langle \nu, \alpha^\vee \rangle = 1$, which is possible because ν is nonzero and 2-restricted. For the Chevalley raising and lowering operators,

$$e_\alpha(a \otimes f_\alpha b) = a \otimes b.$$

The target of (5.2) is Frobenius twisted, so its Lie algebra action is zero. Every module homomorphism to it kills the displayed vector, a contradiction.

A homomorphism from a finite-dimensional simple module to I/k has image in a finite stage of the filtration. Taking the first stage with nonzero intersection with that simple image gives a nonzero map into one of its sections. We have proved

$$\mathrm{Hom}_{\mathbf{G}}(L(\nu), I/k) = 0.$$

Apply this also to the dual of the coefficient module. Tensoring $0 \rightarrow k \rightarrow I \rightarrow I/k \rightarrow 0$ with $L(\nu)$ shows that the map from $H^1(\mathbf{G}, L(\nu))$ to $H^1(\mathbf{G}, L(\nu) \otimes I)$ is injective. The tensor identity for induction and derived Frobenius reciprocity identify this with the asserted restriction map. For split groups this is also the special case $U = k$ of [14, Theorem 7.4]; the argument above includes the diagram twist and the equality case $q = 2$. $\square$

Proposition 5.5 (Lifting a prescribed quadratic form). *Let k have characteristic two and let $\mathbf{G}$, q , and σ be as in Lemma 5.4. Suppose that $V = L(\nu)$ is nontrivial and q -restricted, and that its restriction to $\mathbf{G}^\sigma$ is irreducible and preserves a nonzero quadratic form Q . If V is not the natural module for a group isomorphic to Sp_{2n} , $n \geq 1$, then the rational action of $\mathbf{G}$ preserves the same quadratic form Q .*

Proof. The polar form B of Q is nonzero: otherwise Q is the square of a linear form, and that linear form would be invariant under the nontrivial irreducible finite action. Its radical is invariant, so B is nondegenerate. Thus the finite representation is self-dual. The injective parametrization by q -restricted highest weights in Steinberg's finite-group restriction Theorem [35, Theorem 43] implies that $V \cong V^*$ rationally. Schur's Lemma, first for $\mathbf{G}$ and then for $\mathbf{G}^\sigma$, shows that their spaces of invariant bilinear forms are both one-dimensional and agree. In particular B itself is $\mathbf{G}$ -invariant.

Polarization gives an exact sequence of rational modules

$$0 \longrightarrow (V^*)^{[1]} \longrightarrow \mathrm{Sym}^2(V^*) \longrightarrow \bigwedge^2(V^*) \longrightarrow 0. \quad (5.4)$$

The first map sends a twisted linear form to its square. The obstruction to refining B by a rationally invariant quadratic form is its connecting class

$$c_B \in H^1(\mathbf{G}, (V^*)^{[1]}).$$

This class restricts to zero on $\mathbf{G}^\sigma$, because Q is a refinement there. This description is also [18, Proposition 3.6 and Corollary 3.9].

Let $\mathbf{G}_1$ be the first Frobenius kernel. Its action on $(V^*)^{[1]}$ is trivial. The inflation–restriction sequence therefore begins

$$0 \longrightarrow H^1(\mathbf{G}/\mathbf{G}_1, (V^*)^{[1]}) \longrightarrow H^1(\mathbf{G}, (V^*)^{[1]}) \longrightarrow (H^1(\mathbf{G}_1, k) \otimes (V^*)^{[1]})^{\mathbf{G}/\mathbf{G}_1}.$$

The required first-kernel calculation is

$$H^1(\mathbf{G}_1, k)^{[-1]} \cong \begin{cases} N, & \mathbf{G} \cong \mathrm{Sp}_{2n}, \ n \geq 1, \quad N \text{ its natural module,} \\ 0, & \text{otherwise.} \end{cases}$$

It is reproduced in the proof of [18, Proposition 7.1, p. 411], citing [7, Theorem 3.1(C)(a),(f)]. Here the symplectic case includes $A_1 = C_1$ and $B_2 = C_2$. The exceptional module is $L(\omega_1)$ in C_n numbering, including $n = 1$, and $L(\omega_2)$ in B_2 numbering. It is simple and self-dual. After untwisting, the last invariant space is therefore zero under our hypothesis. Inflation, followed by untwisting the quotient by $\mathbf{G}_1$, gives an isomorphism

$$H^1(\mathbf{G}, V^*) \xrightarrow{\sim} H^1(\mathbf{G}, (V^*)^{[1]}).$$

Write $c_B = F^*c$, where F is ordinary Frobenius. Put $\Gamma = \mathbf{G}^\sigma$ and $M = V^*$. Since F commutes with σ , its restriction F_Γ is an automorphism of Γ . Naturality gives the commutative square

$$\begin{array}{ccc} H^1(\mathbf{G}, M) & \xrightarrow{F^*} & H^1(\mathbf{G}, M^{[1]}) \\ \mathrm{res}_M \downarrow & & \downarrow \mathrm{res}_{M^{[1]}} \\ H^1(\Gamma, M) & \xrightarrow{F_\Gamma^*} & H^1(\Gamma, F_\Gamma^* M). \end{array}$$

The lower arrow is an isomorphism, with inverse obtained by pulling back along F_Γ^{-1} with the transported coefficients. This does not identify M with its Frobenius twist. Hence the vanishing of $\mathrm{res}(c_B)$ implies that of $\mathrm{res}(c)$. This and Lemma 5.4, applied to V^* , imply $c = 0$ and $c_B = 0$. Sequence (5.4) supplies a $\mathbf{G}$ -invariant quadratic refinement Q_0 of B .

Finally $Q - Q_0$ has zero polarization, hence is the square of a linear form ℓ . Its finite invariance implies that ℓ is invariant under $\mathbf{G}^\sigma$. Such a linear form is zero on this nontrivial irreducible module. Therefore $Q_0 = Q$. $\square$

Remark 5.6. The exclusion of the natural symplectic module, including the $A_1 = C_1$ and $B_2 = C_2$ conventions just stated, is required in the inflation argument. The geometric natural modules and the special isogenies in characteristic two are constructed as classical algebraic maps below; Proposition 5.5 is used for the remaining defining-characteristic nonlinear steps. Suzuki and Ree endomorphisms have bounded rank and do not occur here.

5.4 The retained edge maps

Lemma 5.7 (A nonsingular-line stabilizer). *Let (W, Q) be a nonsingular quadratic space of even dimension $2r + 2$ over $\mathbb{F}_q$, where $r \geq 3$ and q is even, and let $v \in W$ satisfy $Q(v) \neq 0$. Write B for the polar form and $\overline{W} = v^\perp / \langle v \rangle$. The kernel of*

$$O(W, Q)_{\langle v \rangle} \longrightarrow \mathrm{Sp}(\overline{W}, B)$$

is the central group generated by the symmetry

$$r_v(x) = x + \frac{B(x, v)}{Q(v)}v.$$

The image is the full symplectic group. The symmetry has nonzero Dickson invariant, and restriction gives

$$\Omega(W, Q)_{\langle v \rangle} \cong \mathrm{Sp}_{2r}(q).$$

In particular this stabilizer has no nontrivial finite unipotent kernel on $\overline{W}$. Over an algebraic closure, the vector stabilizer $\mathrm{SO}(W, Q)_v$ is a smooth connected adjoint group of type B_r . The ambient module W restricted to this group has composition factors k, N_{B_r}, k along the flag $\langle v \rangle < v^\perp < W$.

Proof. A line stabilizer fixes v , because a scalar preserving its nonzero quadratic value has square one, and the field has characteristic two. Choose a complement U to $\langle v \rangle$ in $v^\perp$ and choose t with $B(v, t) = 1$ and $B(t, U) = 0$. The space U is symplectic. An isometry inducing the identity on $\overline{W}$ sends $u \in U$ to $u + \ell(u)v$. Preservation of Q forces $\ell(u)^2 Q(v) = 0$, so it fixes U pointwise. Preservation of B then forces its image of t to be $t + av$. Preservation of Q says $a^2 Q(v) + a = 0$, hence $a = 0$ or $a = Q(v)^{-1}$. These are exactly the identity and r_v .

For completeness, let $s \in \mathrm{Sp}(U)$. The difference $Q(u) + Q(su)$ has zero polarization, so, because the field is perfect, there is a unique linear form ℓ satisfying

$$\ell(u)^2 Q(v) = Q(u) + Q(su).$$

The map $v \mapsto v, u \mapsto su + \ell(u)v$ is an isometry of $v^\perp$. Witt's extension theorem [37, Theorem 7.4, pp. 57–58] extends this isometry to W : the ambient polar radical is zero, so its radical-intersection criterion is automatic even though $v^\perp$ has degenerate polar form. This proves surjectivity. The formula for r_v also shows that it commutes with every isometry fixing v .

The Dickson invariant of an isometry in even characteristic is $\mathrm{rank}(g-1)$ modulo two; in particular it is one on r_v . Its zero kernel is $\Omega(W, Q)$ over a finite field of even order [37, Theorems 11.43 and 11.51, pp. 160, 164–165]; the exceptional group $\Omega^+(4, 2)$ is outside the dimension range here. Each fibre of the displayed surjection has two elements differing by r_v , so exactly one lies in this kernel. The last finite assertion follows.

We also justify the algebraic assertion, including its scheme convention. Put $W_0 = v^\perp$. Restriction defines

$$H = \mathrm{SO}(W, Q)_v \longrightarrow \mathrm{SO}(W_0, Q|_{W_0}).$$

The kernel of restriction from the full orthogonal vector stabilizer is the constant group of order two computed above: over an arbitrary k -algebra its remaining parameter satisfies $a + a^2Q(v) = 0$, an étale equation. Its nonidentity element has Dickson invariant one, so the displayed map has trivial scheme-theoretic kernel. We give its inverse over an arbitrary k -algebra A . Put $c = Q(v)$ and retain the decomposition $W = kv \oplus U \oplus kt$ used above. An element $h \in \mathrm{SO}(W_0)(A)$ has symplectic quotient action, of determinant one; the odd special orthogonal determinant condition therefore makes it fix v . Write $h(u) = su + \ell(u)v$ for $u \in U_A$. There is a unique $w \in U_A$ such that $B(su, w) = \ell(u)$ for all u . The extension fixing v and restricting to h is obtained by setting

$$g(t) = t + w + av, \quad ca^2 + a + Q(w) = 0.$$

The bilinear condition is $B(su, w) + \ell(u) = 0$, and the quadratic condition is exactly the displayed equation. Its derivative is one and its leading coefficient is a unit, so adjoining a solution gives an étale double cover. The two extensions differ by r_v and have opposite Dickson invariants. The unique Dickson-zero extension on this cover agrees on overlaps and descends to A . This is a functorial inverse, proving the asserted scheme isomorphism. The odd special orthogonal group is smooth, connected, and adjoint of type B_r by [15, author version, Theorem C.2.11 and Proposition C.3.10]. This restriction calculation is also the one in the proof of [15, author version, Proposition C.3.1]. Its action on $W_0/\langle v \rangle$ is the purely inseparable isogeny to the symplectic group described in [15, author version, Remark C.3.6]; consequently this quotient is the simple module N_{B_r} . Both outer factors of the flag are trivial. We used the vector stabilizer: the line-stabilizer scheme $\mathrm{SO}(W, Q)_{\langle v \rangle}$ is $H \times \mu_2$. Indeed its scalar on v has square one, and multiplying by the inverse central scalar makes it fix v ; the two factors have trivial intersection. Its reduced subgroup and its finite-field points are the ones just described. $\square$

Proposition 5.8 (Components and rational edge maps). *In the setting of Proposition 4.16, each retained projected subgroup P has a characteristic normal quasisimple subgroup $L = P^{(\infty)}$ and P/L is soluble. In a fixed higher coordinate, these components nest with the projected subgroups and are normalized by their common bottom action. All intermediate components of a shortcut are normalized by its lower coordinate stabilizer. The transported product of an active component maps onto the lower nonabelian socle and is not contained in the common bottom.*

For every retained geometric or defining-characteristic almost-simple edge, its actual inclusion on these components has a rational lift between their simple classical algebraic groups, after passage to simply connected central covers. The lift agrees with the finite inclusion after the specified central quotients.

On a geometric edge, every nontrivial absolute composition factor of the target natural module is a tensor product of Frobenius twists of the source natural simple module or its dual. The number of tensor factors is uniformly bounded by the constants in Proposition 4.16. Trivial factors and repeated copies are allowed. For type B in characteristic two, “natural simple module” means the $2r$ -dimensional quotient of the usual $(2r + 1)$ -dimensional module.

Proof. Finite components and their normalization. We first specify the components in the structures of Proposition 4.1. A nondegenerate block or form subgroup contains the derived classical group on each active block. A tensor subgroup contains the central product of the derived groups on its tensor factors. An extension-field or subfield subgroup contains the derived classical group over its indicated field. In the retained twisted tensor construction, the single Frobenius orbit of the tensor factors has fixed-point group equal to the classical group over the corresponding extension field; its derived group is the active quasisimple group. The almost-simple case contains its quasisimple socle preimage.

The reductions leave one nonabelian active factor. Hence every permitted normalizer preserves it. All other base factors commute with it modulo scalars, and their surviving images are soluble. The retained diagonal, field, graph, and factor-permutation quotients are soluble as well; the exclusion of other nonabelian factors here includes factors in the permutation quotient, not just base factors. Thus the explicitly displayed active derived group is normal and the quotient is soluble. Scalar preimages introduce only central kernels: if C/Z is the displayed perfect active factor, with Z central, then $C = C'Z$ and $C' = C''$, giving its quasisimple derived preimage. For the characteristic-two nonsingular-line case, Lemma 5.7 gives the asserted component directly. For a two-edge projection, we verify the precise hypothesis of Lemma 5.1. Use the common upper-coordinate notation $P_0 \leq P_1$ and $Q_1 \leq R_1 \leq P_1$ from Lemma 3.8. Here Q_1 is soluble, R_1/Q_1 is the middle simple active factor, and P_1/R_1 is soluble. The upper retained edge supplies the quasisimple component $E_1 = P_1^{(\infty)}$. Perfectness gives $E_1 \leq R_1$. Its image in R_1/Q_1 is nontrivial, since Q_1 is soluble, and is normal, hence is the full simple group. Therefore

$$E_1 Q_1 = R_1, \quad E_1 \cap Q_1 = Z(E_1).$$

Both E_1 and Q_1 are normal in P_1 . Thus, for $x \in Q_1$, commutation with x maps E_1 into $E_1 \cap Q_1 = Z(E_1)$. It is a homomorphism to an abelian group, so perfectness makes it trivial. Consequently Q_1 centralizes E_1 . Conjugation on E_1 agrees with conjugation on $E_1/Z(E_1) \cong R_1/Q_1$: an automorphism of the perfect group E_1 acting identically modulo its centre differs from the identity by a homomorphism $E_1 \rightarrow Z(E_1)$ and is therefore the identity.

The original-core product identity in Proposition 2.9 gives $Q_1 \leq P_0$. Indeed its transported product comes from the middle original core, which is contained in D and hence in the lower labelled group. It follows that

$$J = (P_0 \cap R_1)/Q_1$$

is the actual first-edge inner projection of Lemma 3.8. Its retained structure supplies the normal quasisimple component $J^{(\infty)}$ with soluble quotient. This is a structural assertion about this particular J ; the excluded parabolic or module semidirect extensions have not been inferred to split.

Let $\pi : P_1 \rightarrow \text{Aut}(E_1)$ be conjugation. By the centralization and simple-quotient identification just proved, $\pi(P_0 \cap R_1)$ is the inner image of J , isomorphic to J . It is normal in $\pi(P_0)$, and $\pi(P_0)/\pi(P_0 \cap R_1)$ is soluble because it is a quotient of $P_0/(P_0 \cap R_1) \leq P_1/R_1$. Hence the image of $J^{(\infty)}$ is normal quasisimple in $\pi(P_0)$, with soluble quotient. These are exactly the hypotheses of Lemma 5.1, which produces the actual lower component inside E_1 .

Finally, if $P \leq P'$ and L' is the component of P' , the image of the perfect group L in the soluble group P'/L' is trivial. Hence $L \leq L'$. Characteristicity gives normalization by the common bottom. Each intermediate inner projection is normal in its own coordinate stabilizer, which contains the lower coordinate stabilizer; its characteristic component is therefore normalized by that lower stabilizer too.

To check that the active component is not killed, use the notation $Q \leq R \leq P$ of Proposition 2.9. In our multiplicity-one coordinate, R/Q is the nonabelian simple active factor and P/R is soluble. Hence $L \leq R$. Since P/L is soluble, the simple group R/Q has trivial image modulo LQ , and $LQ = R$. Thus L maps onto R/Q . Transporting L over all upper coordinates gives a normal subgroup of the lower labelled group, mapping onto its socle modulo its original bottom core. If this product were contained in the common bottom, normality would put it inside that core, a contradiction.

For each actual finite component L_i , fix its standard cover $U_i = \mathbf{G}_i^{\sigma_i}$ and the surjection $\pi_i : U_i \twoheadrightarrow L_i$ of Lemma 5.3, over the specified simple quotient.

Rational maps for the geometric edges. We have obtained the finite components, their nesting, and their survival under the original bottom-core quotient. We now lift their inclusions. For the geometric cases we describe the maps on natural modules over k , which also proves the assertion about their composition factors. Write N for the source natural simple module.

- (i) A block embedding acts on its active block by g , on a dual block, when present, by $g^{-\top}$, and trivially on omitted spaces. Its constituents are N , N^* , and trivial modules. A form embedding is the natural action with its prescribed invariant bilinear or quadratic form. These are rational matrix maps.
- (ii) For a tensor factor with an omitted space U , the map is $g \mapsto g \otimes 1_U$. On the active factor its natural module is the direct sum of $\dim U$ copies of N , or of a prescribed dual or Frobenius twist of N . The commuting omitted factor changes the multiplicity space, not this formula.
- (iii) Restriction of scalars from $\mathbb{F}_{p^{ab}}$ to $\mathbb{F}_{p^a}$, followed by extension of scalars to k , has the explicit decomposition

$$k \otimes_{\mathbb{F}_{p^a}} \mathbb{F}_{p^{ab}} \cong \bigoplus_{j=0}^{b-1} k, \quad x \mapsto (x, x^{p^a}, \dots, x^{p^{a(b-1)}}).$$

Thus its map is the direct sum of the rational representations $N^{[aj]}$. If a form pairs two summands, the corresponding summand is expressed as the required dual. A subfield inclusion is the same natural matrix map, possibly precomposed with its specified field or diagram automorphism. On natural modules in the ranks in use, a diagram automorphism gives N or N^* ; triality has already been discarded with bounded rank.

- (iv) In a twisted tensor orbit the map is

$$g \mapsto \rho_0(F^{a_0}g) \otimes \cdots \otimes \rho_{b-1}(F^{a_{b-1}}g),$$

where each ρ_j is natural or dual. This is a rational representation and it is the defining matrix construction of the finite subgroup. Its natural module is $\bigotimes_j N_j^{[a_j]}$, $N_j \in \{N, N^*\}$. The orbit lengths, field degrees, and omitted tensor dimensions are bounded by Proposition 4.16.

In characteristic two, two further classical maps must be interpreted literally. The nonsingular-vector stabilizer in a group of type D_{r+1} has connected reduced subgroup of type B_r ; its action on the $2r + 2$ -dimensional natural space has, by Lemma 5.7, the invariant filtration

$$0 < \langle v \rangle < v^\perp < W$$

with factors k, N_{B_r}, k . Its simply connected cover therefore supplies the required B_r -to- D_{r+1} map. The middle symplectic action is an isogeny from this *adjoint* B_r stabilizer to Sp_{2r} . It has no nontrivial kernel on k -points, consistently with Lemma 5.7; it is not a finite unipotent radical. In the following display, instead, $\mathbf{B}_r = \mathrm{Spin}_{2r+1}$ and $\mathbf{C}_r = \mathrm{Sp}_{2r}$ denote the *simply connected* groups. The preceding vector stabilizer is the adjoint quotient of $\mathbf{B}_r$; the kernels of its symplectic action and of the simply connected maps are not being identified. For the two special isogenies

$$s : \mathbf{B}_r \longrightarrow \mathbf{C}_r, \quad t : \mathbf{C}_r \longrightarrow \mathbf{B}_r, \quad st = F_p, \quad ts = F_p,$$

one has

$$s^* N_C = N_B, \quad t^* N_B = N_C^{[1]}$$

by [17, author preprint, §I.3.2]. If the inverse of s on $\mathbb{F}_q$ -points is needed, it is induced by the rational map tF_p^{f-1} , since $q = p^f$. No inverse inseparable morphism is asserted. These formulas preserve the claimed description of simple constituents, including the trivial factors of the full odd orthogonal module.

Rational maps for the nonlinear edges. For a defining-characteristic almost-simple edge, inflate its actual finite irreducible representation along $\pi_i : U_i \twoheadrightarrow L_i$, using the unique linear lift of Lemma 5.3 if the natural action was given projectively. Surjectivity preserves every invariant subspace and every prescribed form. A form similitude multiplier is a character of the perfect group U_i , hence is trivial. Only now apply Steinberg's restriction Theorem [35, Theorem 43]: the resulting U_i -module is the restriction of its q_i -restricted rational simple module, with the specified field and diagram twist. Here $\sigma_i = F_{p^{f_i}} \gamma_i$ uses f_i ordinary Frobenius digits, including in the unitary case; the unitary natural scalar field $\mathbb{F}_{p^{2f_i}}$ does not change this restriction parameter. This yields a rational map to $\mathrm{GL}(V)$ whose determinant is trivial: a connected semisimple group has no nontrivial rational character. Thus it maps to a linear target, including the algebraic group underlying a finite unitary target. In a symplectic target or an orthogonal target in odd characteristic, finite self-duality and the injective highest-weight parametrization identify the invariant rational bilinear form with the finite form, by Schur's Lemma. Its symmetry or alternating property is consequently the prescribed one. For an orthogonal target in characteristic two the remaining nonlinear module satisfies Proposition 5.5, which lifts its actual quadratic form. The natural and special-isogeny cases are the geometric maps already constructed. The image is connected and hence lies in the identity component of the isometry group.

Compatibility through the fixed central covers. These constructions initially give maps into whichever central form of the target is specified by its natural action. A homomorphism from a simply connected semisimple source lifts through a central isogeny of semisimple algebraic groups; see [15, author version, Exercise 6.5.2(iii), p. 236]. Thus they give maps between simply connected covers as claimed. Let $\phi_i : \mathbf{G}_i \rightarrow \mathbf{G}_{i+1}$ be such a map and let $\iota_i : L_i \hookrightarrow L_{i+1}$ be the actual finite inclusion. By construction their induced maps to the target adjoint group agree on U_i . For $u \in U_i$, the target adjoint image of $\phi_i(u)$ is fixed by σ_{i+1} , so

$$\delta(u) = \sigma_{i+1}(\phi_i(u))\phi_i(u)^{-1} \in Z(\mathbf{G}_{i+1})(k).$$

Centrality gives $\delta(uv) = \delta(u)\delta(v)$. Perfectness of U_i forces $\delta = 1$, so $\phi_i(U_i) \leq U_{i+1}$, including when the algebraic centre is nonreduced. The two homomorphisms $\pi_{i+1}\phi_i|_{U_i}$ and $\iota_i\pi_i$ to L_{i+1} agree modulo its centre. Their discrepancy is consequently a homomorphism $U_i \rightarrow Z(L_{i+1})$, and again is trivial. Thus we have the exact commutative diagram

$$\begin{array}{ccc} U_i & \xrightarrow{\phi_i|_{U_i}} & U_{i+1} \\ \pi_i \downarrow & & \downarrow \pi_{i+1} \\ L_i & \xrightarrow{\iota_i} & L_{i+1}. \end{array} \tag{5.5}$$

This proves compatibility with the actual finite inclusions through the fixed central-cover maps. It gives no uniqueness assertion for rational maps agreeing on finite points. $\square$

5.5 Composing a chosen path

Lemma 5.9 (Pathwise lifting). *Fix a path through retained edges with finite nested components $L_0 \leq \dots \leq L_s$. Choose an edge lift $\phi_i : \mathbf{G}_i \rightarrow \mathbf{G}_{i+1}$ from Proposition 5.8 for each edge, with the finite central-cover identifications specified there. Then*

$$\Phi_{i,j} = \phi_{j-1} \cdots \phi_i \quad (i < j)$$

is a rational lift of the actual finite inclusion along that path. If a rational $\mathbf{G}_j$ -module is irreducible on L_i under this finite inclusion, it is irreducible on the algebraic image $\Phi_{i,j}(\mathbf{G}_i)$. A nontrivial such image is a closed connected simple algebraic group; if its rank is strictly smaller than the rank of $\mathbf{G}_j$, it is proper.

Proof. The finite compatibility statements for the edges compose, so the first assertion follows by induction on $j - i$. Every subspace invariant under the algebraic image is invariant under the contained finite image. Finite irreducibility therefore implies algebraic irreducibility, without a density assertion or a bound on the highest weight of the composite representation. The kernel of a nontrivial map from a connected simple algebraic group is finite as a group scheme: its connected reduced normal subgroup cannot be the whole source. Its image is closed and connected, and is simple up to the allowed central form. Inseparability can interchange the types B and C in characteristic two, but does not change rank. The last assertion follows from the rank inequality. $\square$

Remark 5.10. The exponents in each chosen ϕ_i are retained when composing. For example, on $\mathrm{SL}_n(p^f)$ the representations $g \mapsto g \otimes F^a(g)$ and $g \mapsto g \otimes F^{a+f}(g)$ agree, although their rational representations need not agree. Accordingly an independently chosen shortcut map is never identified with a composed path map. The only comparison needed later is their agreement on the actual finite component, to which the irreducibility implication in Lemma 5.9 applies. Also, a strict finite subfield inclusion can have full algebraic image; properness in the later application comes from rank growth.

Corollary 5.11 (Passing irreducibility backwards through geometry). *Suppose that a retained geometric edge has source component L , and that the target natural module restricts absolutely irreducibly and nontrivially to a subgroup $H \leq L$. Then the source natural simple module restricts absolutely irreducibly to H .*

Proof. Work over the algebraic closure. Any proper submodule or nontrivial composition filtration for the L -action would be an H -invariant one. Thus the target module is a single nontrivial composition factor for L . Proposition 5.8 expresses it as a tensor product of twists of the source natural module or its dual. If one factor had a nonzero proper H -invariant subspace, tensoring that subspace with all the other factors would give a nonzero proper invariant subspace of the product. Hence each factor is irreducible on H . Taking a dual or applying a field Frobenius to all matrix entries preserves irreducibility, so the source natural module itself is irreducible on H . $\square$

6 A uniform bound on natural-module composition length

We continue in the setting of Proposition 4.16. The endpoint types have a common characteristic, field parameter $q = p^f$, and classical family. Write n_i for their natural dimensions, using the symplectic realization for a group of type B in characteristic two. The natural scalar field is $\mathbb{F}_q$, except in the unitary case, where it is $\mathbb{F}_{q^2}$. All ranks in this section may be assumed to be at least nine.

For a comparison $i < j$, let $P = P_{ij}$ be the inner projection in the upper simple group T_j , and let $M = M_{ij}$ be the corresponding lower coordinate stabilizer in $\text{Aut}(T_j)$. Proposition 5.8 supplies a quasisimple component $E = E_{ij}$, normal in both P and M , with P/E soluble. In matrix representations we take the indicated central covers. The same stabilizer normalizes the intermediate component on either edge of a shortcut. These are structural conclusions about the remaining edges, not consequences of the composition-length bound to be proved here.

We first pass to a semisimple natural section, losing at most two trivial factors, and then bound three sources of composition length. A soluble centralizer bounds the multiplicity of each absolute simple type. Many distinct types could still occur, so we group them into orbits under descent, the normalizer action, and duality. The commuting-normal test allows at most one large orbit of summands, with a bounded-dimensional remainder. Finally, comparing the source and ambient field shifts along the same two-edge shortcut bounds the number of types in each orbit. This last step makes the length bound independent of f .

We use Lemma 4.9, with its tests (N1) and (N2). Here $P = P_{ij}$, $M = M_{ij}$, and $F_j = \text{core}_{z_j}(D)$ is the original higher core. For an M -invariant $H \leq T_j$, write $\hat{H} \leq N_j/F_j$ for its transported product. A killed product lies in D/F_j , and hence $H \leq P$; the converse is not asserted. For one product the enlargement is $\langle z_i/F_j \rangle \hat{H}$; for two it is $\langle z_i/F_j, \hat{H}_1, \hat{H}_2 \rangle$. Each application checks properness of this enlargement and normality of every tested product in that same enlargement. The pairs used in this section commute exactly in N_j/F_j , so (N2) uses $\bar{K} = 1$. Their full inverse images commute modulo F_j , which is normal in the actual enlargement and contained in D . The transported active component $\hat{E}$ is not killed, by the component-survival conclusion of Proposition 5.8.

6.1 Semisimplicity and the nonsingular-line exception

Lemma 6.1 (Exclusion of invariant parabolics). *After passing to a fixed-spacing subchain, no comparison stabilizer M_{ij} normalizes a proper parabolic subgroup of T_j .*

Proof. This is Proposition 4.16(v), with its fixed-spacing thinning already incorporated in the retained chain. $\square$

Lemma 6.2 (A semisimple natural section). *Let V be the upper natural module in a retained comparison. As a module for the lower quasisimple component, either V is semisimple, or the characteristic is two, V is an even-dimensional quadratic space, and there is an M -invariant nonsingular line R such that*

$$V^\circ = R^\perp/R$$

is a semisimple symplectic module. In the latter case the composition length of V is that of V° plus two. The lower component and its soluble-quotient and normal-subgroup properties pass to V° .

Proof. Work first over the finite natural scalar field. Put

$$S = \text{soc}_E(V), \quad R_0 = \text{rad}_E(V), \quad U = S \cap R_0, \quad W = S + R_0.$$

If V is not semisimple, $R_0 \neq 0$, and a simple submodule of R_0 shows that $U \neq 0$. Also $W < V$: the proper submodule S is contained in a maximal submodule, which necessarily contains R_0 . Socles and radicals are intrinsic, so every semilinear normalizer preserves these constructions.

For a linear group without graph operations, U gives an M -normalized parabolic, contrary to Lemma 6.1. Suppose that M also contains duality operations. The side-preserving subgroup has index two. A duality acts on subspaces by a semilinear correlation, and its transport of a module to the dual exchanges the socle with the annihilator of the radical, and the radical with the annihilator

of the socle. Consequently it exchanges U and W as members of a flag. Thus the flag $U \leq W$ is preserved by all of M ; if $U = W$ it is a one-member flag. Both members are nonzero and proper. This again contradicts Lemma 6.1. This argument uses correlations on subspaces and does not treat a graph automorphism as a linear operator on V .

In a bilinear or Hermitian space, identify the module with its appropriate dual using the preserved form. Duality of socles and radicals gives

$$R_0^\perp = S, \quad S^\perp = R_0.$$

Hence U is totally isotropic. This supplies a forbidden parabolic in the symplectic, unitary, and odd-characteristic orthogonal cases.

It remains to consider a quadratic form Q_V in characteristic two. On a totally polar-isotropic space U its restriction has the form ℓ^2 , for a linear functional ℓ , since the finite field is perfect. Its zero space is an M -invariant singular subspace. Unless U is a nonsingular line, this zero space is nonzero and contradicts Lemma 6.1. We are therefore left with the nonsingular line $R = U$.

The space $R^\perp/R$ has a nondegenerate alternating form. If its E -module were not semisimple, the argument with socle and radical would give a nonzero M -invariant totally isotropic subspace A of that quotient. Its inverse image B in $R^\perp$ is totally polar-isotropic and contains R . On B the quadratic form is the square of a linear functional nonzero on R . Its zero hyperplane is therefore a singular complement to R , projecting isomorphically onto A . This complement is M -invariant and again gives a forbidden parabolic in the original quadratic space. The quotient is semisimple.

The flag $0 < R < R^\perp < V$ adds precisely two one-dimensional composition factors to that quotient. Both are trivial for E because E is perfect. By Lemma 5.7, the inner stabilizer has the canonical finite-group isomorphism

$$\theta : \Omega(V, Q_V)_R \xrightarrow{\sim} \mathrm{Sp}(V^\circ).$$

Every $m \in M$ induces a semilinear similarity m° on V° . Evaluation on quotient vectors gives

$$\theta(mhm^{-1}) = m^\circ \theta(h) (m^\circ)^{-1} \quad (h \in \Omega(V, Q_V)_R).$$

Consequently inverse images under θ preserve normalization by the actual M , inclusions, and exact commutation of inner subgroups. Since $P = M \cap T_j$ preserves R , it lies in this inner stabilizer. Write $P^\circ = \theta(P)$ and $E^\circ = \theta(E)$; then $P^\circ/E^\circ \cong P/E$. Every candidate section subgroup is pulled back through θ before applying a normal test. Every lifted enlargement lies in the proper original line stabilizer, even if its symplectic image is the whole symplectic group. Transport over the original coordinates uses the same normal higher core F_j in (N1) and (N2). There is no additional finite kernel in this inner-group isomorphism.

Finally, extending the finite scalar field to its algebraic closure preserves semisimplicity here. Indeed the semisimple image algebra of the finite group is a product of matrix algebras over finite fields; those fields are separable, and scalar extension preserves semisimplicity of that algebra. $\square$

6.2 Multiplicities and packets

In this subsection V denotes the semisimple space supplied by Lemma 6.2; the two removed factors will be added back at the end. All centralizers and groups on summands are taken in the indicated finite classical realization and then in the inner projective group. Passing through the finite central covers does not change a solubility assertion.

Lemma 6.3 (Soluble centralizer). *The centralizer of the lower quasisimple component in the inner classical group on V is soluble.*

Proof. Write $C = C_{T_j}(E)$ in the original natural realization. The group MC normalizes E and cannot contain T_j : the proper nontrivial E cannot be normal in the simple ambient group. Hence the full preimage of $\bar{U} = \langle z_i/F_j, \hat{E}, \hat{C} \rangle$ is proper in z_j . The two products $\hat{E}, \hat{C}$ are normal in $\bar{U}$ and commute exactly. Their full preimages commute modulo $F_j \trianglelefteq U$, with $F_j \leq D$. The active product $\hat{E}$ is not killed, so (N2) kills $\hat{C}$ and forces $C \leq P$. Since

$$C \cap E \leq Z(E), \quad C/(C \cap E) \hookrightarrow P/E,$$

C is soluble.

For the symplectic section of Lemma 6.2, put $E^\circ = \theta(E)$ and take the inverse image under θ of $C_{\mathrm{Sp}(V^\circ)}(E^\circ)$. The M -equivariance and injectivity of θ show that this inner inverse image and E commute exactly and are normal in their enlargement with M . Their transported products again commute in N_j/F_j , and their original preimages commute modulo the normal subgroup $F_j \leq D$. The enlargement lies in the proper original nonsingular-line stabilizer even if its symplectic image is the full symplectic group. Condition (N2) and the same quotient calculation prove that the symplectic centralizer is soluble. $\square$

Lemma 6.4 (An absolute multiplicity bound). *Every absolute irreducible constituent of V has multiplicity at most 15.*

Proof. We give the linear-algebra details needed for a bound independent of the fields and constituent dimensions. Let U be a simple module over the finite natural field F . Its endomorphism division ring is a finite field F' . On an isotypic summand U^m the commuting endomorphism algebra is $\mathrm{Mat}_m(F')$. After scalar extension, U splits into distinct absolute simples, each once; thus m is exactly their common absolute multiplicity.

In the linear case the centralizer contains $\mathrm{GL}_m(F')$. Its subgroup $\mathrm{SL}_m(F')$ has determinant one on the whole F -space: its determinant there is a power of the norm of its F' -determinant. Thus its derived subgroup survives the ambient special-linear condition.

For a form group, pair the simple isotypic summands by duality. A pair of distinct dual summands again supplies $\mathrm{GL}_m(F')$, acting on the second summand by the inverse adjoint. On a self-paired summand the adjoint operation on $\mathrm{Mat}_m(F')$ is an involution. If it acts nontrivially on F' , the commuting isometries are a unitary group on an m -dimensional multiplicity space. If it fixes F' , they preserve a nonsingular symmetric or alternating bilinear form on that multiplicity space. These descriptions follow by writing the invariant pairing as a matrix of pairings between the m copies of U ; Schur's Lemma makes each matrix entry an element of F' , and the symmetry of the original form is precisely the adjoint condition on that matrix.

There is one useful even-characteristic qualification. A nonsingular symmetric multiplicity form need not be alternating. Over a perfect finite field its diagonal function is the square of a linear functional. The vector representing that functional allows one to split off a nonsingular subspace of dimension one or two; its perpendicular is alternating and nonsingular. Thus a multiplicity space of dimension m still contains a nonsingular alternating subspace of dimension at least $m - 2$.

If $m \geq 16$, every group just described contains a subgroup isomorphic to $\mathrm{SL}_3(F'')$ for a suitable finite field F'' . In a general linear group use a three-dimensional block. In a symplectic, orthogonal, or unitary multiplicity space choose paired totally isotropic three-spaces and let $g \in \mathrm{SL}_3(F'')$ act as g and its inverse adjoint, fixing their perpendicular complement. Finite-field forms of dimension at least 16 have such paired spaces; the anisotropic part has dimension at most two. The preceding paragraph handles the nonalternating symmetric form in characteristic two in the same way.

These embedded groups are perfect and have nonabelian simple projective quotients. Being perfect, their determinants, similitude multipliers, spinor norms where applicable, and Dickson invariants in even-characteristic orthogonal groups are trivial. Therefore they lie in the actual inner classical group, rather than merely its full similitude group. A scalar quotient cannot make their images soluble.

In an orthogonal target of characteristic two, preserving the polar form on a nontrivial isotypic summand also preserves the quadratic form. Indeed two E -invariant quadratic refinements of that polar form differ by ℓ^2 for an E -invariant linear functional ℓ . On a sum of nontrivial simple E -modules there is no such nonzero functional. A commuting polar isometry consequently preserves the given refinement. On the trivial isotypic space use its actual quadratic isometry group; the same paired-three-space construction applies. The trivial space is nondegenerate, since it is orthogonal to every nontrivial isotypic summand.

We have proved that $m \geq 16$ would put a nonsoluble subgroup in the centralizer. Lemma 6.3 excludes this. $\square$

Define a *packet* to be an orbit of absolute simple types under finite-field descent, the action induced by M , and the duality pairing when a form or an ambient correlation is present, intersected with the set of types occurring in V . The sum belonging to a packet is defined over the finite natural field. In a form group it is nondegenerate, because all dual partners belong to the same packet. The packet decomposition and every union of its parts are normalized by M . For a linear graph automorphism this last statement refers to the groups on the parts: the induced adjoint anti-automorphism permutes the central isotypic idempotents and takes the full linear group on a part to the full linear group on its paired part. It therefore normalizes the product groups used below, although its action on subspaces is a correlation.

Lemma 6.5 (At most one large packet). *Two disjoint unions of packets on which E acts nontrivially cannot both have dimension at least 16. Consequently either one packet accounts for all but at most 15 dimensions, or the sum of all nontrivial packets has dimension at most 45.*

Proof. Let V_1, V_2 be two such unions, and put all remaining parts in V_3 . Let H_1, H_2 be the derived groups of the full linear or isometry groups on V_1, V_2 , acting trivially on the other parts. They commute, are normalized by M , and are normal in the enlargement generated by them and M . Its inner part preserves the proper decomposition and cannot contain T_j . Thus $\bar{U} = \langle z_i/F_j, \hat{H}_1, \hat{H}_2 \rangle$ has proper full preimage in z_j . Both H_1, H_2 are nonsoluble: the explicit SL_3 construction in the proof of Lemma 6.4 lies in each derived group and in the ambient inner group.

The two transported products are normal in this $\bar{U}$ and commute exactly; their original preimages commute modulo $F_j \trianglelefteq U$, with $F_j \leq D$. Condition (N2) kills one product and forces, say, $H_1 \leq P$. The kernel of the nontrivial action of E on V_2 is central in E , so $H_1 \cap E \leq Z(E)$. Hence

$$H_1/(H_1 \cap E) \hookrightarrow P/E$$

would make H_1 soluble, a contradiction. On the symplectic section, take inverse images under the finite M -equivariant isomorphism θ from Lemma 6.2. They commute exactly and their enlargement

lies in the proper original line stabilizer, so the same application uses the original higher core F_j , as in Lemma 6.3.

Suppose that a nontrivial packet has dimension at least 16. If another nontrivial packet occurs, apply the first assertion to the large packet and its entire complement, including the trivial packet. The complement has dimension at most 15. If no other nontrivial packet occurs, the complement is the trivial isotypic space, whose dimension is at most 15 by Lemma 6.4. If no packet has dimension 16, but their total dimension were at least 46, accumulate packets until their dimension first reaches 16. The accumulated dimension is at most 30, leaving at least 16 dimensions in the other union. This contradicts what was just proved. Thus in this case the total is at most 45. $\square$

6.3 Field shifts and the size of a packet

The endpoint field agreement is useful only together with the normalizer action on both edges of a shortcut. The quantity to control is the difference between the source field action and the scalar action on the upper natural space.

Lemma 6.6 (Field-shift synchronization). *Let the endpoint exponent be f , and let the intermediate exponent on a remaining shortcut be*

$$f' = \frac{u}{v}f, \quad (u, v) = 1.$$

Put $h = \gcd(f, f') = f/v$. If an element of M induces source field exponent a and upper scalar exponent s , then

$$s - a \equiv 0 \pmod{h}, \quad v(s - a) \equiv 0 \pmod{f}. \quad (6.1)$$

Every packet of absolute constituent types has size at most $2v$.

Proof. We work on simply connected covers and use Frobenius positions modulo the group parameter, rather than the degree of its natural scalar field. Write a classical finite group with parameter p^r as the fixed points of τF_p^r , where τ is a diagram operation. Steinberg's Restriction and Tensor Product Theorems give the unique parametrization

$$\lambda = \lambda_0 + p\lambda_1 + \cdots + p^{r-1}\lambda_{r-1}, \quad \lambda_i \text{ } p\text{-restricted,}$$

of its absolute simple modules; see [35, Theorems 41 and 43] and [34]. For a nontrivial simple module define

$$\text{supp}_r(\lambda) = \{i \in \mathbb{Z}/r\mathbb{Z} : \lambda_i \neq 0\}.$$

For an arbitrary module take the union of these supports over its nontrivial composition factors. A field twist by F_p^b translates support by b . Diagram operations and contragredience preserve positions. When a position wraps modulo r , its restricted weight is transformed by τ^{-1} , which does not change the position.

Consider one edge, with source and target exponents r, t , and let $d = \gcd(r, t)$. Suppose that the nonempty union $\mathcal{E}$ of source digit supports in the target natural module lies in one residue class modulo d . If a normalizer has target scalar exponent b and source field exponent a_0 , its transport of a composition factor has type

$$(L(\lambda) \circ \gamma^{-1})^{[b-a_0]},$$

where γ records the source diagram operation. This formula follows directly from $x(mw) = m(\alpha_m^{-1}(x)w)$ and the p^b -semilinearity of m . It applies to composition factors by transporting a composition series; semisimplicity on an individual edge is not required. An ambient correlation adds

contragredience. Inner-diagonal automorphisms do not affect the absolute module type, because they are algebraic inner conjugations on the cover. Normalization therefore implies

$$\mathcal{E} + (b - a_0) = \mathcal{E}.$$

Since $\mathcal{E}$ is nonempty and contained in a single residue class modulo d , we obtain

$$b - a_0 \equiv 0 \pmod{d}. \quad (6.2)$$

Assume for the moment that this support condition holds for both edges of the shortcut. Their source and target exponents are f, f' in opposite orders, so their common divisor d is the same integer h . For an element of M , denote its induced intermediate field exponent by a_1 . Its existence on the same intermediate component is the normalization conclusion of Proposition 5.8. The natural realization of that automorphism has scalar exponent a_1 , with any graph part recorded separately. Applying (6.2) to the two edges gives

$$a_1 - a \equiv 0 \pmod{h}, \quad s - a_1 \equiv 0 \pmod{h}.$$

Their sum proves (6.1).

To bound a packet, let Φ denote the operation of field twisting on absolute source types, and let Γ be the diagram group. The rank assumption gives $|\Gamma| \leq 2$. These operations commute, and $\Phi^f = \tau^{-1}$. Every stabilizer action lies in $\langle \Phi^h, \Gamma \rangle$. If $\tau = 1$, that group has order at most $2f/h$. If $\tau \neq 1$, the diagram involution already belongs to $\langle \Phi^h \rangle$, whose order is at most $2f/h$. Thus in either case the order is at most $2v$. Descent over the upper natural field has exponent f or $2f$ and belongs to the same group. Duality is the opposition diagram operation, also already included. The packet bound follows.

It remains to verify the support hypothesis for every remaining edge of Proposition 4.16 and Proposition 5.8.

- (a) In a nonlinear edge the natural module is a Frobenius twist of a nonzero restricted module. Its support is a singleton. The starting Frobenius position may be arbitrary; it cancels in (6.2).
- (b) A block, form, or subfield edge has natural or dual-natural nontrivial composition factors in one Frobenius position, allowing repetitions and trivial factors. A tensor edge with one active component has the same property: its other tensor factors are trivial for that perfect component, so they supply repetitions. The characteristic-two nonsingular-line representation has these composition factors as well, with the additional trivial factors already described.
- (c) For restriction of scalars, the source constituents are a direct sum of the Galois conjugates of the natural module. To distinguish group parameters from natural scalar fields, put $\epsilon_s, \epsilon_t = 2$ for a unitary source or target, respectively, and 1 otherwise. A scalar restriction of degree b_{sc} satisfies

$$\epsilon_s r = b_{sc} \epsilon_t t.$$

Its constituent positions are $t_0 + j\epsilon_t t$, so they all lie in one residue class modulo $d = \gcd(r, t)$. For example, restricting the natural matrix field of a unitary component with parameter p^r to a symplectic target with the same parameter has scalar degree two. Over the algebraic closure its module is

$$N \oplus N^{[r]} \cong N \oplus N^*.$$

Both positions are zero modulo r ; this repetition causes no tensor carry because the operation is a direct sum.

- (d) For a retained natural twisted-tensor edge, let b_{cyc} be the prime tensor-cycle length. Lemma 4.3 gives the exact usual-field relation

$$r = b_{\text{cyc}}t.$$

In the finite realization supplied by Proposition 5.8, the cyclic fixed-point construction gives the following actual finite-source modules, up to a common initial Frobenius twist [25, Proposition 2.8(ii) and Section 4(5), p. 449]. Here W is the source natural module in each case.

Linear, symplectic, and orthogonal targets. The module is

$$\bigotimes_{j=0}^{b_{\text{cyc}}-1} W_j^{[jt]},$$

where each W_j is W or W^* as prescribed by the diagram operation. Its positions are jt for $0 \leq j < b_{\text{cyc}}$; diagram changes affect the weights rather than these positions.

Unitary target, odd cycle. The source is $\text{SU}_m(p^r)$ and the module is

$$\bigotimes_{j=0}^{b_{\text{cyc}}-1} W^{[2jt]}.$$

Indeed, the cycle initially gives $W^{[jt]}$ for even j and $(W^*)^{[jt]}$ for odd j . On this finite source, $W^* \cong W^{[r]}$. The integers j for even j and $j + b_{\text{cyc}}$ for odd j are precisely $0, 2, \dots, 2b_{\text{cyc}} - 2$, giving the displayed module. Its positions $2jt$ are distinct modulo r because b_{cyc} is odd.

Unitary target, two-cycle. Since the cycle length is prime, this is the only even case. The source is $\text{SL}_m(p^{2t})$ and its module is

$$W \otimes (W^*)^{[t]},$$

with distinct positions $0, t$ modulo $r = 2t$.

Thus in every case the positions are distinct modulo $r = b_{\text{cyc}}t$ and lie in one residue class modulo $d = \gcd(r, t) = t$. A common initial twist adds the same t_0 to every position and preserves both assertions. Steinberg's Tensor Product Theorem therefore gives the required support condition with no coincident-digit carries. This distinctness comes from the exact finite cyclic construction.

Changes between the type B and type C conventions in characteristic two do not alter the numerical field exponent. The special isogenies are defined over $\mathbb{F}_2$, commute with Frobenius, and have Frobenius as their composites [17, Section I.3.1]. Thus each edge can use the convention in which its displayed module has the indicated support, while the intermediate field action has the same exponent in either convention. $\square$

Theorem 6.7 (Uniform absolute composition length). *In the setting of Proposition 4.16, let B be an absolute bound for the denominators of the intermediate field ratios f'/f on the remaining shortcuts. After a further fixed-spacing thinning, every upper natural module, restricted to a lower active quasisimple component, has at most*

$$L = 30B + 62$$

absolute composition factors, counted with multiplicity. The bound uses neither natural-module irreducibility between endpoints nor an irreducible-triples theorem.

Proof. Apply Lemma 6.2. On its semisimple natural section, Lemma 6.4 bounds each absolute multiplicity by 15, and Lemma 6.6 bounds each packet by $2B$ absolute types.

If there is a large nontrivial packet, it contributes at most $30B$ factors. All other nontrivial packets have total dimension at most 15 by Lemma 6.5. If no packet is large, their total dimension is at most 45. In either case the number of nontrivial composition factors is at most $30B + 45$. The trivial isotypic space has dimension at most 15, again by Lemma 6.4. Thus the semisimple section has length at most $30B + 60$. Adding the possible two factors from the nonsingular-line case gives the displayed bound.

All constants came from the remaining-edge bounds, explicit finite-dimensional linear algebra, and the common cyclic congruence (6.1). They are independent of field exponent, rank, and chain length. $\square$

7 Irreducible restrictions and the chain bound

We finish the proof of Theorem 3.2. The essential representation-theoretic input is the corrected classification of irreducible triples in [11, Theorem 4.1 and Table 3]. The composition-length bound of Section 6 is the starting point. After stabilizing the numbers of trivial and nontrivial factors, a first rank bound forces the natural endpoint modules to restrict irreducibly on earlier endpoints. A second rank bound uses this new irreducibility to limit the source ranks of the nonlinear edges on later shortcuts. The long chain supplies more such ranks than the bound permits.

We develop the two classification consequences together. The first uses bounded composition length of a natural restriction; the second uses an intervening simple group of larger rank whose natural module remains irreducible. These hypotheses control the twisted-diagonal spin families in different ways. Simplicity of the smallest group alone gives no uniform bound on the ambient ranks.

7.1 The two rank bounds

Throughout this section, algebraic groups are defined over an algebraically closed field k of characteristic $p > 0$, and representations are rational. We pass to simply connected covers when defining highest weights. For a classical simple group J , write N_J for its natural simple module and $n(J) = \dim N_J$. In characteristic two, for J of type B_r , this means the $2r$ -dimensional simple quotient of the orthogonal natural module of dimension $2r + 1$. Thus the finite B_r and C_r conventions agree. Write $\ell_H(M)$ for the composition length of an H -module M .

We call a nontrivial simple module *of natural type* if it is a Frobenius twist of a natural simple module or its dual, allowing the central, diagram, and characteristic-two B/C isogenies in the chosen realization. At ranks at least nine, its highest weight on a fixed simply connected source is $p^a \omega_1$, or $p^a \omega_r$ for the dual in type A_r , for some $a \geq 0$. The exclusions of natural type below therefore include the natural module, its dual, and the simple natural quotient in the characteristic-two odd orthogonal realization.

We record explicitly the image convention in the irreducible-triple Theorem. If $\varphi : Y \rightarrow \mathrm{SL}(V)$ is its representation, the table records $(\varphi(Y), \varphi(H), V)$. These images have the same ranks as the original groups. The possible B/C change in characteristic two must be made using this fixed representation; see [11, Lemma 2.1 and Remark 4.2(b)]. Let $\sigma : B_r \rightarrow C_r$ and $\tau : C_r \rightarrow B_r$ be the special isogenies. With Bourbaki numbering, their pullbacks on fundamental weights are

$$\begin{aligned} \sigma^*(\omega_i^C) &= \omega_i^B & (i < r), & & \sigma^*(\omega_r^C) &= 2\omega_r^B, \\ \tau^*(\omega_i^B) &= 2\omega_i^C & (i < r), & & \tau^*(\omega_r^B) &= \omega_r^C. \end{aligned} \tag{7.1}$$

Both composites are Frobenius; these are the root-isogeny formulas of [17, Chapter I, Section 3.2]. In particular,

$$\sigma^* N_{C_r} = N_{B_r}, \quad \tau^* N_{B_r} = N_{C_r}^{[1]}. \quad (7.2)$$

Here $[a]$ denotes the twist by p^a . Consequently the length and irreducibility of a natural restriction are preserved on passing to the image groups in the table. Frobenius twists preserve these properties by transporting invariant subspaces by the entrywise field automorphism. This argument does not require the subgroup embedding to be defined over the prime field.

For reference, the complete part of Table 3 with a connected simple subgroup of rank $r \geq 9$ is displayed in Table 1. We omit restrictions on p and on the weight of V that only remove cases and hence cannot enlarge the set of ranks. The two rows with an unbounded number of blocks retain their essential condition $p = 2$.

Rows of Table 3	Subgroup image	Ambient image
I_1, I'_1	C_r	A_{2r-1}
I_2, I_3	B_r	A_{2r}
I_4, I_5	D_r	A_{2r-1}
I_6	A_r	$A_{(r^2+r-2)/2}$
I_7	A_r	$A_{(r^2+3r)/2}$
IV_1, IV'_1	B_r	D_{r+1}
IV'_2	diagonal B_r in $B_r B_r$	D_{2r+1}
S_6	diagonal B_r in B_r^t	$D_{tr+1}, t \geq 2, p = 2$
MR_4	D_r	C_r
MR_5	diagonal B_r in B_r^t	$B_{tr}, t \geq 2, p = 2$

Table 1: The high-rank simple-subgroup rows of the corrected irreducible-triple table. Diagonal projections may involve different Frobenius morphisms.

Here and below “diagonal” allows the twisted projections explained in [11, Remark 6.1(c)]. A simple group projecting onto several simple blocks has an isogeny onto each block, so their ranks all equal r . The exceptional projection of an A_1 into a B_2 block cannot occur when $r \geq 9$. Every row not displayed either has subgroup rank at most eight or specifies an actual nonsimple product. This proves the exhaustion asserted in Table 1; one cannot replace a product in an unprimed row by an arbitrary diagonal subgroup without again testing irreducibility.

Put

$$\mathcal{R}(r) = \left\{ r, r+1, 2r-1, 2r, 2r+1, \frac{r^2+r-2}{2}, \frac{r^2+3r}{2} \right\}. \quad (7.3)$$

All rows outside S_6 and MR_5 have ambient rank in this set, which has at most seven elements.

Lemma 7.1 (Rank bound with composition length). *Let $H < Y$ be connected classical simple algebraic groups with $r = \text{rk}(H) \geq 9$. Suppose V is a nontrivial irreducible, p -restricted, tensor-indecomposable Y -module which is not of natural type, and $V|_H$ is irreducible. If $\ell_H(N_Y) \leq c$ for an integer $c \geq 1$, then*

$$\text{rk}(Y) \in \mathcal{R}(r) \cup \{tr, tr+1 : 2 \leq t \leq c\}.$$

In particular the number of possible ambient ranks is at most $2c+7$. The assertion is unchanged by central covers and the isogenies in (7.1).

Proof. Apply the corrected irreducible-triple Theorem to V and use Table 1. Only S_6 and MR_5 require an additional argument. Each of their t blocks contributes a nontrivial constituent of the

natural restriction, of dimension $2r$. In MR_5 , passing from the odd orthogonal natural module to its simple quotient removes one trivial constituent and leaves all t nontrivial constituents. In S_6 , there are also two trivial constituents, which do not affect the lower bound t on length. Different Frobenius twists may change the isomorphism types of these constituents, but not their dimensions or their number counted with multiplicity. Thus $t \leq c$. The image conventions and (7.2) justify the same calculation on the original groups. No complete-reducibility assertion is used. $\square$

The length hypothesis cannot simply be dropped. For example, in characteristic two fix $H = B_r$ and embed it in $B_r^t < B_{tr}$ by $g \mapsto (g, F(g), \dots, F^{t-1}(g))$. The spin module of B_{tr} restricts to

$$\bigotimes_{a=0}^{t-1} L_H(\omega_r)^{[a]},$$

which is irreducible by Steinberg's Tensor Product Theorem. The ambient spin module is restricted, nonnatural, and tensor-indecomposable, whereas t is arbitrary. This is the simple diagonal case of MR_5 .

Lemma 7.2 (Rank bound with an intervening group). *Let $H < X < Y$ be connected classical simple algebraic groups with $r = \text{rk}(H) \geq 9$ and $s = \text{rk}(X) > r$. Suppose $N_X|_H$ is irreducible. If V is a nontrivial irreducible, p -restricted, tensor-indecomposable Y -module, not of natural type, and $V|_H$ is irreducible, then*

$$\text{rk}(Y) \in \mathcal{R}(r).$$

The same conclusion holds when $X = Y$ and $N_Y|_H$ is irreducible.

Proof. Use the same representation $\varphi : Y \rightarrow \text{SL}(V)$ throughout the proof. For (Y, H, V) , every row of Table 1 outside S_6 and MR_5 gives the conclusion. Suppose one of these two rows occurs. Then $p = 2$, the image of H is B_r , the image of Y is D_m or B_m , and V is a half-spin or spin module, respectively. Let W be the orthogonal natural module of this fixed ambient image, on a cover when necessary. Every nontrivial composition factor of $W|_H$ has dimension $2r$.

Since $H < X$, irreducibility on H implies irreducibility on X . Apply the irreducible-triple Theorem a second time to (Y, X, V) , keeping the ambient group and representation unchanged. The theorem requires restrictedness and tensor-indecomposability on Y , so it makes no such assumption about the X -module $V|_X$. If the ambient image is B_m , the only row with a simple subgroup of rank at least ten is MR_5 . If it is D_m and V is half-spin, the only such rows are IV_1 , IV'_1 , IV'_2 , and S_6 . In all these cases the image of X is B_s and every nontrivial composition factor of $W|_X$ is a Frobenius twist of N_X , of dimension $2s$.

By hypothesis and (7.2), each such constituent remains irreducible on H . Restricting a composition series for $W|_X$ and refining it over H therefore exhibits an H -composition factor of dimension $2s$. This contradicts the preceding dimension $2r$, since $s > r$. The argument works for nonsplit natural restrictions as well as for direct sums.

There is no extra case from MR_4 : its ambient image is C_m , whereas the image of the fixed representation here is B_m or D_m . One cannot change that image between the two applications. Equations (7.1)–(7.2) preserve the natural irreducibility hypothesis on passing to these image groups, including an inseparable isogeny. They do not replace the fixed spin representation V by a natural representation.

Finally, if $X = Y$, the irreducibility of $N_Y|_H$ directly excludes S_6 and MR_5 , since their natural restrictions have at least two nontrivial composition factors. $\square$

7.2 From bounded length to natural irreducibility

We now use the finite components and algebraic lifts supplied by Proposition 5.8 and Lemma 5.9. Fix the last endpoint of a finite retained chain and one of its socle coordinates. Proposition 5.8 realizes the earlier endpoint components in this coordinate as nested quasisimple subgroups: the terminal derived groups of the nested inner projections nest. If $P_i \leq P_j$ are two such projections and $Q_j \leq R_j \leq P_j$ are the core and active preimages of Lemma 3.8, their original pair comparison is $(P_i \cap R_j)/Q_j$. The two-step component argument in Proposition 5.8 identifies its terminal derived subgroup with the conjugation image of the lower component on the higher one; the kernel is central. Thus this common finite realization carries the pair restrictions bounded in Theorem 6.7.

Write L_i for the finite component at a chain endpoint, $\mathbf{G}_i$ for its simply connected algebraic source, and N_i for its natural simple module over k . In the notation $N_j|_{L_i}$, take the natural representation on the standard finite cover U_j and pull it back along the finite map $U_i \rightarrow U_j$ from (5.5), or a composite of those maps. Two such finite lifts inducing the same map to L_j differ by a homomorphism from the perfect group U_i to the central kernel of π_j , and hence are equal. The finite pair maps therefore agree with their composites, so these restrictions compose. Algebraic restrictions are taken along one chosen composed path at a time. We make no identification of independently chosen shortcut lifts.

Lemma 7.3 (Stabilized composition factors). *Suppose that, for every $i < j$ in a chain of quasisimple components, the restriction $N_j|_{L_i}$ has exactly a nontrivial and b trivial composition factors. Assume $a \geq 1$ and that every nontrivial simple L_j -constituent restricts nontrivially to each earlier L_i . For every $i < j < k$, each nontrivial composition factor of $N_k|_{L_j}$ restricts irreducibly and nontrivially to L_i .*

Proof. Let $U_1, \dots, U_a$ be the nontrivial L_j -constituents, counted with multiplicity. If $U_t|_{L_i}$ had only trivial composition factors, the image of the perfect group L_i would be conjugate into an upper unitriangular group and hence would be both perfect and solvable. It would be trivial, contrary to the nontriviality hypothesis. Thus, writing a_t, b_t for its nontrivial and trivial composition counts, we have $a_t \geq 1$ and $b_t \geq 0$. Additivity of composition multiplicities along restriction gives

$$a = \sum_{t=1}^a a_t, \quad b = b + \sum_{t=1}^a b_t.$$

It follows that $a_t = 1$ and $b_t = 0$ for every t , as required. $\square$

Proposition 7.4 (Natural endpoint restrictions). *In the remaining configurations of Proposition 4.16, arbitrarily long chains have arbitrarily long subchains on which every natural endpoint module restricts absolutely irreducibly to each earlier endpoint.*

Proof. By Theorem 6.7, fix an absolute integer c bounding $\ell_{L_i}(N_j)$ for every retained comparison $i < j$. The pair of nontrivial and trivial composition counts has at most $(c+1)^2$ possible values. Finite Ramsey applied to these pair colors gives arbitrarily long subchains with constant counts (a, b) . These restrictions are nontrivial, and $a \geq 1$ follows because a perfect group cannot act nontrivially with only trivial composition factors. Moreover, a nontrivial simple representation of a quasisimple group has central kernel. Its restriction along the noncentral earlier component is therefore nontrivial. The hypotheses of Lemma 7.3 hold.

For each $j < k$, realize every nontrivial constituent U of $N_k|_{L_j}$ as the restriction of the corresponding restricted finite-group highest weight module of $\mathbf{G}_j$. Steinberg's Tensor Product Theorem decomposes that rational simple module into Frobenius twists of p -restricted simple

modules; see [35, Theorem 41]. If a restricted factor is tensor decomposable, continue factoring it until its nontrivial factors are tensor-indecomposable. This terminates since every factor has smaller positive dimension. Every resulting factor is p -restricted: its dominant highest weight is a summand of a p -restricted dominant weight.

Color a triple $i < j < k$ red if some such U has a nontrivial terminal factor V , not of natural type, and blue otherwise. In a red triple, Lemma 7.3 says that $U|_{L_i}$ is irreducible. Hence each tensor factor $V^{[d]}$ is irreducible on L_i . Use a path lift from $\mathbf{G}_i$ to $\mathbf{G}_j$ and replace its image by its p^d -Frobenius image. The module V is irreducible on this closed connected simple subgroup: any proper rational invariant subspace would also be invariant under the corresponding finite component. The subgroup has rank r_i and is proper because the endpoint ranks increase.

The natural restriction on this subgroup has algebraic composition length at most c . Indeed, restriction from an algebraic group to a finite subgroup is exact and cannot reduce composition length, and Frobenius transport preserves length. The finite restriction here is a twist of $N_j|_{L_i}$. Lemma 7.1 consequently bounds the number of possible middle ranks r_j by $2c + 7$, for fixed i .

A red homogeneous chain with more than $2c + 9$ vertices is impossible: fix its first and last vertices i, k and vary its middle vertex j . The distinct ranks r_j would exceed that bound. A second application of finite Ramsey therefore gives arbitrarily long blue homogeneous subchains.

Fix $i < j < k$ in a blue subchain and choose a nontrivial constituent U of $N_k|_{L_j}$. There is at least one nontrivial terminal tensor factor, and every such factor is now of natural type. Since $U|_{L_i}$ is irreducible, a natural or dual natural Frobenius factor restricts irreducibly to L_i . Undoing its twist, dual, and the isogenies (7.1) shows that $N_j|_{L_i}$ is irreducible. Delete the last vertex of the blue subchain, so that a suitable k exists for every remaining pair $i < j$. All modules and restrictions were taken over k , so the resulting finite restrictions are absolutely irreducible. $\square$

7.3 Restricted natural factors and shortcut tails

The word “restricted” in the next lemma concerns one fixed algebraic source. A representation with many different Steinberg digits need not have bounded tensor length. The retained nonlinear edges of Proposition 4.16, however, have one digit after a common Frobenius twist is removed.

Lemma 7.5 (A restricted product of natural factors). *Let J be a classical simple algebraic group of rank at least nine. Let M be a nontrivial p -restricted simple J -module. If every nontrivial tensor-indecomposable factor of M is of natural type, then $M \cong N_J$ or N_J^* . In particular there is exactly one nontrivial terminal factor and $\dim M = n(J)$.*

Proof. A tensor factor of a simple module is simple, since any proper nonzero submodule of the factor would tensor to one of the whole module. Highest weights add in an irreducible tensor product. The highest weights of the natural-type factors have the form $p^a\omega_1$, or $p^a\omega_r$ in type A_r for the dual. Equation (7.1) shows explicitly that the special isogenies add no further natural weights. Since their sum is p -restricted and all fundamental-weight coefficients are nonnegative, every exponent a is zero. The terminal factors are therefore N_J or N_J^* .

Any two of these factors have reducible tensor product. For two identical factors, the kernel of the flip minus the identity is a nonzero proper invariant subspace: it contains $v \otimes v$, while $v \otimes w - w \otimes v \neq 0$ for independent v, w . This also proves properness in characteristic two. For opposite factors, $N_J \otimes N_J^* \cong \text{End}(N_J)$ contains the proper invariant line of scalar endomorphisms. The same arguments apply to dual identical factors. Tensoring such a proper submodule with all remaining factors contradicts simplicity of M . There is thus exactly one terminal factor. $\square$

Lemma 7.6 (Pulling irreducibility through a natural tail). *Consider an edge between the finite components associated with algebraic sources J and K . Suppose every nontrivial absolute composition factor of the target natural module, restricted to the source, is a tensor product of natural-type J -modules. If that target module is absolutely irreducible on an earlier noncentral quasisimple component L , then $N_J|_L$ is absolutely irreducible. This applies to each retained geometric edge and to every one-digit edge whose restricted terminal factors are all of natural type.*

Proof. Irreducibility on L implies irreducibility on the source component, so the target restriction has a single absolute composition factor. It is nontrivial because the target natural dimension exceeds one. By hypothesis this factor is a tensor product of natural-type modules. Irreducibility of its further restriction forces each factor to restrict irreducibly. Undoing a twist, a dual, and the natural isogenies yields the assertion for N_J . For retained geometric edges this also follows from Corollary 5.11. For the stated one-digit edges it follows from Lemma 7.5, after restoring the common twist. $\square$

7.4 The final contradiction

Completion of the proof of Theorem 3.2. Suppose tested chains with different endpoint labels have unbounded length. Apply Proposition 3.13 and Proposition 4.16. We obtain arbitrarily long configurations with strictly increasing endpoint ranks, with their minimum rank tending to infinity, and with two-edge shortcuts on all pairs. There are absolute constants $c_0 > 0$, $C \geq 1$, and $B \geq 1$ such that the natural endpoint dimensions satisfy

$$n_j \geq c_0 n_i^2 \quad (i < j), \quad (7.4)$$

and every retained geometric edge satisfies

$$n_{\text{target}} \leq C n_{\text{source}}^B. \quad (7.5)$$

Every other edge is a defining-characteristic irreducible representation with one restricted digit after removing a common twist. At least one such nonlinear edge occurs in each shortcut. Proposition 5.8 and Lemma 5.9 supply the actual component homomorphisms and their algebraic path lifts. By Proposition 7.4, pass to arbitrarily long subchains with all natural endpoint restrictions absolutely irreducible.

Color a pair of endpoints red if its chosen shortcut has a one-digit edge with a nontrivial restricted tensor-indecomposable factor not of natural type. Color the pair blue otherwise. Finite Ramsey leaves arbitrarily long homogeneous configurations of one of these two kinds. We exclude both.

First consider a blue configuration. On every one-digit edge, Lemma 7.5 bounds the target natural dimension by the source natural dimension; restoring a Frobenius twist does not change dimension. Thus every edge of every shortcut satisfies (7.5), enlarging C, B if necessary. Composing at most two edges gives the uniform bound

$$n_j \leq A n_i^E \quad (i < j), \quad A = C^{B+1}, \quad E = B^2. \quad (7.6)$$

Choose a fixed integer d with $2^d > E$. Along d consecutive comparisons of a retained chain, (7.4) gives

$$n_d \geq c_0^{2^d-1} n_0^{2^d},$$

whereas the shortcut from its first to its last endpoint gives $n_d \leq A n_0^E$. These inequalities contradict each other as the minimum rank, and hence n_0 , tends to infinity. This proves the bounded-natural-factor case without imposing a bound on the number of Steinberg digits of an arbitrary composite endpoint representation.

Now consider a red homogeneous configuration. Fix an early endpoint H of rank $r \geq 9$. Choose eight later pair comparisons with disjoint ordered index intervals, leaving a gap between successive intervals. For each such comparison, let X be its lower endpoint, and choose the last edge in its shortcut that has a restricted factor not of natural type. Let Y be the algebraic source of that edge and Z its target. Every edge after it has a natural tail in the sense of Lemma 7.6: it is either geometric or a one-digit edge with only natural restricted terminal factors. The upper endpoint natural module is irreducible on the finite component of H . Applying that lemma backwards along the tail shows that N_Z is irreducible on the same component.

Remove the common Frobenius twist of the Y -representation on N_Z , and choose a nontrivial restricted tensor-indecomposable factor V not of natural type. Irreducibility of the whole restriction implies irreducibility of the corresponding twisted factor on the finite H . Compose the actual algebraic maps along the path from H through X to Y , and apply the removed Frobenius morphism to both lower images. Then V is irreducible on the resulting connected simple image of H . Its rank is r , and the resulting image of X has its original rank $s > r$. Its natural simple module remains irreducible on the image of H , by the endpoint conclusion and (7.2). Finite irreducibility implies each of these rational irreducibility statements, since a rational invariant subspace would also be invariant under the actual finite image.

If the algebraic image of X is proper in Y , apply Lemma 7.2. If that image is all of Y , the map from X to Y is an isogeny. This can occur even when the finite groups are strictly nested, for example on a subfield edge. The pullback of N_Y is a natural simple module or its dual, up to Frobenius twist and the isogenies in (7.2), by the root-datum classification of isogenies [15, Definition 6.1.8, Example 6.1.9 and Theorem 6.1.16(1)]. Indeed, for compatible simple roots an isogeny has a root bijection δ and powers q_i of the characteristic. Write the Cartan matrices in the row-coroot convention $A_{ij}^X = \langle \alpha_j, \alpha_i^\vee \rangle$ and $A_{ab}^Y = \langle \beta_b, \beta_a^\vee \rangle$. The compatibility equation is $q_j A_{ij}^X = q_i A_{\delta(i)\delta(j)}^Y$. Pairing the pullback of the natural fundamental weight with the source simple coroots gives q_i at the node mapped to the natural node and zero elsewhere. In the retained ranks this node is the natural node, or its type- A dual; the only change of root lengths is the characteristic-two B/C case in (7.2). Since N_X is irreducible on the lower H , it follows that N_Y is irreducible on the corresponding image of H . The $X = Y$ case of Lemma 7.2 therefore applies. Thus every one of the eight chosen source ranks belongs to $\mathcal{R}(r)$. They are nevertheless distinct. Along an algebraic edge the rank cannot decrease, because a maximal torus maps with finite kernel into a torus of the image. Hence each chosen source rank lies between the ranks of the two endpoints of its comparison. The gaps between the ordered comparison intervals, and strict increase of endpoint ranks, make these eight intervals disjoint in rank. This contradicts $|\mathcal{R}(r)| \leq 7$.

Both homogeneous alternatives are impossible. All Ramsey bounds and all path lengths used here depend only on the absolute constants in the preceding lemmas. In particular only fixed finite paths are needed before taking the large-rank limit. The assumed unbounded family of tested chains cannot exist, proving the asserted absolute chain bound and completing Theorem 3.2. $\square$

8 A finite lattice carrying all the tests

We now construct the lattice to which Theorem 3.2 will be applied. All choices in this section precede the choice of a representing group. In particular, the length of a test chain will be chosen using the absolute bound in that Theorem.

Throughout this section, an *original* vertex means a vertex of one of the two small lattices specified below, before any insertion. An original atom of a Boolean lattice need not remain an atom after insertions; its original meets and joins will, however, be preserved.

8.1 Two original components and their requested tests

There are two original components, each with endpoints 0, 1. For each of the requested pairs below, we shall insert a chain and the tests from Definition 3.1. A forward pair is read in the displayed order, with bottom 0; a reverse pair is read in the dual order, with bottom 1.

The *Boolean component* is the Boolean lattice B_4 of subsets of a fixed four-element set. Its forward requested pairs are

$$(v, 1) \quad (0 < v < 1 \text{ original}). \quad (8.1)$$

Its reverse requested pairs are all

$$(u, v) \quad (0 < v < u < 1 \text{ original}), \quad (8.2)$$

where a reverse pair is read in the reverse order: its test starts at u and ends at v . Thus every reverse test is contained in a forward interval with two proper original endpoints.

The *detector component* has a distinguished original vertex X . We choose its lower half so that every coatom below X contains one of two designated vertices d_1, d_2 . It is the horizontal sum of the two chains

$$0 < p_i < d_i < q_i < X \quad (i = 1, 2), \quad (8.3)$$

and its upper half is the horizontal sum of the two chains

$$X < e_i < z_i < f_i < 1 \quad (i = 1, 2). \quad (8.4)$$

The two halves are joined at X , and we put $z = z_1$. Its forward requested pairs are

$$(d_1, 1), \quad (d_2, 1), \quad (X, 1), \quad (8.5)$$

and its only reverse requested pair is (z, X) . Both $[0, X]$ and $[X, 1]$ are already fenced, with two separate components containing comparable pairs.

8.2 The lattice properties required in the reduction

The forward requests will equate the indicated simple labels to the global top label. The reverse requests will equate labels inside reversed upper filters. These local representations are the ones that will arise in Section 9; a representation of the entire dual lattice is not needed. The construction will prove the following statement while preserving the original Boolean operations.

Proposition 8.1 (The decorated lattice). *There is a finite nonempty lattice L with the following properties. The properties hold in either orientation of L , using the corresponding Boolean and detector components.*

- (i) *The lattice contains a bounded copy of the original B_4 . Every proper nonzero original Boolean vertex v has both $[0, v]$ and $[v, 1]$ fenced. The global interval $[0, 1]$ is fenced.*
- (ii) *There are original detector vertices d_1, d_2, X, z with $0 < d_i < X < z < 1$. The intervals $[0, d_i]$, $[0, X]$, $[X, 1]$ and $[z, 1]$ are fenced. Every coatom of $[0, X]$ is above a d_i .*
- (iii) *In any finite subgroup-interval representation $L \cong [D, G]$, the simple labels of every proper nonzero original Boolean vertex, of d_1, d_2 , and of X equal the simple label of G .*

- (iv) Let $0 < v < u < 1$ be comparable original Boolean vertices. In any finite subgroup-interval representation

$$[v, 1]^{\text{op}} \cong [A, U_v],$$

the labels of the vertices corresponding to u and v , taken relative to the common bottom A , are defined and equal. The same assertion holds for the detector pair (z, X) in any representation $[X, 1]^{\text{op}} \cong [A, U_X]$.

In (iv), no representation of the entire dual lattice is assumed. If several such upper filters have compatible realizations with a common bottom A , the conclusions hold in that compatible system.

The remainder of this section constructs this lattice. We first show how to add the chain tests while protecting their covering relations, then combine the forward and reverse tests and choose their length.

8.3 Private insertions and their persistence properties

Let $a < b$ in a finite lattice P , and let Q be a finite bounded lattice whose bottom and top are identified with a and b . A *private insertion of Q into $[a, b]$* retains the old orders and introduces no comparisons between an old vertex and a fresh interior vertex except those forced through a and b .

Lemma 8.2 (Private insertion). *A private insertion gives a finite lattice and preserves all old meets and joins. If w is a fresh interior vertex and v is an old vertex, then*

$$v \leq w \iff v \leq a, \quad w \leq v \iff b \leq v. \quad (8.6)$$

An old cover $c \prec d$ is destroyed by the insertion if and only if the insertion has endpoints $a = c$, $b = d$ and has a nonempty interior.

Proof. The prescribed order is transitive: every route from an old vertex into the fresh interior passes through a , and every route back passes through b . This proves (8.6) and antisymmetry. For an old vertex v incomparable with a fresh interior vertex w , the cross operations are

$$v \wedge w = v \wedge a, \quad v \vee w = v \vee b,$$

where the operations on the right are those of P . In the comparable cases, the smaller and larger vertices give the meet and join. For two fresh vertices, the operations in Q remain their operations: an old common lower bound lies below a , and an old common upper bound lies above b . An analogous argument shows that an old meet or join cannot be changed by a new bound. This proves the lattice assertion.

If a fresh vertex w lies strictly between the old cover $c \prec d$, then (8.6) gives $c \leq a < b \leq d$. The cover condition forces $a = c$ and $b = d$. Conversely, an insertion with those endpoints and a fresh interior vertex destroys the cover. $\square$

A *chain insertion with k interior vertices* means a private insertion of a chain with exactly k fresh interior vertices. A *fencing insertion* means two separate chain insertions, each with exactly two fresh interior vertices, into the same interval. In particular, these are nontrivial fences in the sense used in Lemma 2.2.

Lemma 8.3 (Persistent fences). *Suppose the proper part of an interval $[a, b]$ has two different components in its comparability graph, each containing two strictly comparable vertices. Then $[a, b]$ is fenced. This property persists under any finite sequence of private insertions into the surrounding lattice. In particular, a fencing insertion makes an interval fenced permanently.*

Proof. Vertices in different components of the proper comparability graph have meet a and join b . Indeed, a common lower bound strictly above a , or a common upper bound strictly below b , would connect them. For a given interior vertex, choose the two comparable vertices in one of the two specified components different from its own. They are its required complements.

Consider an insertion into $[c, d]$. A fresh vertex can belong to $[a, b]$ only if $a \leq c < d \leq b$, by Lemma 8.2. If both c, d are proper vertices of $[a, b]$, they lie in the same comparability component. All comparisons from the new interior to the old proper part then enter that component. The same is true if just one of c, d is proper. If $c = a$ and $d = b$, the new interior has no comparisons with the old proper part. Thus no insertion merges two old components. The two reserved comparable pairs survive, proving persistence. A fencing insertion supplies two such separate components. $\square$

We shall make several insertions in a *batch*. This means a finite list of private insertions all of whose endpoints were present before the batch. Perform them in any fixed order. All their fresh vertices are distinct, and Lemma 8.2 applies at every step. This convention makes each batch a completely specified finite operation; it does not call for repeating the batch on its new vertices.

8.4 The shape of one chain test

Fix an integer $N \geq 2$ and two existing vertices $a < b$. The following five batches constitute the *shape construction* for a test on (a, b) . All sets of vertices appearing in a later batch are the named sets from the earlier batches, not all vertices then present.

- (1) Insert a chain with $N - 1$ fresh interior vertices and name its vertices

$$a = x_0 < x_1 < \cdots < x_N = b.$$

- (2) For each $0 \leq i < j \leq N$, insert one fresh vertex v_{ij} privately into $[x_i, x_j]$. The designated shortcut is $x_i \prec v_{ij} \prec x_j$.

- (3) Make fencing insertions into every interval in the following finite set:

$$\{[x_i, x_j] : 0 \leq i < j \leq N\} \cup \{[a, v_{ij}] : 1 \leq i < j \leq N\}.$$

- (4) Define the set of upper targets

$$\mathcal{U} = \{x_j : 1 \leq j \leq N\} \cup \{v_{ij} : 1 \leq i < j \leq N\}. \quad (8.7)$$

For every $u \in \mathcal{U}$, insert privately into $[a, u]$ the bounded lattice with fresh vertices $m_u, t_{u,1}, \dots, t_{u,16}$ and relations

$$a < m_u < t_{u,k} < u \quad (1 \leq k \leq 16),$$

with the $t_{u,k}$ pairwise incomparable. Thus $a < m_u$ and $[m_u, u]$ is exactly M_{16} at this stage.

- (5) For every pair (a, u) with $u \in \mathcal{U}$, make two separate chain insertions with one interior vertex. Also do so for every pair (c, b) with

$$c \in \mathcal{C} = \{x_i : 0 \leq i < N\} \cup \{v_{ij} : 0 \leq i < j < N\}. \quad (8.8)$$

Repeated pairs may be listed just once. The two new vertices for a pair (c, d) are its reserved coatom witnesses: they are coatoms below d , and their meet is c .

The vertices x_i and v_{ij} are called the *principal vertices* of the test. The vertices $m_u, t_{u,k}$ and the reserved coatom witnesses are not principal vertices and are not used as endpoints of any subsequent shape insertion for this test.

Lemma 8.4 (A finite shape construction). *The five batches are finite and have the following properties.*

- (i) *Every designated shortcut has both edges as covers.*
- (ii) *Every $[x_i, x_j]$ is fenced. For every $u \in \mathcal{U}$, $[a, u]$ is fenced, a is a meet of coatoms below u , and $a < m_u$ with $[m_u, u] \cong M_{16}$.*
- (iii) *Every vertex in $\mathcal{C}$ is a meet of coatoms below b . Each v_{iN} is itself a coatom below b .*

No interval $[a, v_{0j}]$ is included in the extra target operations. The construction introduces at most $27N^2$ vertices.

Proof. Each shortcut edge is a cover immediately after its one-vertex insertion. By the cover criterion in Lemma 8.2, a later shape insertion could destroy it only by using that exact edge as its endpoint pair. The listed pairs never do so. The only apparent possibilities would be $[a, v_{0j}]$, which is excluded from (8.7), and $[v_{iN}, b]$, which is excluded from (8.8). This proves (i).

All pair fences are present in the third batch and persist by Lemma 8.3. The chain targets x_j already have the fence on $[a, x_j]$ from that batch; the other upper targets are listed there expressly. Each reserved coatom witness is fresh when inserted and is not an endpoint of a subsequent shape insertion. Its edge to the upper endpoint therefore survives. Its meet with its companion remains the lower endpoint, by Lemma 8.2.

When an M_{16} branch is inserted, its interval $[m_u, u]$ has no other vertices: an old vertex above m_u must be above u . A subsequent shape insertion could add a vertex inside this interval only if both its endpoints lay in $[m_u, u]$. None of $m_u, t_{u,k}$ is used subsequently as a shape endpoint. The remaining possible endpoint is u , which cannot by itself form a nondegenerate interval inside $[m_u, u]$. Thus this interval remains exactly M_{16} . These arguments give (ii) and (iii).

For completeness, the numbers of new vertices in the first four batches are respectively

$$N - 1, \quad \frac{N(N + 1)}{2}, \quad 4N^2, \quad \frac{17N(N + 1)}{2}.$$

The fifth batch adds at most $2N(N + 1)$ vertices. Their sum is $15N^2 + 12N - 1 \leq 27N^2$. □

The exclusions involving v_{0j} and v_{iN} are necessary. Inserting a fence into either of their designated cover edges would destroy the two-edge shortcut. The target set (8.7) is exactly the one required by Theorem 3.2; no test at a target v_{0j} is needed there.

8.5 Combining the forward and reverse tests

Here is the complete construction for either original component C . First perform the five shape batches for all its forward requested pairs, using fresh vertices for every pair. One may perform a batch for all pairs before passing to the next batch. Then apply the same five batches in the dual lattice for all reverse requested pairs, again with disjoint fresh vertices. In each case a private insertion is made in the order currently being used; its dual is a private insertion in the forward lattice.

Let C_N^{sh} be the finite lattice now obtained, and freeze its vertex set. Define two subsets of this frozen set:

$$\mathcal{F}_C = \bigcup_{(a,1) \text{ forward requested}} [a, 1]_{C_N^{\text{sh}}}, \quad (8.9)$$

$$\mathcal{R}_C = \bigcup_{(u,v) \text{ reverse requested}} [v, u]_{C_N^{\text{sh}}}. \quad (8.10)$$

The following is a *single final batch*, with every endpoint chosen from this frozen set:

- make a fencing insertion into $[0, t]$ for every $t \in \mathcal{F}_C \setminus \{1\}$;
- make a fencing insertion into $[t, 1]$ for every $t \in \mathcal{R}_C$.

In the detector omit the first insertion when $t = X$, since $[0, X]$ is already fenced. No insertion into $[0, X]$ is made anywhere in the construction. Denote the resulting component by C_N .

We emphasize that the new vertices of this final batch are not added to $\mathcal{F}_C$ or $\mathcal{R}_C$. The next lemma proves that no such closure operation is needed.

Lemma 8.5 (Simultaneous tests and the final batch). *For either component C_N , all the following statements hold.*

- (i) *Every forward shape test retains all conclusions of Lemma 8.4. Its principal vertices other than 1 have fenced intervals from 0.*
- (ii) *Every reverse shape test retains all conclusions of Lemma 8.4 in the reverse order. Every vertex in each of its comparison intervals has a fenced interval from the reverse base 1.*
- (iii) *For a forward chosen chain and any $i < j$, every vertex $t \in [x_i, x_j]$, except possibly $t = 1$, satisfies one of the following alternatives:*
 - (a) $[0, t]$ is fenced;
 - (b) *there is a vertex s with $x_i \leq s < t$, with $[0, s]$ fenced, such that every vertex of $[0, t]$ is comparable to s .*
- (iv) *In the detector the coatoms of $[0, X]$ are exactly q_1, q_2 . In particular, each contains one of d_1, d_2 below it. The intervals $[0, X]$ and $[X, 1]$ remain fenced.*

The construction is finite. If C has q_0 vertices and altogether h requested pairs in the two directions, then

$$|C_N| \leq 9(q_0 + 27hN^2).$$

Proof. We first record how the two shape constructions are separated. Every fresh vertex of a forward shape construction lies below no proper original vertex. This follows inductively from (8.6): its upper insertion endpoint is either 1 or an earlier fresh forward vertex with the same property. In contrast, every vertex of a reverse shape construction lies in a forward interval $[v, u]$ with $0 < v < u < 1$ original. Thus every vertex of $\mathcal{R}_C$ lies below a proper original upper bound, and no fresh forward vertex belongs to $\mathcal{R}_C$.

These observations also show that the shape constructions do not spoil each other's designated intervals. Different forward tests have disjoint fresh interiors; their only shared shape endpoints are original vertices and 1. Hence neither uses a fresh endpoint of the other's designated cover. No reverse shape insertion can use such a forward endpoint as its lower endpoint, since it lies below a proper original upper bound. For reverse tests, two different tests again have disjoint fresh interiors and share only original endpoints. Lemma 8.2 therefore protects their covers. The same argument protects the exact M_{16} intervals: a shape insertion could enter one only if its two endpoints lay inside it, whereas its fresh bottom and atoms are never shape endpoints. Fences persist by Lemma 8.3.

We next check the final batch. A new vertex of $[0, t]$ belongs to no comparison interval $[a, b]$ with $a > 0$, by (8.6). A new vertex of $[t, 1]$ can belong to such a comparison only if $b = 1$ and $a \leq t$. The lower endpoints of the latter insertions are in $\mathcal{R}_C$, and consequently lie below proper original upper bounds.

In particular, a reverse fence cannot subdivide a fresh forward coatom edge $c \prec 1$: that would require its lower endpoint to be exactly c , which lies below no proper original vertex. A forward cover with a proper upper endpoint cannot be subdivided by an insertion ending at 1. Thus all forward shortcut edges and all reserved coatom witnesses survive. A forward M_{16} interval with proper upper endpoint cannot be entered by a reverse fence. If its upper endpoint is 1, its fresh bottom lies below no proper original vertex; hence no point of $\mathcal{R}_C$ lies above that bottom. So this M_{16} interval also remains exact. Forward base insertions cannot enter any of these positive-bottom intervals.

All reverse comparison intervals have two proper original bounds in the forward order. Neither kind of final insertion can add a vertex to such an interval: an insertion beginning at 0 cannot do so, and an insertion ending at 1 cannot do so. Their covers, witnesses and exact M_{16} intervals are therefore unchanged. The final reverse fences label every vertex in each reverse comparison interval by the definition of $\mathcal{R}_C$. All required forward principal vertices belong to $\mathcal{F}_C$; the sole omitted proper target is the detector X , whose lower interval was fenced at the outset. This proves (i) and (ii).

For (iii), an old vertex in a forward comparison belongs to $\mathcal{F}_C$, so its lower interval is fenced, with the stated exception for the global top. The only possible new vertex is an interior vertex w of a final reverse fencing insertion into $[s, 1]$. If $x_i \leq w$, then $x_i \leq s$ by (8.6). Every vertex below w either lies below s or lies on the same private chain above s . This assertion also includes vertices of other insertions in the final batch: a point from another branch can be below w only by passing through the old endpoint s . Thus s is comparable to every vertex of $[0, w]$. Moreover $s \in \mathcal{F}_C$, since $x_i \leq s < 1$, and its lower interval is fenced. This is exactly alternative (b), including the requirement that the bottleneck lie above the lower chosen chain vertex. It proves closure of the normal-group tests without any new round of insertions.

For (iv), no shape insertion introduces a point strictly below X . Forward fresh points lie below no proper original vertex, and reverse shape insertions in the detector are contained in $[X, z]$. A final insertion ending strictly below X introduces only points below that endpoint, so it cannot create a new coatom of $[0, X]$. A final insertion ending above X introduces no new point below X . The only remaining possibility would be an insertion into $[0, X]$, which was omitted. Hence the original covers $q_i \prec X$ are precisely the coatoms of the lower half. Its two separate original branches,

and those of the upper half, still give fences by Lemma 8.3.

Finally, Lemma 8.4 bounds the size before the final batch by $q = q_0 + 27hN^2$. Each of $\mathcal{F}_C$ and $\mathcal{R}_C$ has at most q members, and each fencing insertion adds four vertices. The final size is at most $q + 4q + 4q = 9q$. This also proves finiteness directly from the displayed batches. $\square$

8.6 Assembly and proof of the lattice properties

Let B_N and E_N denote respectively the decorated Boolean and detector components just constructed. Define L_N to be the horizontal sum

$$L_N = B_N \boxplus B_N^{\text{op}} \boxplus E_N \boxplus E_N^{\text{op}}, \quad (8.11)$$

identifying only the common endpoints 0, 1. Interiors of different components are incomparable, with cross meet 0 and cross join 1. The order reversal exchanging each component with its opposite is an order anti-isomorphism of L_N . Thus L_N is self-dual.

Every component has two comparable proper original vertices. The global interval $[0, 1]$ is therefore fenced: for a vertex in one component, take a comparable pair in a different component. A comparison with a proper starting point, or a principal interval with a proper upper endpoint, contains no interior vertex from a different component. Thus horizontal summation preserves every local property proved above, and supplies the previously deferred label at the global top.

Proof of Proposition 8.1. Let B be the absolute chain threshold of Theorem 3.2, and choose the single integer $N = \max\{2, \lceil B \rceil\}$. Construct $L = L_N$ by the finite operations above. This choice is independent of every possible representing group. Finiteness follows from Lemma 8.5 and (8.11); no limiting lattice is used.

All original lattice operations survive by Lemma 8.2. Every proper original Boolean vertex is an endpoint of a forward requested pair and of a reverse requested pair, so its two principal intervals receive the required fences. The detector statements follow from Lemma 8.5. This gives (i) and (ii).

For (iii), take a finite representation $[D, G]$. By Lemma 2.3, the global fence and the required principal fences provide all forward labels. In every forward requested test, the sequence $x_0 < \dots < x_N$ and its private shortcuts have the shape required by Theorem 3.2, including its nonconsecutive shortcuts. The interval $[x_i, x_j]$ is fenced; every target in (8.7) has its bottom saturation, fence, and exact positive-bottom M_{16} interval; and all the specified principal vertices are meets of coatoms below the largest endpoint, with v_{iN} themselves coatoms. Finally, Lemma 8.5(iii), together with the global fence, verifies condition T5 at every vertex of every comparison: either its lower interval is fenced or it has the specified labelled bottleneck above the lower chain vertex. In the latter case, Corollary 2.4 supplies the required soluble-normal and commuting-normal consequences. All hypotheses of the chain Theorem hold, and $N \geq B$ forces equality of its endpoint types. The forward requested pairs are precisely (8.1) and (8.5), giving the asserted equalities.

For (iv), fix one of the indicated actual upper-filter representations. Its least lattice vertex is the former 1, and its group bottom is A . The entire reverse test for (u, v) lies inside this upper filter. Its principal labels come from intervals $[w, 1]^{\text{op}}$, which are fenced by the final reverse batch. All vertices of its comparison intervals have those fences, by Lemma 8.5(ii). The reverse shape construction supplies all the other hypotheses of the same chain Theorem. Consequently the two endpoint labels, now taken relative to A , are equal. The proof for (z, X) is identical. If the actual intervals are compatible, their common subgroups and their labels are the same on overlaps; the argument makes no additional choice of a representation and thus applies simultaneously.

Finally, the order reversal exchanging the paired components in (8.11) supplies the same Boolean and detector data in the other orientation. This proves the last assertion. $\square$

We record explicitly the Boolean incidence used in the final gluing argument; Figure 2 shows its proper faces. In the reverse original B_4 , fix an original atom x and let $a_1, \dots, a_3$ be the other original atoms. Put

$$u_i = x \vee a_i \quad (1 \leq i \leq 3),$$

with operations in that reverse Boolean copy. Each $u_i \vee u_j$ for $i \neq j$ has rank three and is proper, while $\bigvee_{i=1}^3 u_i$ is the reverse top. Every proper original Boolean vertex is the join of the original atoms below it. These equalities are still equalities in the full decorated lattice, since all insertions preserve old operations. Thus the local proper joins and the incompatible total join required later are present in the single finite lattice of Proposition 8.1.

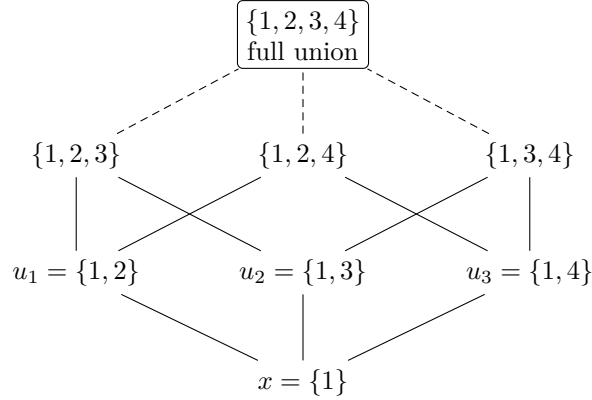


Figure 2: The original Boolean faces containing x . Each pair among u_1, u_2, u_3 has a proper union, but their total union is full. All displayed subset relations survive the private insertions. In the extension configuration of Proposition 9.7, solid edges correspond to restrictions of compatible homomorphisms on actual nested domains. The dashed edges reach the full union, for which a common extension is forbidden.

9 From subgroup intervals to extensions

We now apply the decorated lattice of Proposition 8.1. The reductions in this section use its tests in both orientations. The simple group attached to the top of the original group interval will be denoted by T . The simple group attached to an interval of extension domains will later be denoted by R ; these groups need not be isomorphic.

Lemma 9.1. *Let T be a finite nonabelian simple group and let $\text{Inn}(T) \leq H \leq \text{Aut}(T)$. Every nontrivial normal subgroup of H contains $\text{Inn}(T)$. In particular, H has trivial soluble radical and has characteristic socle $\text{Inn}(T)$.*

Proof. An automorphism σ centralizing every inner automorphism satisfies $c_{\sigma(t)} = c_t$ for every $t \in T$. Since $Z(T) = 1$, this implies $\sigma(t) = t$ for every t , so $C_{\text{Aut}(T)}(\text{Inn}(T)) = 1$. If $K \trianglelefteq H$, then

$$[K, \text{Inn}(T)] \leq K \cap \text{Inn}(T).$$

Simplicity makes the intersection either trivial or all of $\text{Inn}(T)$. In the first case the centralizer calculation gives $K = 1$. This proves the assertions. $\square$

Lemma 9.2. *A nontrivial fenced interval cannot be an interval in a finite soluble group. Consequently, if $\text{Inn}(T) \leq B \leq H \leq \text{Aut}(T)$, the interval $[B, H]$ is not nontrivially fenced.*

Proof. Lemma 2.3 would give a nonabelian simple composition factor of the soluble top group. For the second assertion, quotient by $\text{Inn}(T)$, which is contained in the bottom B . The resulting interval lies in the soluble group $\text{Out}(T)$, by Schreier's Theorem. $\square$

9.1 Minimality in both orientations

Let $\mathcal{L}$ be the lattice in Proposition 8.1. Suppose that one of $\mathcal{L}$ and $\mathcal{L}^{\text{op}}$ is a finite group interval. Among all representations of either orientation choose

$$[D, G] \cong \mathcal{M}, \quad \mathcal{M} \in \{\mathcal{L}, \mathcal{L}^{\text{op}}\},$$

with $|G|$ minimal. Minimizing over both orientations is the same kind of reduction used in the subgroup-interval and signalizer setting of [2, Theorem 3 and Section 6]; see also the comparison in [28, Section 4]. We give the needed complementary-socle argument in full. Quotienting by $\text{core}_G(D)$ preserves the interval; hence this core is trivial. The global fence and Lemma 2.3 give

$$S = \text{soc}(G) = \prod_{i \in I} T_i \cong T^m, \quad G = DS, \quad (9.1)$$

where T is nonabelian simple and D acts transitively on the factor set I . Proposition 8.1 applies to the chosen orientation $\mathcal{M}$.

Proposition 9.3. *In this minimal representation, $D \cap S = 1$.*

Proof. Put $E = D \cap S$. For every $Y \in [D, G]$, the identity $G = DS$ gives

$$Y = D(Y \cap S).$$

Thus $Y \mapsto Y \cap S$ is an isomorphism from $[D, G]$ to the interval of D -invariant subgroups between E and S ; its inverse is $K \mapsto DK$.

First suppose that E is subdirect in S . By the subdirect-power description in Lemma 2.5, its simple factors are full diagonal strips on a partition Π of I . Every subgroup between E and S is subdirect. Its strips split the blocks of Π , and their diagonal identifications are forced by their containment of E . Therefore the preceding interval is anti-isomorphic to the lattice of D -invariant equivalence relations refining Π .

Fix $i_0 \in I$, let $A = D_{i_0}$, and let B be the stabilizer in D of the block of Π containing i_0 . The block correspondence for a transitive action identifies these equivalence relations with $[A, B]$: the block containing i_0 associated with $U \in [A, B]$ is $U i_0$, and all its other blocks are its D -translates. Conversely, the setwise stabilizer of the block containing i_0 recovers U . It follows that $\mathcal{M}^{\text{op}} \cong [A, B]$. But $|B| \leq |D| < |G|$, contrary to the choice of G over both orientations.

The coordinate projections of E have equal order, since D is transitive on I . If they are proper and nontrivial, Proposition 2.7, applied with the possibly unlabelled lower endpoint D , and the global fence imply that E is the product of those projections. Choose a proper original Boolean vertex Y of the decorated lattice. Its forward label equals the top label, so $Y \cap S$ is subdirect by Proposition 2.6. It is proper, because $Y < G$.

A product of nontrivial coordinate subgroups cannot lie in a proper subdirect subgroup: the latter has a diagonal link between two coordinates, whereas the former contains an element that is nontrivial in one of those coordinates and trivial in the other. This contradicts $E \leq Y \cap S$. The only remaining possibility is that all coordinate projections of E are trivial, and then $E = 1$. $\square$

9.2 The extension dictionary

We give the parametrization with its group actions and order relations explicit. It is the induced-coordinate form of [28, Proposition 2.2 and Lemma 2.3].

Proposition 9.4. *Let a finite group D act on $S = \prod_{i \in I} T_i$, where all T_i are isomorphic to a nonabelian simple group T , and suppose that its action on I is transitive. Fix $i_0 \in I$, set $A = D_{i_0}$, identify T_{i_0} with T , and let $\alpha : A \rightarrow \text{Aut}(T)$ be the induced action.*

The D -invariant subdirect subgroups of S are in bijection with pairs (U, β) , where

$$A \leq U \leq D, \quad \beta : U \longrightarrow \text{Aut}(T), \quad \beta|_A = \alpha.$$

Write $S(U, \beta)$ for the corresponding subgroup. Then

$$S(U, \beta) \leq S(V, \gamma) \iff V \leq U \text{ and } \beta|_V = \gamma. \quad (9.2)$$

In particular, the interval of D -invariant subgroups from $S(U, \beta)$ to S is anti-isomorphic to $[A, U]$, by restricting β .

Proof. Use left conjugation for the given action and let $\pi_0 : S \rightarrow T$ be the chosen coordinate projection. The map

$$s \longmapsto f_s, \quad f_s(d) = \pi_0(d \cdot s),$$

identifies S with

$$\mathcal{F}_\alpha = \{f : D \rightarrow T : f(ad) = \alpha(a)(f(d)) \text{ for } a \in A, d \in D\}.$$

Indeed, values on representatives of $A \backslash D$ are independent coordinates of S . Under this identification the action is $(g \cdot f)(d) = f(dg)$.

An extension β defines

$$\mathcal{F}_\beta = \{f : D \rightarrow T : f(ud) = \beta(u)(f(d)) \text{ for } u \in U, d \in D\}.$$

This is a D -invariant subgroup of $\mathcal{F}_\alpha$, with independent values on the cosets in $U \backslash D$. Consequently $\mathcal{F}_\beta \cong T^{[D:U]}$, and it projects onto every coordinate of $\mathcal{F}_\alpha$. Set $S(U, \beta) = \mathcal{F}_\beta$ in these coordinates.

Conversely, let $K \leq \mathcal{F}_\alpha$ be a D -invariant subdirect subgroup. Its diagonal-strip partition is invariant under the transitive action on $A \backslash D$. The block containing A has stabilizer U for some $A \leq U \leq D$. Projection at 1 identifies the diagonal factor on that block with T . The induced action of U on this factor gives a homomorphism $\beta : U \rightarrow \text{Aut}(T)$ extending α . For $f \in K$ and $u \in U$ one has $f(u) = \beta(u)(f(1))$. Applying this identity to $d \cdot f$ gives

$$f(ud) = \beta(u)(f(d)).$$

Thus $K \leq \mathcal{F}_\beta$. Both groups have one freely varying diagonal factor for each coset of U , so they are equal.

Containment of two subdirect groups reverses refinement of their strip partitions. Hence $S(U, \beta) \leq S(V, \gamma)$ forces $V \leq U$. For $v \in V$, varying $f(1)$ over T in the two defining equations then gives $\beta(v) = \gamma(v)$. The converse follows immediately from those equations. Finally, every subgroup containing a subdirect subgroup is subdirect, so all members of the stated interval have already been included. $\square$

Apply Proposition 9.4 to (9.1). By Proposition 9.3, the full interval $[D, G]$ is the lattice of D -invariant subgroups of S , via $Y \mapsto Y \cap S$. Whenever $Y \cap S$ is subdirect, write (U_Y, β_Y) for its extension. In particular,

$$[Y, G]^{\text{op}} \cong [A, U_Y]. \quad (9.3)$$

For nested filters these are compatible actual group intervals: the larger domain's homomorphism restricts to the smaller domain's homomorphism. No representation of the entire dual lattice is asserted by (9.3).

9.3 The detector and the common image

Proposition 9.5. *In the minimal representation fixed above, the induced factor-stabilizer action α satisfies $\alpha(A) \geq \text{Inn}(T)$.*

Proof. Let X be the detector vertex of Proposition 8.1. Its forward type equals T , so $E_X = X \cap S$ is subdirect in S . The action of D on the simple factors of E_X is transitive. Its factor-stabilizer action is $\beta_X : U_X \rightarrow \text{Aut}(T)$ by the construction in Proposition 9.4.

Suppose that this action preserved a subgroup $1 < P < T$. Transporting P among the factors of E_X would give a D -invariant product Q of proper nontrivial coordinate subgroups of E_X . Thus DQ is strictly between D and X . Choose a coatom Y of $[D, X]$ containing DQ . The detector's coatom property gives a designated vertex $d \leq Y$ whose intersection $d \cap S$ is subdirect in the global coordinates. Hence $Y \cap S$ projects onto every simple factor of E_X : projection onto such a factor can be identified with projection onto any one global coordinate in its diagonal support. Thus $Y \cap S$ is a proper subdirect subgroup of E_X containing the product Q , which is impossible by the diagonal-link argument in Proposition 9.3.

The automorphism lemma recalled by Pálffy [28, Lemma 2.11] states that a subgroup of $\text{Aut}(T)$ not containing $\text{Inn}(T)$ preserves some proper nontrivial subgroup of T . It follows that

$$\beta_X(U_X) \geq \text{Inn}(T).$$

The reverse detector interval is the actual interval $[A, U_X]$ in (9.3), and it is fenced. Set $K = \ker \beta_X$. Lemma 2.2 gives either $AK = U_X$ or $K \leq A$. The first possibility gives $\beta_X(U_X) = \beta_X(A) = \alpha(A)$, as required.

In the second possibility $K \leq C_X = \text{core}_{U_X}(A)$, and $[A, U_X]$ is represented as

$$[\alpha(A), \beta_X(U_X)].$$

Write $H = \beta_X(U_X)$ and $B = \alpha(A)$. If $\text{core}_H(B)$ is nontrivial, it contains $\text{Inn}(T)$ by Lemma 9.1. Quotienting by that core leaves a soluble top, contradicting Lemma 9.2. Otherwise $C_X = K$, and $U_X/C_X \cong H$ has socle T with multiplicity one. The reverse detector supplies a strictly smaller labelled domain in $[A, U_X]$ whose simple type is equal to that of U_X/C_X . The equality is supplied by the reverse tests in Proposition 8.1, applied through (9.3). Proposition 2.6 would make its positive socle multiplicity strictly smaller than one. This final contradiction excludes $K \leq A$. $\square$

Lemma 9.6. *With α as in Proposition 9.5, let $A < U \leq D$ and let $\beta : U \rightarrow \text{Aut}(T)$ extend α . Suppose that $[A, U]$ is fenced. Then*

$$U = A \ker \beta, \quad \beta(U) = \alpha(A).$$

Proof. The normal-subgroup test gives the asserted supplementation unless $\ker \beta \leq A$. In the latter case $[A, U]$ is isomorphic to $[\alpha(A), \beta(U)]$. Proposition 9.5 puts the inner group in its bottom, contrary to Lemma 9.2. Supplementation gives the equality of images. $\square$

Proposition 9.7 (The Boolean extension configuration). *If either orientation of the decorated lattice has a finite-group interval representation, there are finite groups $A \leq D$, finite nonabelian simple groups T, R , a homomorphism $\alpha : A \rightarrow \text{Aut}(T)$, and extensions*

$$\beta_J : U_J \rightarrow \text{Aut}(T) \quad (\emptyset \neq J \subsetneq \{1, \dots, 4\})$$

with the following properties. Set $U_\emptyset = A$ and $\beta_\emptyset = \alpha$.

(E1) *For nonempty proper J , $[A, U_J]$ is fenced. If $C_J = \text{core}_{U_J}(A)$, the group U_J/C_J has unique minimal normal subgroup*

$$N_J/C_J \cong R^{I_J}, \quad U_J = AN_J.$$

(E2) *If $J \subsetneq K \subsetneq \{1, \dots, 4\}$, then $U_J < U_K$ and $\beta_K|_{U_J} = \beta_J$. For nonempty J , the equal-type transport of Proposition 2.6 applies: in particular*

$$\frac{U_J \cap N_K}{C_K} \cong \frac{N_J}{C_J}, \quad N_J = (U_J \cap N_K)C_J,$$

and the canonical surjections $I_K \rightarrow I_J$ are U_J -equivariant and compose. Whenever a nonempty family of proper subsets has proper union K , its domains generate U_K .

(E3) *All images $\beta_J(U_J)$ equal $H = \alpha(A) \geq \text{Inn}(T)$.*

(E4) *If a family of proper nonempty subsets has union $\{1, \dots, 4\}$, its homomorphisms have no common extension to the subgroup of D generated by their domains.*

Proof. Use the minimal representation already chosen. Index the original Boolean component in its reverse order by subsets J of $\{1, \dots, 4\}$. Let Y_J be the corresponding forward group, so $Y_\emptyset = G$ and $Y_{\{1, \dots, 4\}} = D$. Every proper nontrivial Y_J has the top forward type and hence a subdirect intersection with S . Proposition 9.4 provides (U_J, β_J) .

The reverse labels and tests in Proposition 8.1 apply in the actual filters (9.3). They give the fences and equal simple types on comparable proper Boolean nodes. All the nonempty proper nodes therefore have one common simple type R : connect two atoms through their proper two-atom join, and connect any other node to an atom below it. Lemma 2.3 and Proposition 2.6 prove (E1) and the transport assertions in (E2). The order relation and restrictions follow from (9.2).

For the assertion about a proper join, let K be the union of a nonempty family of index sets, and put $V = \langle U_J : J \text{ in the family} \rangle$. Then $A \leq V \leq U_K$, and $\beta_K|_V$ is an extension of all the β_J . Its subdirect group is contained in every $Y_J \cap S$ and contains $Y_K \cap S$. Original Boolean meets are preserved in the decorated lattice, so the intersection of those Y_J is exactly Y_K . The subdirect groups are equal; the dictionary gives $V = U_K$.

Proposition 9.5 and Lemma 9.6 give (E3). Finally, a common extension for a family with full union would give a nontrivial subdirect subgroup of S contained in all the $Y_J \cap S$. Their intersection is $D \cap S = 1$, because their reverse Boolean join is the top. This proves (E4). $\square$

10 Boolean gluing of extensions

We prove that the configuration in Proposition 9.7 cannot exist. Throughout this section, T is the target simple group of the homomorphisms, whereas R is the simple type of the domain socles. We retain the notation $A, D, H, U_J, \beta_J, C_J, N_J, I_J$ of that proposition. All extension domains lie in the finite group D .

The extensions will glue if their domain cores have images containing $\text{Inn}(T)$. We first prove this criterion: a least normal subgroup of A with that image is normalized by every domain, and conjugation on its simple quotient supplies the common extension. We then force the criterion on the Boolean faces. For an atom x , the normal image $\alpha(C_x)$ is either trivial or contains the inner group. In the trivial case, compatible lifts from the three rank-two faces through x produce a different common extension, again forbidden by (E4). Hence every atom core retains the inner group; intersections pass this property to the larger proper faces.

10.1 A criterion for extending over generated domains

We begin with the elementary fact that makes least subgroups with a prescribed perfect image available.

Lemma 10.1 (Intersections retaining a perfect image). *Let $f : M \rightarrow H_0$ be a homomorphism of finite groups, let $P \leq H_0$ be perfect, and let $B_1, \dots, B_s \trianglelefteq M$, with $s \geq 1$, satisfy $P \leq f(B_i)$ for every i . Then*

$$P \leq f\left(\bigcap_{i=1}^s B_i\right).$$

Proof. For two normal subgroups,

$$f(B_1 \cap B_2) \supseteq f([B_1, B_2]) = [f(B_1), f(B_2)] \supseteq [P, P] = P.$$

Induction proves the assertion for the finite family. □

The following criterion applies to any finite family of extension domains. It will be used at the end with two rank-three Boolean faces whose union is full.

Lemma 10.2 (A common core and its conjugation extension). *Let $A \leq U_i \leq D$ be a finite nonempty family of subgroups of a finite group. Let T be nonabelian simple and suppose that homomorphisms*

$$\alpha : A \rightarrow \text{Aut}(T), \quad \beta_i : U_i \rightarrow \text{Aut}(T)$$

satisfy

$$\beta_i|_A = \alpha, \quad \alpha(\text{core}_{U_i}(A)) \geq \text{Inn}(T).$$

Put $K = \langle U_i : i \rangle$. Then $\alpha(\text{core}_K(A)) \geq \text{Inn}(T)$, and there is a unique homomorphism $\beta : K \rightarrow \text{Aut}(T)$ extending all the β_i .

Proof. Write $J = \text{Inn}(T)$ and let $\iota : T \rightarrow J$ be the isomorphism $t \mapsto c_t$. Consider

$$Q = \bigcap \{B \trianglelefteq A : \alpha(B) \geq J\}.$$

The family is nonempty. Since J is perfect, Lemma 10.1 shows that its finite intersection still maps over J . The subgroup $\alpha^{-1}(J)$ belongs to the family, so in fact

$$\alpha(Q) = J. \tag{10.1}$$

Thus Q is the least normal subgroup of A with image containing J .

Fix i and put $C_i = \text{core}_{U_i}(A)$. The hypothesis gives $Q \leq C_i$. Make the analogous construction in C_i :

$$Q_i = \bigcap \{B \trianglelefteq C_i : \alpha(B) \geq J\}.$$

The same intersection argument, using $C_i \cap \alpha^{-1}(J)$, gives $\alpha(Q_i) = J$. For $u \in U_i$ and $c \in C_i$,

$$\alpha(ucu^{-1}) = \beta_i(u)\alpha(c)\beta_i(u)^{-1}.$$

Since $C_i \trianglelefteq U_i$ and $J \trianglelefteq \text{Aut}(T)$, conjugation by U_i permutes the defining family for Q_i . Thus $Q_i \trianglelefteq U_i$, and in particular $Q_i \trianglelefteq A$. Minimality of Q gives $Q \leq Q_i$. Conversely, $Q \leq C_i$, $Q \trianglelefteq C_i$, and (10.1) give $Q_i \leq Q$. Consequently $Q_i = Q$ for every i .

It follows that $Q \trianglelefteq K$ and $Q \leq \text{core}_K(A)$, proving the common-core assertion. Set $L = Q \cap \ker \alpha$. Since $L = Q \cap \ker \beta_i$ for every i , the subgroup L is normalized by each U_i and therefore by K . The map

$$\vartheta : Q/L \longrightarrow T, \quad \vartheta(qL) = \iota^{-1}(\alpha(q))$$

is an isomorphism. Transporting the conjugation action of K on Q/L through ϑ defines a homomorphism $\beta : K \rightarrow \text{Aut}(T)$. For $u \in U_i$ and $q \in Q$,

$$\alpha(uqu^{-1}) = \beta_i(u)\alpha(q)\beta_i(u)^{-1} = \iota(\beta_i(u)(\vartheta(qL))).$$

As ϑ is onto, this proves $\beta|_{U_i} = \beta_i$. These restrictions determine β uniquely because the U_i generate K . $\square$

10.2 An atom whose core is killed

For $B \subseteq I_J$, let

$$N_J(B) = q_J^{-1} \left(\prod_{i \in B} S_{J,i} \right), \quad q_J : N_J \longrightarrow S_J = N_J/C_J, \quad (10.2)$$

where $S_{J,i}$ are the simple direct factors of S_J . Thus $N_J(\emptyset) = C_J$. For comparable nonempty index sets $J \subset K$, write $p_{KJ} : I_K \rightarrow I_J$ for the canonical factor map.

Lemma 10.3. *If x is an original Boolean atom and $\alpha(C_x) = 1$, then β_x factors through the permutation action of U_x on I_x .*

Proof. By (E3), $U_x = A \ker \beta_x$. Since $A < U_x$ and $C_x \leq A$, the kernel is not contained in C_x . The assumption makes β_x a homomorphism on U_x/C_x , so its nontrivial kernel there contains the unique minimal normal subgroup N_x/C_x . Consequently $\beta_x(N_x) = 1$.

Let P be the kernel of the action of U_x on I_x . Modulo C_x , the faithful conjugation action on S_x embeds P/N_x into $\text{Out}(R)^{I_x}$. To justify faithfulness, the centralizer of S_x in U_x/C_x is normal and disjoint from S_x ; uniqueness of the nonabelian minimal normal subgroup makes it trivial. Schreier's Theorem therefore makes P/N_x soluble. Its image under β_x is a soluble normal subgroup of $H = \alpha(A)$. Lemma 9.1 makes that image trivial, so $P \leq \ker \beta_x$, as claimed. $\square$

Fix an atom x . To rule out $\alpha(C_x) = 1$, we shall show that this assumption produces a subgroup L normalized by A , generated by subgroups on which the local extensions are trivial. The remaining condition for the formula $\theta(a\ell) = \alpha(a)$ to define an extension on AL is $A \cap L \leq \ker \alpha$. By Lemma 10.3, it is enough to prove that every element of $A \cap L$ fixes the factor set I_x . The three proper faces through x provide the local generators; their proper pairwise joins will let us compare different factor labels.

Set

$$\mathcal{U}_x = \{x \cup \{a\} : a \in \{1, \dots, 4\} \setminus x\}.$$

This family has three members. Distinct $u, v \in \mathcal{U}_x$ have a proper join $w = u \cup v$ of rank three. For $j \in I_x$ and $y \supseteq x$, write

$$F_{y,j} = p_{yx}^{-1}(j), \quad F_{x,j} = \{j\}.$$

All factor maps in this notation are defined on proper nodes. Figure 2 displays these three faces and their proper pairwise joins.

A lift of the fibre $F_{u,j}$ must lie over the corresponding support $F_{u \cup v,j}$ in each proper pair join $u \cup v$. We therefore intersect its support preimages in all those joins. The perfect-image intersection lemma ensures that these simultaneous requirements do not lose any factor of the desired fibre.

Lemma 10.4 (Compatible lifts of the fibres). *For $u \in \mathcal{U}_x$ and $j \in I_x$, define*

$$L_{u,j} = N_u(F_{u,j}) \cap \bigcap_{\substack{v \in \mathcal{U}_x \\ v \neq u}} (U_u \cap N_{u \cup v}(F_{u \cup v,j})). \quad (10.3)$$

These subgroups have the following properties.

- (i) *They are normal in N_u , and $q_u(L_{u,j}) = \prod_{i \in F_{u,j}} S_{u,i}$.*
- (ii) *Conjugation by U_x permutes the choices according to its action on I_x . In particular, C_x normalizes every $L_{u,j}$.*
- (iii) *$U_u = \langle A, L_{u,j} : j \in I_x \rangle$.*
- (iv) *For $j \neq k$, one has*

$$[L_{u,j}, L_{v,k}] \leq \begin{cases} C_{u \cup v}, & u \neq v, \\ C_u, & u = v. \end{cases}$$

Both right-hand sides are contained in C_x .

- (v) *If $\alpha(C_x) = 1$, then $\beta_u(L_{u,j}) = 1$ for all u, j .*

Proof. Fix a proper pair join $w = u \cup v$. Equal-type transport gives

$$N_u = (U_u \cap N_w)C_u. \quad (10.4)$$

The first factor on the right acts by inner automorphisms on N_w/C_w , hence preserves every coordinate subproduct there. The second factor satisfies $C_u \leq C_x \leq U_x$. Since C_x acts trivially on I_x , equivariance of p_{wx} says that C_u fixes the set $F_{w,j}$. It too normalizes $N_w(F_{w,j})$. Thus $U_u \cap N_w(F_{w,j})$ is normal in N_u and lies in N_u .

In N_w/C_w , the subdirect group $(U_u \cap N_w)/C_w$ has its diagonal strips indexed by I_u . Their supports are the fibres of p_{wu} . Because $p_{wx} = p_{ux}p_{wu}$, intersecting with the $F_{w,j}$ -support product selects exactly the strips indexed by $F_{u,j}$. The isomorphism from this subdirect group to N_u/C_u therefore shows that $U_u \cap N_w(F_{w,j})$ maps onto $\prod_{i \in F_{u,j}} S_{u,i}$ under q_u . The first group in (10.3) is normal in N_u and has the same image. This image is perfect. Lemma 10.1 proves (i).

Every domain occurring in (10.3) contains U_x , and all the factor maps are U_x -equivariant. Its conjugation therefore permutes the whole definition with j , proving (ii). The images in (i) generate N_u/C_u , and $C_u \leq A$; since $U_u = AN_u$, this proves (iii).

For $u \neq v$, the two lifts in (iv) lie in disjoint coordinate support products of $N_{u \cup v}/C_{u \cup v}$. They commute in that quotient. For $u = v$ the identical argument takes place in N_u/C_u . Core inclusion along $U_x \leq U_u \leq U_{u \cup v}$ gives the asserted containment in C_x .

Finally suppose that $\alpha(C_x) = 1$. Then $\beta_u(C_u) = 1$. The kernel of β_u supplements A and is not contained in C_u , so it contains N_u by the unique-minimal-normal-subgroup argument in Lemma 10.3. This proves (v). $\square$

We next construct subgroups B_k which distinguish the labels $k \in I_x$ and are normalized by every lift with a different label. Different labels commute only modulo C_x , and C_x need not be normal in the group generated by all the lifts. The following finite core argument keeps a full simple image without requiring that normality.

Lemma 10.5 (Taking cores without losing a simple image). *Let $R_0 \leq D$, let $C \trianglelefteq R_0$, and suppose that R_0/C is nonabelian simple. Let $L_1, \dots, L_s \leq D$ be normalized by R_0 . Suppose that, for every t , there is a subgroup $D_t \trianglelefteq R_0$ such that*

$$D_t C = R_0, \quad D_t^{L_t} = D_t, \quad [D_t, L_t] \leq D_t \cap C.$$

Then, for $W = \langle R_0, L_1, \dots, L_s \rangle$, the subgroup $\text{core}_W(R_0)$ maps onto R_0/C . No normalization of C by L_t is assumed.

Proof. Let $\pi : R_0 \rightarrow R_0/C$ be the quotient map. Start with $B = R_0$. At every stage B will be normal in R_0 and satisfy $\pi(B) = R_0/C$. For a fixed t , replace B by

$$F_t(B) = \bigcap_{\ell \in L_t} (B \cap D_t)^\ell, \quad K^\ell = \ell^{-1} K \ell. \quad (10.5)$$

The subgroup $B \cap D_t$ still maps onto R_0/C by Lemma 10.1. Put $K = B \cap D_t$. The subgroup K is normal in D_t , because $B \trianglelefteq R_0$ and $D_t \leq R_0$. Since L_t normalizes D_t , every conjugate K^ℓ lies in D_t and is normal in D_t . For $d \in D_t$ and $\ell \in L_t$, the assumption $[D_t, L_t] \leq D_t \cap C$ gives $\pi(d^\ell) = \pi(d)$. Consequently $\pi(K^\ell) = \pi(K) = R_0/C$ for every ℓ . This applies the quotient map only to elements of $D_t \leq R_0$; it does not require C to be normalized by L_t . Applying Lemma 10.1 inside D_t shows that $F_t(B)$ has full image.

The new subgroup is normal in R_0 . Indeed, R_0 normalizes B , D_t , and L_t . For $r \in R_0$ one therefore has $(K^\ell)^r = (K^r)^{\ell^r} = K^{\ell^r}$, and $\ell^r \in L_t$. Thus R_0 permutes precisely the conjugates intersected in (10.5). The operation is descending, since the identity conjugate occurs. Apply $F_1, \dots, F_s$ in repeated rounds. Finiteness forces stabilization, and at a stable round every intermediate inclusion is equality. The resulting subgroup B_* is therefore normalized by each L_t as well as by R_0 . Thus $B_* \trianglelefteq W$ and $B_* \leq R_0$, while $\pi(B_*) = R_0/C$. Since $B_* \leq \text{core}_W(R_0)$, the asserted core also has full image. $\square$

Lemma 10.6 (Subgroups detecting the factor labels). *Put $L_j = \langle L_{u,j} : u \in \mathcal{U}_x \rangle$ and $R_k = N_x(\{k\})$. There are subgroups $B_k \leq R_k$, for $k \in I_x$, such that*

- (i) $q_x(B_k) = S_{x,k}$;
- (ii) every L_j with $j \neq k$ normalizes B_k ;
- (iii) C_x normalizes every B_k , and A permutes the B_k in the same way as it permutes I_x .

In particular, the subgroups B_k are pairwise distinct.

Proof. For $u \in \mathcal{U}_x$ set

$$D_{u,k} = U_x \cap N_u(F_{u,k}).$$

The transported subdirect group $(U_x \cap N_u)/C_u$ maps isomorphically onto N_x/C_x . Selecting its strips over k shows that

$$D_{u,k} \leq R_k, \quad D_{u,k} C_x = R_k, \quad D_{u,k} \cap C_x = C_u. \quad (10.6)$$

It is normal in R_k : all elements of $R_k \leq N_x$ fix the factor labels I_x , and the support-preimage construction is equivariant.

For $j \neq k$, both $D_{u,k}$ and $L_{u,j}$ lie in N_u and have disjoint supports modulo C_u . Thus

$$[D_{u,k}, L_{u,j}] \leq C_u.$$

Since $C_u \leq D_{u,k}$, this also proves that $L_{u,j}$ normalizes $D_{u,k}$. Moreover R_k normalizes $L_{u,j}$ by Lemma 10.4(ii), since R_k fixes every label of I_x . We can therefore apply Lemma 10.5 to R_k , its normal subgroup C_x , and all the lifts $L_{u,j}$ with $j \neq k$, using the subgroup $D_{u,k}$ for the lift labelled by (u, j) .

Define canonically

$$W_k = \langle R_k, L_j : j \neq k \rangle, \quad B_k = \text{core}_{W_k}(R_k).$$

That lemma proves (i); the definition proves (ii). Since $C_x \leq R_k \leq W_k$, it normalizes B_k . Conjugation by A permutes the pairs (W_k, R_k) and hence their cores, proving (iii). Their images in N_x/C_x are the distinct nontrivial coordinate factors, so the B_k are distinct. $\square$

To use the subgroups B_k , we must collect all letters with one chosen label at the end of a word in the lifts. The errors lie in C_x , which normalizes each individual lift. The following word calculation records exactly the form needed.

Lemma 10.7 (Collection with labelled factors). *Let $C \leq D$ normalize subgroups $P_\lambda \leq D$ indexed by a finite set. Assign each index a label $c(\lambda) \in I$. Suppose that*

$$[P_\lambda, P_\mu] \leq C \quad \text{whenever } c(\lambda) \neq c(\mu).$$

Put $L_j = \langle P_\lambda : c(\lambda) = j \rangle$ and $L = \langle L_j : j \in I \rangle$. For every $k \in I$, every element $g \in L$ can be written

$$g = pqc, \quad p \in \langle L_j : j \neq k \rangle, \quad q \in L_k, \quad c \in C.$$

Proof. Write g as a word whose letters belong to the individual P_λ ; inverses are allowed because these are subgroups. Interchange adjacent letters with different labels to move the k -labelled letters to the end. Each interchange introduces a commutator in C . Move that error to the right through the remaining word, using

$$cl = (clc^{-1})c.$$

The conjugated letter remains in its original P_λ , so its label does not change. Each interchange reduces the number of pairs consisting of a k -letter preceding a non- k -letter. The process terminates with the asserted expression. This is a word calculation; it does not form a quotient by C or require $C \leq L$. $\square$

Proposition 10.8. *For every original Boolean atom x , $\alpha(C_x) \geq \text{Inn}(T)$.*

Proof. The subgroup $\alpha(C_x)$ is normal in $\beta_x(U_x) = H$. Lemma 9.1 reduces the assertion to excluding $\alpha(C_x) = 1$. Assume that equality and use the lifts and detecting subgroups just constructed. Put

$$L = \langle L_j : j \in I_x \rangle.$$

The group A normalizes L .

Let $a \in A \cap L$. We show that a fixes every member of I_x . For subgroup conjugation use the right-action notation $K^g = g^{-1}Kg$. If a sent k to $h \neq k$, equivariance would give $B_k^a = B_h$. Lemmas 10.4 and 10.7, with $C = C_x$ and the individual $L_{u,j}$ as labelled subgroups, give

$$a = pqc, \quad p \in \langle L_j : j \neq k \rangle, \quad q \in L_k, \quad c \in C_x.$$

The element p normalizes B_k , q normalizes B_h , and c normalizes both. It follows successively that

$$B_k^{qc} = B_h, \quad B_k^q = B_h, \quad B_k = B_h,$$

contradicting Lemma 10.6. Thus a fixes the factor set. Lemma 10.3 now gives $\alpha(a) = 1$.

Consequently $A \cap L \leq \ker \alpha$. Because A normalizes L , the formula

$$\theta : AL \longrightarrow \text{Aut}(T), \quad \theta(al) = \alpha(a)$$

defines a homomorphism. It is well defined by the intersection inclusion; multiplication is respected since $L \trianglelefteq AL$. For each $u \in \mathcal{U}_x$, Lemma 10.4(iii),(v) says that U_u is generated by A and lifts killed by both θ and β_u . Hence $\theta|_{U_u} = \beta_u$.

The three sets $u \in \mathcal{U}_x$ have full union $\{1, \dots, 4\}$. Their generated domain is AL , so this common extension contradicts (E4). The assumption $\alpha(C_x) = 1$ is impossible. $\square$

10.3 Compatible joins and the contradiction

Lemma 10.9 (Cores under a compatible join). *For every nonempty proper original Boolean node z ,*

$$C_z = \bigcap_{x \subseteq z, |x|=1} C_x.$$

Consequently $\alpha(C_z) \geq \text{Inn}(T)$.

Proof. The assertion is immediate for an atom, so suppose that $|z| \geq 2$. Every C_x in the intersection contains C_z . Pass to U_z/C_z and use bars for the images of A, U_x, C_x . Put

$$N = N_z/C_z, \quad E_x = (U_x \cap N_z)/C_z.$$

The transport isomorphism in (E2) gives $E_x \cap \overline{C}_x = 1$. Both are normal in $\overline{U}_x$, so $[E_x, \overline{C}_x] = 1$.

Since $U_z = AN_z$ and $A \leq U_x$, one has $\overline{U}_x = \overline{A}E_x$. Let $M = \langle E_x : x \subseteq z, |x| = 1 \rangle$. Each E_x is normalized by $\overline{A}$. The compatible-join generation in (E2) therefore gives

$$\overline{U}_z = \overline{A}M,$$

so $M \trianglelefteq \overline{U}_z$. It is a nontrivial subgroup of the minimal normal subgroup N , and hence $M = N$. Thus $\bigcap_x \overline{C}_x$ centralizes N . The centralizer is normal in $\overline{U}_z$ and intersects N in $Z(N) = 1$. If it were nontrivial it would contain the unique minimal normal subgroup N , a contradiction. The intersection is therefore trivial, which proves the equality of cores.

Each C_x is normal in A and maps over $\text{Inn}(T)$ by Proposition 10.8. Apply Lemma 10.1 with $P = \text{Inn}(T)$ to obtain the image assertion. $\square$

Theorem 10.10. *The Boolean extension configuration of Proposition 9.7 does not exist.*

Proof. Take the two proper Boolean nodes

$$z = \{1, 2, 3\}, \quad w = \{2, 3, 4\}.$$

Lemma 10.9 gives $\alpha(C_z), \alpha(C_w) \geq \text{Inn}(T)$. Lemma 10.2 therefore extends β_z and β_w to $\langle U_z, U_w \rangle$. Since $z \cup w = \{1, \dots, 4\}$, this contradicts (E4). $\square$

Proof of Theorem 1.2. Choose the finite decorated lattice of Proposition 8.1. If either it or its dual had a finite-group interval representation, Proposition 9.7 would produce the configuration excluded by Theorem 10.10. Hence this lattice has no such representation, proving Theorem 1.2. $\square$

Theorem 1.1 now follows from Theorem 1.2 and the global Pálffy–Pudlák equivalence [29, Theorem 2].

References

- [1] Michael Aschbacher. On the maximal subgroups of the finite classical groups. *Inventiones Mathematicae*, 76(3):469–514, 1984. [DOI](#).
- [2] Michael Aschbacher. On intervals in subgroup lattices of finite groups. *Journal of the American Mathematical Society*, 21(3):809–830, 2008. [DOI](#).
- [3] Robert Baddeley and Andrea Lucchini. On representing finite lattices as intervals in subgroup lattices of finite groups. *Journal of Algebra*, 196(1):1–100, 1997. [DOI](#).
- [4] Robert W. Baddeley. A new approach to the finite lattice representation problem. *Periodica Mathematica Hungarica*, 36(1):17–59, 1998. [DOI](#).
- [5] Alberto Basile. *Second Maximal Subgroups of the Finite Alternating and Symmetric Groups*. PhD thesis, Australian National University, Canberra, April 2001. Posted in 2008 as [arXiv:0810.3721v1](#).
- [6] Christopher P. Bendel, Daniel K. Nakano, Brian J. Parshall, Cornelius Pillen, Leonard L. Scott, and David I. Stewart. Bounding cohomology for finite groups and Frobenius kernels. *Algebras and Representation Theory*, 18(3):739–760, 2015. [DOI](#). Preprint locators refer to [arXiv:1208.6333v1](#), titled *Bounding extensions for finite groups and Frobenius kernels*.
- [7] Christopher P. Bendel, Daniel K. Nakano, and Cornelius Pillen. Extensions for Frobenius kernels. *Journal of Algebra*, 272(2):476–511, 2004. [DOI](#).
- [8] Ferdinand B rner. A remark on the finite lattice representation problem. In *Contributions to General Algebra*, volume 11, pages 5–38. Verlag Johannes Heyn, Klagenfurt, 1999.
- [9] John N. Bray, Derek F. Holt, and Colva M. Roney-Dougal. *The Maximal Subgroups of the Low-Dimensional Finite Classical Groups*, volume 407 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 2013. [DOI](#); [book PDF](#).
- [10] Timothy C. Burness, Martin W. Liebeck, and Aner Shalev. Generation of second maximal subgroups and the existence of special primes. *Forum of Mathematics, Sigma*, 5:e25, 2017. 41 pp. [DOI](#).
- [11] Timothy C. Burness and Donna M. Testerman. Irreducible subgroups of simple algebraic groups—a survey. In C. M. Campbell, C. W. Parker, M. R. Quick, E. F. Robertson, and C. M. Roney-Dougal, editors, *Groups St Andrews 2017 in Birmingham*, volume 455 of *London Mathematical Society Lecture Note Series*, pages 230–260. Cambridge University Press, Cambridge, 2019. [DOI](#). Locators refer to the corrected [arXiv:1803.01174v2](#) (2018).
- [12] Mika l Cavallin and Donna M. Testerman. A new family of irreducible subgroups of the orthogonal algebraic groups. *Transactions of the American Mathematical Society, Series B*, 6:45–79, 2019. [DOI](#); [published article PDF](#).
- [13] Claude Chevalley. D monstration d’une hypoth se de M. Artin. *Abhandlungen aus dem Mathematischen Seminar der Universit t Hamburg*, 11:73–75, 1935. [DOI](#).
- [14] Edward Cline, Brian Parshall, Leonard Scott, and Wilberd van der Kallen. Rational and generic cohomology. *Inventiones Mathematicae*, 39:143–163, 1977. [DOI](#).

- [15] Brian Conrad. Reductive group schemes. In *Autour des schémas en groupes, Vol. I*, volume 42–43 of *Panoramas et Synthèses*, pages 93–444. Société Mathématique de France, Paris, 2014. Page locators refer to the [author version](#) (390 pp.).
- [16] William DeMeo. Interval enforceable properties of finite groups, 2014. [arXiv:1205.1927v4](#), April 7, 2014.
- [17] Michael F. Dowd and Peter Sin. On representations of algebraic groups in characteristic two. *Communications in Algebra*, 24(8):2597–2686, 1996. [DOI](#); section and page locators use the [author version](#).
- [18] Skip Garibaldi and Daniel K. Nakano. Bilinear and quadratic forms on rational modules of split reductive groups. *Canadian Journal of Mathematics*, 68(2):395–421, 2016. [DOI](#); [published PDF](#).
- [19] George Grätzer and E. Tamás Schmidt. Characterizations of congruence lattices of abstract algebras. *Acta Scientiarum Mathematicarum (Szeged)*, 24:34–59, 1963. [Introduction scan](#), pp. 34–37.
- [20] Joke Hästö. Growth of cross-characteristic representations of finite quasisimple groups of Lie type. *Journal of Algebra*, 407:275–306, 2014. [DOI](#); [arXiv:1112.3941v2](#). Table locators use this version.
- [21] Jens Carsten Jantzen. *Representations of Algebraic Groups*, volume 107 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, second edition, 2003. [DOI](#).
- [22] Peter B. Kleidman and Martin W. Liebeck. *The Subgroup Structure of the Finite Classical Groups*, volume 129 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 1990. [DOI](#).
- [23] Vicente Landazuri and Gary M. Seitz. On the minimal degrees of projective representations of the finite Chevalley groups. *Journal of Algebra*, 32(2):418–443, 1974. [DOI](#).
- [24] Michael J. Larsen and Richard Pink. Finite subgroups of algebraic groups. *Journal of the American Mathematical Society*, 24(4):1105–1158, 2011. [DOI](#); [published offprint](#).
- [25] Martin W. Liebeck and Gary M. Seitz. On the subgroup structure of classical groups. *Inventiones Mathematicae*, 134(2):427–453, 1998. [DOI](#); [author-uploaded text](#).
- [26] Frank Lübeck. Small degree representations of finite Chevalley groups in defining characteristic. *LMS Journal of Computation and Mathematics*, 4:135–169, 2001. [DOI](#); [published PDF](#).
- [27] George F. McNulty. A juggler’s dozen of easy problems. *Algebra Universalis*, 74(1-2):17–34, 2015. [DOI](#).
- [28] Péter P. Pálffy. Subgroups of twisted wreath products. In C. M. Campbell, C. W. Parker, M. R. Quick, E. F. Robertson, and C. M. Roney-Dougal, editors, *Groups St Andrews 2017 in Birmingham*, volume 455 of *London Mathematical Society Lecture Note Series*, pages 455–468. Cambridge University Press, Cambridge, 2019. [DOI](#); [author version](#).
- [29] Péter Pál Pálffy and Pavel Pudlák. Congruence lattices of finite algebras and intervals in subgroup lattices of finite groups. *Algebra Universalis*, 11:22–27, 1980. [DOI](#); [article scan](#).

- [30] Pavel Pudlák and Jiří Tůma. Every finite lattice can be embedded in a finite partition lattice. *Algebra Universalis*, 10:74–95, 1980. [DOI](#).
- [31] Vladimir Reprnitskiĭ and Jiří Tůma. Intervals in subgroup lattices of countable locally finite groups. *Algebra Universalis*, 59(1–2):49–71, 2008. [DOI](#); [author preprint](#).
- [32] Mark Schaffer. Twisted tensor product subgroups of finite classical groups. *Communications in Algebra*, 27(10):5097–5166, 1999. [DOI](#).
- [33] Gary M. Seitz. The maximal subgroups of classical algebraic groups. *Memoirs of the American Mathematical Society*, 67(365):iv+286, 1987. [DOI](#).
- [34] Robert Steinberg. Representations of algebraic groups. *Nagoya Mathematical Journal*, 22:33–56, 1963. [DOI](#); [published PDF](#).
- [35] Robert Steinberg. *Lectures on Chevalley Groups*. Yale University, Department of Mathematics, New Haven, CT, 1967. Notes prepared by John Faulkner and Robert Wilson; [original Yale scan](#), with acknowledgment dated June 1968. Locators use original pagination.
- [36] Robert Steinberg. Generators, relations and coverings of algebraic groups, II. *Journal of Algebra*, 71(2):527–543, 1981. [DOI](#); [original article PDF](#).
- [37] Donald E. Taylor. *The Geometry of the Classical Groups*, volume 9 of *Sigma Series in Pure Mathematics*. Heldermann Verlag, Berlin, 1992. Locators refer to the [author PDF corrected April 2021](#).
- [38] Ewald Warning. Bemerkung zur vorstehenden arbeit. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 11:76–83, 1935. [DOI](#).